Assortment Optimization in the Presence of Focal Effect: Operational Insights and Efficient Algorithms

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Abstract: The assortment provided by the seller can influence customers' evaluation of item utility. A possible consequence is that certain items in an assortment become "focal items" to customers, and customers overevaluate their utilities. We call such a phenomenon the *focal effect*. Kovach and Tserenjigmid (2022) recently propose a focal Luce model (FLM) to describe customers' choices in the presence of the focal effect. The merit of the FLM lies in its flexibility to model different consumer psychology that leads to different choice behaviors. In this paper, we use the FLM to capture several scenarios where the focal effect occurs and consider the associated assortment optimization problems. The first scenario we consider is when the focal effect arises from item ranking, and customers prefer items that appear at certain positions in the ranking. This scenario captures the case where customers have a preference for the cheapest items in the assortment, and the compromise effect. The second scenario describes the case where the focal effect is on some predetermined items. An example of this scenario is choice overload, where customers are more likely to choose the no-purchase option when the assortment size is larger. We characterize the optimal assortment structure under each scenario and give the underlying operational insights. We find that the assortment optimization problems under these two scenarios can be solved in polynomial time with some practical assumptions. The polynomial-solvability extends even to the more challenging joint assortment and pricing optimization problems. Finally, we conduct numerical experiments to evaluate the FLM's performance using both synthetic and real datasets. The results show that the FLM performs well in predicting customer choice behavior and the corresponding optimal assortment generates higher profit than benchmark assortments when the focal effect exists.

Key words: Focal effect, assortment optimization, focal Luce model

1 Introduction

Assortment optimization is one of sellers' most important techniques for boosting profit. The problem involves choosing a subset (i.e., assortment) from a universe of items for display to maximize the total expected profit. A key ingredient of the problem is the *choice model*, which describes the customer choice behavior and computes the purchase probability of each item in the assortment. When the choice model accurately reflects real-world customer behavior, the optimal assortment is more likely to generate higher profit (Chen et al. 2022).

When customers are presented with an assortment, their perception of an item's utility can be significantly shaped by the composition of that assortment. This paper studies situations where certain items within an assortment become "focal items", which may be over-evaluated by customers. We call this phenomenon the *focal effect*. The focal effect has been observed in various scenarios. For instance, Cao et al. (2022) find that in the airline industry, the cheapest available fare class of a flight often receives most of the bookings, regardless of the fare class. Moreover, the fraction of customers who choose the cheapest fare class is often much greater than that predicted by the multinomial logit (MNL) model calibrated from data. In this case, the cheapest fare class is the focal item and its price ranking among available offerings gives rise to the focal effect. In another example, Dean (2008) finds that customers are more likely to choose the status quo option when presented with larger assortments. When the status quo option is the no-purchase option, this behavior is known as choice overload (Chernev et al. 2015). In this scenario, the difficulty in evaluating numerous options makes the status quo option a focal item for decision-making.

The Luce model, also known as the MNL model, is the most widely used choice model in the literature (Train 2009, Gallego et al. 2019). However, it cannot explain the focal effect because it assumes that the attraction value of each item is fixed in any assortment, and the choice probability of an item is proportional to its attraction value. As a result, in the presence of the focal effect, it underestimates the choice probability of the focal items (Cao et al. 2022).

Kovach and Tserenjigmid (2022) propose a focal Luce model (FLM) to characterize the focal effect by modifying the MNL model. Specifically, two elements are central to capturing the focal effect: (1) the identification of the set of focal items in each assortment and (2) the quantification of the focal items' attraction value increment. Based on these considerations, Kovach and Tserenjigmid (2022) introduce two functions: *the focal function* to identify the focal items in any assortment, and *the distortion function* to quantify the attraction value increment of focal items. With focal items' attraction values modified, FLM uses a probability computation rule similar to the MNL model, where the choice probability is proportional to the modified attraction value.

Since Kovach and Tserenjigmid (2022) do not specify the explicit forms of these functions, the FLM serves as a flexible modeling framework. In this paper, we leverage the FLM to characterize various scenarios where the focal effect occurs and study assortment optimization problems in these contexts.

In the first scenario, we consider the case where the focal effect arises from the ranking of items. For instance, when items are sorted in a certain manner, customers may prefer the most "attractive" items in an assortment. If the attraction values have a rank the same as the items' prices from low to high, then the focal items are the cheapest items in the assortment. Similarly, if items are sorted by sales amount, the focal items are those with the highest sales. We specify the focal function and distortion function to capture this scenario. Under the model where top-ranked items are focal, our analysis shows that there exists a "truncated-profit-ordered" optimal assortment. Specifically, when considering the attraction value ranking, the optimal assortment comprises two subsets: a subset containing items whose attraction values are larger than a threshold and a profit-ordered subset (which may be empty) chosen from the remaining items whose attraction values are smaller than the same threshold. Compared to the profit-ordered optimal assortment under the MNL model, the truncated-profit-ordered assortment includes not only items with high profits but also items whose attraction values are high. With this property, we devise an efficient algorithm to solve the associated assortment optimization problem.

In some cases, the focal item may not occupy the top rank but instead the middle rank. This phenomenon, known as the compromise effect (Simonson and Tversky 1992), occurs when customers perceive the middle option as a balanced compromise between extremes. We also adapt the FLM to capture this scenario and develop an efficient algorithm to find the optimal assortment.

The second scenario where the focal effect emerges is when the focal item has distinct attributes or obvious advantages over other items in the assortment. In such a scenario, the focal item is predetermined and does not depend on the assortment. For instance, the "status quo bias" is observed when customers tend to select the default option to simplify their decision-making process, especially when faced with a large assortment. In another example, an item may become focal because of the seller's advertisement, marketing efforts, or strategic placement of the item (Zhang 2006, Mehta et al. 2008, Terui et al. 2011). We also adopt FLM to capture this scenario, and we let the focal item be fixed. We find that the optimal assortment satisfies a *floating threshold property*, meaning that an item will be included in the optimal assortment if its profit exceeds a threshold, which varies among items. When the level of focal effect increases, this property provides insights into how the optimal assortment will change. For the assortment optimization problem, in the general case, we prove that it is NP-hard. However, when the distortion function has certain structures, we characterize the transition in the complexity of assortment optimization from polynomial-time solvable to NP-hard.

An interesting example in the fixed focal item scenario is choice overload, where customers prefer to choose the no-purchase option when the assortment size is large. We analyze the property of the optimal assortment in this case and find that when choice overload intensifies, the optimal assortment size may expand rather than shrink. The intuition behind this counterintuitive result is that, as the impact of choice overload gets stronger, the seller needs to increase the total purchase probability to mitigate its effect. Consequently, including highly attractive items becomes beneficial. These items are usually excluded from the optimal assortment when the choice overload effect is weak because their profits are not large enough. Thus, the optimal assortment may expand when the choice overload becomes more significant.

We also study the joint assortment and pricing optimization problem under the aforementioned scenarios. We model price as a continuous decision variable and use different techniques to analyze this problem. We show that this more complex problem can be transformed into a one-dimensional root-finding problem and efficiently solved. Additionally, we identify several structural properties when the focal effect intensifies. For instance, when the choice overload effect gets stronger, the seller should decrease the optimal price, and the optimal assortment size depends on the property of the distortion function.

In addition to theoretical results, we conduct numerical experiments using synthetic and real datasets. Our findings indicate that the FLM is better able to predict customer choice behavior than the MNL model and achieves comparable performance with the general attraction model. Moreover, it also improves the seller's profit in the presence of the focal effect.

To summarize, we apply the FLM to different scenarios where the focal effect arises and consider the assortment optimization problems and joint assortment and pricing optimization problems. Though finding the optimal assortment is generally NP-hard, our analysis shows that by properly defining the focal function and distortion function, the FLM well captures customer choice behavior, and the decision problems admit computational tractability. The numerical studies show that the FLM has strong predictive power on customers' choices and helps improve the seller's profit.

In the rest of this section, we review the related literature. The paper is then organized as follows. We introduce the FLM in Section 2 and use examples to illustrate how the model works. In Section 3 and Section 4, we apply the FLM to capture two different scenarios where the focal effect occurs. We provide operational insights about the seller's optimal assortments and develop efficient algorithms for assortment optimization problems. In Section 5, we investigate the joint assortment and pricing optimization problems under these scenarios. Section 6 shows the performance of the FLM on two real datasets. Finally, we summarize our key findings and insights in Section 7.

1.1 Related Literature

Assortment optimization is one of the most crucial problems studied in profit management. To capture customer substitution behavior, utility-based models are widely used, such as the MNL model (Train 2009, Luce 2012). Talluri and van Ryzin (2004) consider the corresponding assortment optimization problem and show that there exists a profit-ordered optimal assortment, which can be found in polynomial time. This tractability is further extended to the robust optimization setting

(Rusmevichientong and Topaloglu 2012, Désir et al. 2023) and totally unimodular constrained setting (Davis et al. 2013). However, the MNL model suffers from the independence of irrelevant alternatives (IIA) property (Train 2009), which limits its application. To address this drawback, more sophisticated models are proposed, and the associated assortment optimization problems are studied. For example, assortment optimization problems under the nested logit model (Davis et al. 2014, Gallego and Topaloglu 2014, Kunnumkal 2023), the mixed MNL model (Méndez-Díaz et al. 2014, Rusmevichientong et al. 2014) and the Markov chain choice model (Blanchet et al. 2016, Feldman and Topaloglu 2017) have been considered. Recently, Berbeglia et al. (2022) conduct a comprehensive empirical study to compare different choice models.

However, all the aforementioned choice models fail to describe the well-observed phenomenon where the assortment influences items' utilities. To capture such focal effect, Kovach and Tserenjigmid (2022) propose the FLM. They provide behavioral foundations for the FLM and discuss how to identify focal items with transaction data. Furthermore, they demonstrate that by incorporating structured focal functions and distortion functions, the FLM encompasses a wide range of models that reflect bounded rationality.

There have been other choice models considering scenarios similar to the focal effect. For example, Gallego et al. (2015) propose a general attraction model (GAM), where the attraction value of the no-purchase option is amplified by a value depending on items not included in the assortment. Cao et al. (2022) propose the spiked MNL (SMNL) model to capture the phenomenon where customers prefer the cheapest item in an assortment.

In certain situations, multiple items in the assortment may have their utilities affected by other offered items. For example, Wang and Wang (2017) incorporate the *network effect* into the choice model to capture this phenomenon, where each item's utility is a function of its market share. Later, Lo and Topaloglu (2019) study the MNL model with *the synergy effect*, where an item's *attraction value* depends on those of other provided items. Following a similar spirit, Maragheh et al. (2020) and Najafi et al. (2023) propose different contextual MNL (CMNL) models that assume the *utility* of each item in an assortment is affected by all other offered items. However, these models need to characterize the relation between all item pairs, which requires estimating $\Theta(n^2)$ parameters. Recently, Bai et al. (2023) propose an endogenous context-dependent MNL model, where the utility of each item depends on both the item's intrinsic utility and the deviation of the intrinsic utility from the expected maximum utility among all the items in the assortment. They only need to estimate one more parameter than MNL, and thus the model can be easily estimated.

There have also been related studies in non-parametric choice models. For example, Chen and Mišić (2022) propose the decision forest model to capture the irrational choice behavior. They show that the model can represent all possible discrete choice models. Furthermore, Akchen and Mišić

(2021) study the assortment optimization problem under this model. Berbeglia and Venkataraman (2018) also propose a generalized stochastic preference model that incorporates the non-rationality into the stochastic preference (or equivalently ranking-based) choice model.

The FLM is similar to the consider-then-choose model, which adopts a two-stage framework to model customer choice behavior. Specifically, each customer first selects a *consideration set* from a given assortment and then only chooses items from the consideration set. Various methods have been proposed in the literature to form the consideration set. For example, Wang and Sahin (2018) incorporate the search cost and assume that the customer forms the consideration set by choosing a subset that maximizes the expected utility after subtracting the total search cost. Aouad et al. (2021) model the formation of the consideration set as a process of applying screening rules onto items and propose three different models with different screening rules. Both Gallego and Li (2024) and Aouad et al. (2024) assume that each item is included in the consideration set independently with a fixed probability. Bai et al. (2024) assume that the consideration set consists of items with top-K largest utilities. In this paper, we use a similar way to form the focal set. However, the customers are allowed to purchase items outside of the focal set, and the attraction values of items in the focal set are amplified.

In addition, the formation of the focal set in the FLM is also related to the literature on parallel search (where customers consider a prespecified number of items simultaneously to make a choice, see, e.g., Vishwanath 1992, Dukes and Liu 2016), sequential search (where customers evaluate items sequentially until they reach an optimal stopping time to make a choice, see, e.g., Weitzman 1979, Kosilova and Alptekinoglu 2024), and rational inattention (where customers need to pay a cost to learn the random utility of each item, see, e.g., Matějka and McKay 2015, Maćkowiak et al. 2023). However, the FLM differs from these streams of literature in two ways. First, it allows customers to purchase items outside of the focal set. Second, the focal set is not formed sequentially and does not incur extra cost.

Finally, we notice that there is a concurrent work by Guan et al. (2023), which also studies the assortment optimization problem under the FLM. In comparison, their work concentrates on a threshold focal set, while we study the cases where the focal effect arises from item ranking and where focal items are fixed. We also characterize the complexity of the assortment optimization problem under various distortion functions and provide more operational insights into the seller's assortment and pricing decisions in the presence of the focal effect.

2 Focal Luce Model

In this section, we introduce the FLM and formulate the seller's assortment optimization problem. Let $\mathcal{N} = \{1, 2, \dots, n\}$ be a finite set of items, and 0 be the no-purchase option. We use $\mathcal{S}_+ = \mathcal{S} \cup \{0\}$

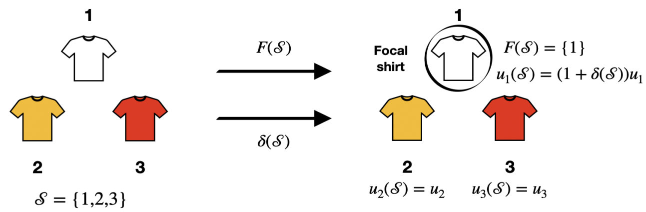

Figure 1 Illustration of the FLM (Saito 1996). In this model, the white shirt becomes the focal item. The intrinsic attraction value for each item is u_i, and $u_i(\mathcal{S})$ denotes item i's attraction value in assortment $\mathcal{S}$.

to denote the extended item set for all $\mathcal{S} \subseteq \mathcal{N}$. Each item $i \in \mathcal{N}$ is associated with price p_i, cost c_i, and profit $r_i := p_i - c_i \geq 0$. For the no-purchase option, we let $p_0 = c_0 = 0$.

Before introducing FLM, we first review the classical Luce model, i.e., the MNL model. In the MNL model, each item $i \in \mathcal{N}_+$ has an intrinsic utility v_i. The customer's random utility for item $i \in \mathcal{N}_+$ is $v(i) = v_i + \epsilon_i$, where $\{\epsilon_i\}_{i \in \mathcal{N}_+}$ are i.i.d. standard Gumbel random variables. Given an assortment $\mathcal{S}$, the customer purchases item $i \in \mathcal{S}_+$ with probability $\mathbb{P}(i, \mathcal{S}) = \mathbb{P}(v(i) \geq v(j), \forall j \in \mathcal{S}_+, j \neq i) = u_i / \sum_{j \in \mathcal{S}_+} u_j$, where $u_i := \exp(v_i)$ is the attraction value of item i. We normalize the attraction values by u_0 in this paper, and thus $u_0 = 1$.

Based on the MNL model, to capture the focal effect, the FLM introduces a *focal function* $F(\cdot)$ and a *distortion function* $\delta(\cdot)$. Specifically, the focal function $F : 2^{\mathcal{N}_+} \to 2^{\mathcal{N}_+}$ selects focal items from a given assortment $\mathcal{S}$, thereby forming a *focal set*. In the airline industry example in Section 1, for a given assortment $\mathcal{S}$, the focal set $F(\mathcal{S})$ contains the cheapest fare class. In the choice overload example, we have $F(\mathcal{S}) = \{0\}$ because the no-purchase option is the focal item.

The distortion function $\delta : 2^{\mathcal{N}_+} \to \mathbb{R}_{++}$ quantifies the attraction value increment of items in the focal set. Given $\mathcal{S}$, FLM assumes that the customer's utility for item i is $v(i, \mathcal{S}) = v_i + \log(1 + \delta(\mathcal{S}) \cdot \mathbf{1}\{i \in F(\mathcal{S})\}) + \epsilon_i$, where $\mathbf{1}\{\cdot\}$ indicates whether an item is in the focal set or not. If $i \notin F(\mathcal{S})$, then its utility under the FLM is the same as that under the MNL model. If $i \in F(\mathcal{S})$, then its utility increases by $\log(1 + \delta(\mathcal{S})) > 0$. Thus, the choice probability of item $i \in \mathcal{S}$ is given by

$$\mathbb{P}(i, \mathcal{S}) = \mathbb{P}(v(i, \mathcal{S}) \geq v(j, \mathcal{S}), \forall j \in \mathcal{S}_+, j \neq i) = \frac{u_i(\mathcal{S})}{\sum_{j \in \mathcal{S}_+} u_j(\mathcal{S})},$$

where $u_i(\mathcal{S}) := u_i(1 + \delta(\mathcal{S}) \cdot \mathbf{1}\{i \in F(\mathcal{S})\})$. Note that u_i is the fixed attraction value of item i, and $u_i(\mathcal{S})$ is the attraction value of item i with the focal effect for a given assortment $\mathcal{S}$. When the focal set is empty or contains all items in $\mathcal{S}_+$, FLM reduces to the MNL model. Figure 1 gives an illustration of the "focal machine" in the FLM.

Under the FLM, the focal set is different from the consideration set in the consider-then-choose model since the attraction values of items in the focal set are amplified by the distortion function, while the consideration set does not allow items' attraction values to change. Moreover, the FLM allows customers to purchase items outside of the focal set, which is supported by some empirical study (Hedgcock et al. 2016). In contrast, the consider-then-choose model assumes customers only purchase within the consideration set.

The seller aims to maximize the total expected profit by providing an optimal assortment to customers. Therefore, the seller solves the following assortment optimization problem:

$$\max_{\mathcal{S}\subseteq\mathcal{N}} f(\mathcal{S}) := \frac{\sum_{i\in\mathcal{S}} r_i u_i(\mathcal{S})}{\sum_{i\in\mathcal{S}_+} u_i(\mathcal{S})}. \tag{1}$$

We denote the optimal assortment by $\mathcal{S}^*$.

Since the FLM does not prescribe specific forms to $F(\mathcal{S})$ and $\delta(\mathcal{S})$, it serves as a modeling framework. Hence, throughout the paper, we vary different exogenous $F(\mathcal{S})$ and $\delta(\mathcal{S})$ to characterize different consumer psychology in practice. We also devise efficient algorithms for the seller's decision problems and provide managerial insights for each case. In the remainder of the section, we give some concrete examples, which will be generalized in the following sections.

EXAMPLE 1 (MAXIMIZING ATTRACTION VALUE). Suppose that items in $\mathcal{N}$ are labeled based on their attraction values in descending order, i.e., $u_1 \ge u_2 \ge \cdots \ge u_n$. Let the focal function be

$$F(\mathcal{S}) = \{i \in \mathcal{S} : u_i \ge u_j, j \in \mathcal{S}\}, \forall \mathcal{S} \subseteq \mathcal{N}, \tag{2}$$

which implies that the focal set of the assortment $\mathcal{S}$ contains the item(s) with the highest attraction value. This focal function is a natural choice because the item with the largest attraction value excels over other items in all attributes in many scenarios, and thus it is the most attractive item.

The focal function (2) reflects customers' preference for "extreme" items. This behavior is due to the customer's extremeness-seeking psychology (Gourville and Soman 2007) and is independent of $\mathcal{S}$. Hence, we can set the distortion function to be a constant, i.e., let

$$\delta(\mathcal{S}) = \delta, \text{ for all } \mathcal{S} \subseteq \mathcal{N}. \tag{3}$$

When the attraction value ranking is the same as price rank from low to high, i.e., items are ranked with $p_1 \le p_2 \le \cdots \le p_n$, the model defined by (2) and (3) captures the scenario described in Cao et al. (2022). They show that in the airline industry, the fraction of customers who choose the cheapest available fare class is much greater than that predicted by the MNL model calibrated with real data. To characterize this phenomenon, they propose a spiked MNL model.[1] ◁

[1] Cao et al. (2022) also assume a constant distortion on the cheapest item's attraction value when conducting the numerical experiments.

EXAMPLE 2 (COMPROMISE EFFECT). Suppose that items are sorted by some criteria, such as sales amount or display position. Let $\sigma(i,\mathcal{S})$ be the rank of item i in set $\mathcal{S}$ for all $\mathcal{S}\subseteq\mathcal{N}$, we consider the case where the focal function is given by

$$F(\mathcal{S})=\{i:\sigma(i,\mathcal{S})=\min\{|\mathcal{S}|,\tau\},i\in\mathcal{S}\},\text{ for all }\mathcal{S}\subseteq\mathcal{N},\tag{4}$$

where $\tau\in\{1,2,\ldots,n\}$. When $\tau=1$ and σ is the attraction value ranking, it reduces to Example 1. When $\tau\geq 2$, the focal item in $\mathcal{S}$ is no longer the item ranking first. Instead, it may be the middle-ranked item in $\mathcal{S}$. In this case, the focal function (4) captures the well-known compromise effect in customer choice behavior, which is empirically observed by Simonson and Tversky (1992) and studied by Berbeglia and Venkataraman (2018). This phenomenon describes that the customers are more likely to choose the middle option when presented with an assortment that includes multiple items with varying degrees of quality or price. This effect occurs because the middle option appears as a balanced compromise between extreme items and is thus perceived as the most reasonable choice. Both Example 1 and this case show that when items are sorted by certain criteria, customers may have a preference for items that take certain positions in the ranking. ◁

EXAMPLE 3 (CHOICE OVERLOAD). Consider the case where the focal set contains only the no-purchase option. Then, the focal function is

$$F(\mathcal{S})=\{0\},\ \text{ for all }\mathcal{S}\subseteq\mathcal{N}.\tag{5}$$

This captures choice overload, which refers to the phenomenon that customers are less likely to make a purchase when faced with a larger assortment. This effect has been observed in multiple industries (e.g., Scheibehenne et al. 2010). To capture the customers' dynamic preference for the no-purchase option, we can let $\delta(\mathcal{S})\leq\delta(\mathcal{S}')$ for all $\mathcal{S}\subseteq\mathcal{S}'$. It means that the attraction value of the no-purchase option under the FLM will increase when more items are included in the assortment. Chernev and Hamilton (2009) show that customers' cognitive cost is proportional to the assortment size. Thus, it is also reasonable to make $\delta(\mathcal{S})$ as a function of the assortment size. ◁

In the following sections, we study the decision problems under the FLM. The assortment optimization problem is discussed in Section 3 and Section 4, and the joint assortment and pricing optimization problem is discussed in Section 5.

3 Assortment Optimization with Ranking-Based Focal Items

In this section, we consider the scenario where there is a ranking of items such that focal items are those appearing at certain positions in this ranking. We first specify different models that generalize Example 1 and Example 2 to capture this scenario, then we devise efficient algorithms for the assortment optimization problems and explore insights provided by these models.

3.1 Top-K Focal Model

The customers usually pay more attention to the top-ranked items. To capture this case, we label items in $\mathcal{N}$ based on a ranking function σ. Formally, the items are labeled such that $\sigma(1,\mathcal{N}) < \sigma(2,\mathcal{N}) < \cdots < \sigma(n,\mathcal{N})$, where $\sigma(i,\mathcal{S})$ is the rank of item i in set $\mathcal{S}$ for all $\mathcal{S} \subseteq \mathcal{N}$. Then, the focal function is defined as follows:

$$F(\mathcal{S}) = \{i : \sigma(i,\mathcal{S}) \le \min\{K, |\mathcal{S}|\}, i \in \mathcal{S}\}, \forall \mathcal{S} \subseteq \mathcal{N}, \tag{6}$$

meaning that the focal set contains at most K top-ranked items. We do not specify the criterion that defines the item ranking, and it can be price, sales amount, attraction value, display position, etc. For example, when items are labeled such that $u_1 > u_2 > \cdots > u_n$ and $K = 1$, it reduces to the model in Example 1. When the items are sorted by price from low to high, then the focal set contains the K cheapest items in the assortment.

As discussed in Example 1, we first let the distortion function to be $\delta(\mathcal{S}) = \delta$ for all $\mathcal{S} \subseteq \mathcal{N}$. Later, we will consider a more general distortion function that depends on $\mathcal{S}$. With the focal function and the distortion function specified above, we show the optimal assortment has a special structure. We first review the definition of profit-ordered assortment as preparation.

Definition 1 (profit-ordered Assortment in a Set). Given an item set $\mathcal{N}$, an assortment $\mathcal{S}$ is a *profit-ordered assortment* in $\mathcal{N}$ when there exists a scalar $\theta > 0$ such that $r_i \ge \theta$ for all $i \in \mathcal{S}$ and $r_i < \theta$ for all $i \in \mathcal{N} \setminus \mathcal{S}$.

For the assortment optimization problem under the MNL model, it is well-known that there exists a profit-ordered assortment that is optimal (Talluri and van Ryzin 2004). Under the top-K focal model, we prove that there exists a "K-truncated-profit-ordered" optimal assortment.

Definition 2 (K-Truncated-profit-Ordered Assortment). Let $\mathcal{R}_i := \{\ell \in \mathcal{N} : \ell > i\}$ contain items ranking after i. An assortment $\mathcal{S}$ is a K-truncated-profit-ordered assortment if either of the two cases occurs: (1) $|\mathcal{S}| \le K$; (2) $|\mathcal{S}| > K$ and $\mathcal{S} = \mathcal{S}_{i_K}^{\mathrm{RO}} \cup \{i_1, i_2, \ldots, i_K\}$ with $i_1 < i_2 < \cdots < i_K$, where $\mathcal{S}_{i_K}^{\mathrm{RO}}$ is a profit-ordered assortment in $\mathcal{R}_{i_K}$, and $\{i_1, i_2, \ldots, i_K\}$ is the focal set in $\mathcal{S}$.

The above definition states that, if an assortment is K-truncated-profit-ordered, it is either of size no more than K or can be decomposed into a set including K focal items and a profit-ordered assortment chosen from a truncated set $\mathcal{R}_{i_K}$. Here, the truncated set $\mathcal{R}_{i_K}$ includes items in $\mathcal{N}$ with ranks after the least-favored focal item i_K.

Proposition 1. *Suppose that the focal function is of form (6) and the distortion function is a constant. Then, there exists a K-truncated-profit-ordered optimal assortment to (1).*

Proposition 1 shows that, unlike the MNL model, the optimal assortment under the FLM may not necessarily be profit-ordered, but instead there must exist a K-truncated-profit-ordered assortment that is optimal. This highlights the importance of providing assortments that align with consumer preferences. For instance, when the ranking is based on items' attraction values, low-profit but high-attraction-value items may be included in the optimal assortment and become focal items.

Now, we consider the assortment optimization problem under this setting, and we give a polynomial-time algorithm for it. We start from the case where $|\mathcal{S}^*| \leq K$. In this case, all items in $\mathcal{S}^*$ are focal items, and we only need to solve the following problem and include its optimal solution into the candidate set of $\mathcal{S}^*$,

$$\max_{\mathcal{S}\subseteq\mathcal{N}} \left\{ \frac{\sum_{i\in\mathcal{S}}(1+\delta)r_i u_i}{1+\sum_{i\in\mathcal{S}}(1+\delta)u_i} : |\mathcal{S}| \leq K \right\}. \tag{7}$$

This problem shares the same structure as the cardinality-constrained assortment optimization problem under the MNL model where items' attraction values are $u'_i = (1+\delta)u_i$ for all $i \in \mathcal{N}$. Davis et al. (2013) have shown that this problem can be solved through a linear program. Thus, problem (7) can also be solved through a linear program.

When $|\mathcal{S}^*| > K$, we enumerate all possible choices of the largest-labeled focal item in $\mathcal{S}^*$ and use k to denote its label. Note that we must have $k \in \mathcal{R}_{K-1} = \{i \in \mathcal{N} : i > K-1\}$. Otherwise, item k cannot be the largest-labeled focal item in $\mathcal{S}^*$. Meanwhile, we have $k \neq n$ because the case $k = n$ implies that $|S^*| = K$. Then, we divide $\mathcal{S}^* \setminus \{k\}$ into two non-overlapping subsets — $\mathcal{S}_1^*$ containing focal items and $\mathcal{S}_2^*$ containing non-focal items. The optimization problem can be written as

$$\begin{aligned} \max_{\mathcal{S}_1\subseteq\mathcal{L}_k, \mathcal{S}_2\subseteq\mathcal{R}_k} \quad & \frac{(1+\delta)r_k u_k + (1+\delta)\sum_{i\in\mathcal{S}_1} r_i u_i + \sum_{j\in\mathcal{S}_2} r_j u_j}{1+(1+\delta)u_k + (1+\delta)\sum_{i\in\mathcal{S}_1} u_i + \sum_{j\in\mathcal{S}_2} u_j} \\ \text{s.t.} \quad & |\mathcal{S}_1| = K-1, \end{aligned} \tag{8}$$

where $\mathcal{L}_k := \{i \in \mathcal{N} : i < k\}$ contains items with labels smaller than k. To solve the above problem, we introduce the next lemma.

LEMMA 1. *Let $u'_i = (1+\delta)u_i$ for all $i \in \mathcal{L}_k \cup \{k\}$, and $u'_j = u_j$ for all $j \in \mathcal{R}_k$. Denote by $(x_k^*, \boldsymbol{x}^*, \boldsymbol{y}^*, z^*)$ an optimal solution to the following problem,*

$$\begin{aligned} \max_{x_k, \boldsymbol{x}\in\mathbb{R}^{|\mathcal{L}_k|}, \boldsymbol{y}\in\mathbb{R}^{|\mathcal{R}_k|}, z} \quad & r_k x_k + \sum_{i\in\mathcal{L}_k} r_i x_i + \sum_{j\in\mathcal{R}_k} r_j y_j \\ \textit{s.t.} \quad & x_k + \sum_{i\in\mathcal{L}_k} x_i + \sum_{j\in\mathcal{R}_k} y_j + z = 1, \\ & x_k/u'_k = z, \\ & 0 \leq x_i/u'_i \leq z \quad \forall i \in \mathcal{L}_k, \\ & \sum_{i\in\mathcal{L}_k} x_i/u'_i = (K-1)z, \\ & 0 \leq y_j/u'_j \leq z \quad \forall j \in \mathcal{R}_k, \\ & x_k > 0, z > 0, \boldsymbol{x} \geq 0, \boldsymbol{y} \geq 0. \end{aligned} \tag{9}$$

Then the set $(\mathcal{S}_1^*, \mathcal{S}_2^*)$ *with* $\mathcal{S}_1^* = \{i : x_i^*/u_i' = z^*, \text{for all } i \in \mathcal{L}_k\}$ *and* $\mathcal{S}_2^* = \{j : y_j^*/u_j' = z^*, \text{for all } j \in \mathcal{R}_k\}$ *is an optimal solution to* (8).

Lemma 1 implies that problem (8) can be solved via a linear program in polynomial time. According to this lemma, we enumerate item k in $\mathcal{R}_{K-1} \setminus \{n\}$ and solve (8) for each k. Then, we include the optimal solutions to (8) for different k into the candidate set of $\mathcal{S}^*$. Afterward, we find the optimal assortment by letting it be the assortment in the candidate set with the largest expected profit. Algorithm 1 illustrates the details, and the time complexity is provided in Theorem 1.

Algorithm 1: Algorithm for solving (1) under top-K focal model and constant distortion

Input: K, $\{u_i\}_{i\in\mathcal{N}_+}$, $\{r_i\}_{i\in\mathcal{N}}$, $F(\mathcal{S})$ specified in (6), $\delta(\mathcal{S}) = \delta$.

Initialize: $\mathcal{SOL} \leftarrow \emptyset$;

Solve (7) via linear programming, denote its optimal solution by $\mathcal{S}_{\leq K}^*$;

Update $\mathcal{SOL} \leftarrow \mathcal{SOL} \cup \mathcal{S}_{\leq K}^*$;

Identify set $\mathcal{R}_{K-1}$;

for $k \in \mathcal{R}_{K-1} \setminus \{n\}$ **do**

  Identify sets $\mathcal{L}_k$ and $\mathcal{R}_k$;

  Solve (8) via Lemma 1, denote its optimal solution by $(\mathcal{S}_{k,1}^*, \mathcal{S}_{k,2}^*)$;

  $\mathcal{S}_k^* \leftarrow \mathcal{S}_{k,1}^* \cup \mathcal{S}_{k,2}^* \cup \{k\}$ and update $\mathcal{SOL} \leftarrow \mathcal{SOL} \cup \mathcal{S}_k^*$;

Output: The optimal solution $\mathcal{S}^* \in \arg\max_{\mathcal{S}\in\mathcal{SOL}} f(\mathcal{S})$ with optimal value $f(\mathcal{S}^*)$.

THEOREM 1. *Suppose that the focal function is of form* (6) *and the distortion function is a constant, Algorithm 1 returns the optimal assortment by solving* $n - K + 1$ *linear programs. The time complexity is* $O((n - K + 1)\phi_{LP})$*, where* ϕ_{LP} *is the time complexity of solving a linear program with* $O(n)$ *variables and constraints.*

The efficiency of Algorithm 1 relies on the polynomial-time algorithm for solving the constrained assortment optimization under the MNL model. Davis et al. (2013) show that this problem is polynomially solvable as long as the constraint matrix is *totally unimodular* (TU). Thus, when problem (8) includes additional constraints that preserve the TU property, the corresponding assortment optimization problem remains efficiently solvable.

We also consider a more general $\delta(\mathcal{S})$ to describe such focal effect, beyond the constant distortion function. In particular, we find that when $\delta(\mathcal{S})$ is modular in the focal set, i.e., $\delta(\mathcal{S}) = \rho_0 + \sum_{i\in F(\mathcal{S})} \rho_i$ for some non-negative ρ_i, the K-truncated profit-ordered property still holds. This implies that the assortment optimization problem can be solved efficiently as stated in Theorem 2.

THEOREM 2. *Suppose that the focal function is of form (6) and the distortion function is modular in the focal set, there exists a K-truncated profit-ordered optimal assortment. Moreover, the optimal assortment can be obtained in time $O(n^{K+1})$.*

3.2 Arbitrarily-Located Focal Model

In this subsection, we extend Example 2 to a more general setting. Specifically, given an assortment $\mathcal{S}$, we assume that the focal item ranks $h(|\mathcal{S}|)$ in $\mathcal{S}$ for some function $h:\mathbb{Z}_+ \mapsto \mathbb{Z}_{++}$, i.e., the focal function $\mathcal{F}(\mathcal{S})$ is given by

$$F(\mathcal{S}) = \{i : \sigma(i,\mathcal{S}) = h(|\mathcal{S}|)\}, \text{where } 1 \leq h(|\mathcal{S}|) \leq |\mathcal{S}|. \tag{10}$$

When $h(|\mathcal{S}|) = \tau$ for all $\mathcal{S} \subseteq \mathcal{N}$, it describes the customers' exclusive attention to the item ranked at τ within $\mathcal{S}$, regardless of the assortment size. For example, the customers may focus on the second item returned by the online shopping platform after a search because the first one is usually a sponsored search result and unreliable. In another example, when $h(|\mathcal{S}|) = \lceil |\mathcal{S}|/2 \rceil$, the model captures the phenomenon where customers focus on the middle-ranked item after inspecting the assortment's cardinality, which aligns with the compromise effect.

To find the optimal assortment under this model, we use a method similar to that in Section 3.1. But the difference is that we need to discuss the cardinality of the optimal assortment in this setting. Also, given $|\mathcal{S}^*| = C$, we need to enumerate all possible choices of the focal item that ranks $h(C)$. Assume it is item k. Then, we can still divide $\mathcal{S}^*$ into two disjoint sets $\mathcal{S}_1 \subseteq \mathcal{L}_k$ and $\mathcal{S}_2 \subseteq \mathcal{R}_k$ such that $|\mathcal{S}_1| = h(C) - 1$ and $|\mathcal{S}_2| = C - h(C)$. Then, we optimize for $(\mathcal{S}_1, \mathcal{S}_2)$ by solving a linear program since the cardinality constraint is a TU constraint. The time complexity is given in the following proposition.

PROPOSITION 2. *Suppose that the focal function is of form (10) and the distortion function is a constant, then there exists an algorithm returning the optimal assortment in the time complexity of $O(n^2\phi_{LP})$, where ϕ_{LP} is the time complexity of solving a linear program with $O(n)$ variables and constraints.*

When the distortion function is relaxed from a constant to a modular function in the focal set, the corresponding assortment optimization problem can still be solved efficiently. This also holds if TU constraints are considered. Meanwhile, the truncated-profit-ordered property is maintained when we take $h(|\mathcal{S}|) = \tau$ for all $\mathcal{S} \subseteq \mathcal{N}$.

PROPOSITION 3. *Suppose that the focal function is of form (10) with $h(|\mathcal{S}|) = \tau$ and the distortion function is modular in the focal set, then there exists a τ-truncated-profit-ordered optimal assortment.*

The above analysis considers that the focal set only contains one arbitrary-located focal item. We generalize the model to the case of multiple arbitrary-located focal items and find that the assortment optimization problem can also be solved efficiently. The detailed analysis of this case is provided in EC.4.1.

In summary, we consider the assortment optimization problem under the FLM when the focal effect arises due to item ranking in this section. In the next section, we consider the assortment optimization problem when the focal effect arises from a different mechanism.

4 Assortment Optimization with Fixed Focal Items

In this section, we use the FLM to capture the scenario where the customers focus on several fixed items regardless of the assortment $\mathcal{S}$, which is a generalization of Example 3. Formally, we assume that there exists a predetermined set $\mathcal{N}^* \subseteq \mathcal{N}_+$ consisting of focal items, and the focal function $F(\mathcal{S})$ is defined as follows:

$$F(\mathcal{S}) = \mathcal{S}_+ \cap \mathcal{N}^*, \text{ for all } \mathcal{S} \subseteq \mathcal{N}. \tag{11}$$

When $\mathcal{N}^*$ only contains the no-purchase option, it reduces to the choice overload case in Example 3. Throughout this section, we label the items such that $r_1 \geq r_2 \geq \cdots \geq r_n$. We first revisit Example 3 to investigate operational insight in the choice overload case. Then we show how to solve the assortment optimization problem.

4.1 Choice Overload and Expanded Assortment

We focus on the choice overload case with $\mathcal{N}^* = \{0\}$ in this subsection and explore more insights that the FLM can provide. We use a distortion function satisfying the following Assumption 1 to characterize choice overload. It implies that all items exert the same influence on $\delta(\mathcal{S})$, and thus the distortion function only depends on the assortment size.

ASSUMPTION 1 **(Homogeneous distortion function)**. *There exists a function $g\colon \mathbb{Z}_+ \to \mathbb{R}_+$ such that for all $\mathcal{S} \subseteq \mathcal{N}$, $\delta(\mathcal{S}) = g(|\mathcal{S}|)$.*

In the next example, we show that the optimal assortment depends on the distortion function under the choice overload effect.

EXAMPLE 4. Suppose that $r_i = r$ and $u_i = u$ for all $i \in \mathcal{N}$. The focal function $F(\mathcal{S}) = \{0\}$ for all $\mathcal{S} \subseteq \mathcal{N}$, and the distortion function satisfies Assumption 1. We also assume that $g(|\mathcal{S}|)$ is increasing in $|\mathcal{S}|$. Then, problem (1) becomes

$$\max_{\mathcal{S} \subseteq \mathcal{N}} f(\mathcal{S}) = \frac{\sum_{i \in \mathcal{S}} r_i u_i}{1 + \delta(\mathcal{S}) + \sum_{i \in \mathcal{S}} u_i} = \frac{ru|\mathcal{S}|}{1 + g(|\mathcal{S}|) + u|\mathcal{S}|}.$$

Thus, in this case, maximizing the expected profit is equivalent to maximizing the purchase probability. Figure 2 illustrates how the purchase probability changes with the assortment size under different distortion functions. It can be seen that the purchase probability will first increase and then decrease with respect to the assortment size, which matches the choice overload effect.

When the assortment consists of a few items, customers experience low cognitive cost. Hence, the purchase probability increases as more items are added to the assortment. However, when the assortment size is already large, the choice overload effect intensifies if the assortment further expands. The cognitive burden on customers increases and results in a decrease in purchase probability. The optimal assortment given by the FLM strikes a balance between mitigating the choice overload effect and increasing purchase probability. In contrast, the MNL model suggests the optimal assortment $\mathcal{S}^*_{\text{MNL}} = \mathcal{N}$, failing to account for choice overload. ◁

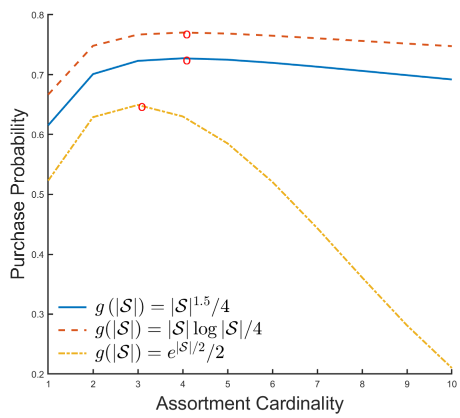

Figure 2 Assortment cardinality v.s. purchase probability under different $g(|\mathcal{S}|)$.

Next, we explore more insights under the choice overload case. Specifically, we are interested in how the optimal assortment changes as the choice overload effect gets more severe. Intuitively, one may guess that the optimal assortment should shrink to alleviate choice overload. However, the following example shows that it is not always the case. In EC.6.1, we identify a more general condition where the optimal assortment will shrink.

EXAMPLE 5 (EXPANDED ASSORTMENT). Consider a FLM with two items. The focal function $F(\mathcal{S}) = \{0\}$. Let $u_1 = 2$, $u_2 = 3$, $r_1 = 1, r_2 = 0.6$, and the distortion function

$$\delta_1(\mathcal{S}) = \begin{cases} 1, & \text{if } |\mathcal{S}| = 1 \\ 2, & \text{if } |\mathcal{S}| = 2 \end{cases}.$$

We let $f(\mathcal{S})$ be the expected profit of assortment $\mathcal{S}$ under the FLM for all $\mathcal{S} \subseteq \mathcal{N}$. We have $f(\{1\}) = 0.5 > f(\{1,2\}) = 0.475 > f(\{2\}) = 0.36$. Thus the optimal assortment is $\mathcal{S}^*_1 = \{1\}$.

Now we consider a larger distortion function

$$\delta_2(\mathcal{S}) = \begin{cases} 2.5, & \text{if } |\mathcal{S}| = 1 \\ 3.5, & \text{if } |\mathcal{S}| = 2 \end{cases}.$$

Note that $\delta_2(\mathcal{S}) = \delta_1(\mathcal{S}) + 1.5$. In this case, we have $f(\{1,2\}) = 0.4 > f(\{1\}) = 0.36 > f(\{2\}) = 0.27$. Thus, the optimal assortment under $\delta_2(\mathcal{S})$ is $\mathcal{S}_2^* = \{1,2\}$, and $|\mathcal{S}_1^*| < |\mathcal{S}_2^*|$. It means the optimal assortment becomes bigger as $\delta(\cdot)$ grows.

To understand this result, we consider the effect of increasing the distortion function from $\delta_1(\mathcal{S})$ to $\delta_2(\mathcal{S})$. This increase leads to a decrease in the expected profit of each assortment. One method to counteract this profit loss is to increase the purchase probability. Although adding new items to the assortment increases the attraction value of the no-purchase option, it also increases the total purchase probability. One can verify that when the assortment expands from $\{1\}$ to $\{1,2\}$, the total purchase probability increases from $1/2$ (under $\delta_1(\mathcal{S})$) to $11/19$ (under $\delta_2(\mathcal{S})$). The positive effect of the increased total purchase probability dominates the negative effect of the larger attraction value for the no-purchase option, thus resulting in a larger optimal assortment. $\triangleleft$

4.2 Floating Threshold Property

In this subsection, we let the prespecified set $\mathcal{N}^* = \{i_{\mathrm{F}}\}$, where $i_{\mathrm{F}} \in \mathcal{N}_+$ is not necessarily the no-purchase option. We consider the characterization of the optimal assortment. Under the MNL model, there exists a profit threshold $\nu > 0$ such that item i belongs to the optimal assortment if and only if $r_i \geq \nu$. In the following proposition, we prove a similar yet different result for the FLM.

PROPOSITION 4 **(Floating Threshold Property)**. *Suppose that the focal function is of form (11) and $\mathcal{N}^* = \{i_F\}$. Let $\mathcal{S}^*$ denote the optimal assortment to (1), and $f(\cdot)$ be defined in (1). We have:*

(1) When the focal item $i_F \neq 0$, we have $i_F \in \mathcal{S}^*$ if and only if $r_{i_F} \geq f_{MNL}(\mathcal{S}^*_{MNL})$, where

$$\mathcal{S}^*_{MNL} \in \arg\max_{\mathcal{S} \subseteq \mathcal{N} \setminus \{i_F\}} \frac{\sum_{i \in \mathcal{S}} u_i r_i}{1 + \sum_{i \in \mathcal{S}} u_i}, \text{ and } f_{MNL}(\mathcal{S}^*_{MNL}) = \frac{\sum_{i \in \mathcal{S}^*_{MNL}} u_i r_i}{1 + \sum_{i \in \mathcal{S}^*_{MNL}} u_i}.$$

Furthermore, if $i_F \in \mathcal{S}^*$, then $r_{i_F} \geq f(\mathcal{S}^*) \geq f(\mathcal{S}^*_{MNL}) = f_{MNL}(\mathcal{S}^*_{MNL})$.

(2) If a non-focal item $i \in \mathcal{S}^*$, then

$$r_i \geq f(\mathcal{S}^*) - \frac{u_{i_F}\left(\delta(\mathcal{S}^*) - \delta(\mathcal{S}^* \setminus \{i\})\right)\left(r_{i_F} - f(\mathcal{S}^*)\right)}{u_i}.$$

(3) If a non-focal item $i \notin \mathcal{S}^*$, then

$$r_i \leq f(\mathcal{S}^*) - \frac{u_{i_F}\left(\delta(\mathcal{S}^* \cup \{i\}) - \delta(\mathcal{S}^*)\right)\left(r_{i_F} - f(\mathcal{S}^*)\right)}{u_i}.$$

Note that $\mathcal{S}^*_{\text{MNL}}$ is the optimal assortment among the non-focal items under the MNL model. Proposition 4-(1) gives the necessary and sufficient condition for the focal item $i_{\text{F}} \in \mathcal{S}^*, i_{\text{F}} \neq 0$ under the FLM, which is similar to the condition under the MNL model. If $i_{\text{F}} \in \mathcal{S}^*$, we find that $r_{i_{\text{F}}} \geq f(\mathcal{S}^*) \geq f_{\text{MNL}}(\mathcal{S}^*_{\text{MNL}})$. Meanwhile, if the profit of focal item i_{F} is larger than the optimal expected profit given by the non-focal items, then it should be included in the optimal assortment. The addition of i_{F} will bring a profit increase under the FLM since the distortion function will amplify $u_{i_{\text{F}}}$ and does not affect other attraction values. Proposition 4-(1) can help us determine whether to incorporate i_{F} in $\mathcal{S}^*$. If $i_{\text{F}} \notin \mathcal{S}^*$, then assortment optimization under the FLM reduces to the one under the MNL model. Therefore, in the following, we focus on the case where $i_{\text{F}} \in \mathcal{S}^*$.

Claims (2) and (3) provide conditions for whether to include a non-focal item in the optimal assortment. Under the FLM, the threshold to include a non-focal item i depends on the sum of the optimal profit and an additional term depending on the attraction value u_i. Thus, the threshold differs among items.

When $i_{\text{F}} \neq 0$ and $\delta(\mathcal{S}^*) - \delta(\mathcal{S}^* \backslash \{i\}) > 0$, adding item i can capture more attention towards focal item i_{F} and reinforce its prominence within the assortment. Such a situation is similar to the "decoy effect" observed in Huber et al. (1982), where item i_{F} can be seen as the "target item" whose demand goes up when an inferior item is added to the assortment. Beyond the result in Huber et al. (1982), Proposition 4-(2) also shows how items' attraction values affect the level of decoy effect. The threshold for item i being included in $\mathcal{S}^*$ is lower than $f(\mathcal{S}^*)$, and increasing in u_i. Thus, less attractive items are easier to get into the optimal assortment compared to more attractive items.

When $i_{\text{F}} \neq 0$ and $\delta(\mathcal{S}^*) - \delta(\mathcal{S}^* \backslash \{i\}) < 0$, adding item i to the assortment decreases the level of focal effect and market share of item i_{F}. In this case, the threshold for item i being included in the optimal assortment is larger than $f(\mathcal{S}^*)$ and decreasing in u_i. Hence, it is only advantageous to include item i in $\mathcal{S}^*$ when it has a large attraction value and a high profit.

Proposition 4-(3) is about the threshold of excluding non-focal item i in the optimal assortment. Similar to Proposition 4-(2), this threshold can be either larger or smaller than $f(\mathcal{S}^*)$, depending on the marginal change of the focal effect of item i_{F}.

When $i_{\text{F}} = 0$, the model characterizes choice overload. When $\delta(\mathcal{S})$ is increasing in $\mathcal{S}$, i.e., $\delta(\mathcal{S}_2) \geq \delta(\mathcal{S}_1)$ for all $\mathcal{S}_1 \subseteq \mathcal{S}_2 \subseteq \mathcal{N}$, if a non-focal item $i \in \mathcal{S}^*$, then its profit must surpass $f(\mathcal{S}^*)$ by a positive amount, which varies among items. As u_i increases, the threshold decreases, meaning that items with higher attraction values are easier to be included in the optimal assortment. This observation implies that the optimal assortment tends to include fewer items with high attraction values when choice overload occurs. This finding echoes the result presented in Chernev and Hamilton (2009), which shows that customers prefer smaller assortments of highly attractive items.

4.3 Complexity of Assortment Optimization

In this subsection, we consider the complexity of assortment optimization problems under the FLM when the focal function $F(\mathcal{S}) = \mathcal{S}_+ \cap \mathcal{N}^*$, $\forall \mathcal{S} \subseteq \mathcal{N}$. We show that the problem complexity depends on the form of the distortion function. When the distortion function takes a general form, the assortment optimization is NP-hard. However, when the distortion function is equipped with some additional structures, the assortment optimization problem is fixed-parameter tractable.

THEOREM 3. *When the focal function $F(\mathcal{S}) = \mathcal{S}_+ \cap \mathcal{N}^*$ for all $\mathcal{S} \subseteq \mathcal{N}$, there exists a set of parameters with $\mathcal{N}^* = \{i_F\}$ and $\delta(\mathcal{S}) = g\left(\sum_{i\in\mathcal{S}} \rho_i\right)$ where $g: \mathbb{R} \to \mathbb{R}$ and $\{\rho_i\}_{i\in\mathcal{N}}$ are distortion parameters of items, such that no polynomial algorithm exists for the corresponding assortment optimization problem (1) unless $P = NP$.*

Theorem 3 shows that when each item has a different distortion parameter ρ_i and $\delta(\cdot)$ is a function of $\sum_{i\in\mathcal{S}} \rho_i$, finding the optimal assortment is NP-hard. In the rest of this subsection, we show that the number of distinct values in the set $\{\rho_i\}_{i\in\mathcal{N}}$ plays a crucial role in the problem's complexity. Specifically, when there are L different values in $\{\rho_i\}_{i\in\mathcal{N}}$, we can design a fixed-parameter tractable algorithm whose time complexity depends on L to solve the assortment optimization problem.

We focus on the case where $0 \neq \mathcal{N}^*$. The case where $0 \in \mathcal{N}^*$ can be handled similarly. Formally, suppose that the L different values are $\theta_1, \theta_2, \ldots, \theta_L$. For each $\ell \in [L]$, let $\mathcal{N}_\ell = \{i \in \mathcal{N} : \rho_i = \theta_\ell\}$ be the set containing items whose distortion parameters are θ_ℓ. Thus, given $\mathcal{S}$, the distortion function can be rewritten as $\delta(\mathcal{S}) = g(\sum_{\ell=1}^{L} \rho_\ell |\mathcal{S}_\ell|)$, where $\mathcal{S}_\ell = \mathcal{S} \cap \mathcal{N}_\ell$ and $\cup_{\ell\in[L]} \mathcal{S}_\ell = \mathcal{S}$. Then, we solve the assortment optimization problem by discussing the cardinality of optimal assortment $\mathcal{S}^*$ and how many items in $\mathcal{N}_\ell$ belong to $\mathcal{S}^*$ for all $\ell \in [L]$. Let $Q := |\mathcal{S}| \leq n$ and decompose Q into L non-negative integers $Q_1, Q_2, \ldots, Q_L$ such that $\sum_{\ell\in[L]} Q_\ell = Q$. Given Q and conditioning on that the size of $\mathcal{S}_\ell$ equals Q_ℓ for all $\ell \in [L]$, we obtain the following problem with respect to the tuple $(Q_1, Q_2, \ldots, Q_L)$:

$$\begin{aligned} \max_{\mathcal{S}_\ell \subseteq \mathcal{N}_\ell, \forall \ell \in [L]} \quad & \frac{(1 + g(\sum_{\ell=1}^{L} \rho_\ell Q_\ell)) \sum_{\ell\in[L]} \sum_{i\in\mathcal{S}_\ell\cap\mathcal{N}^*} r_i u_i + \sum_{\ell\in[L]} \sum_{i\in\mathcal{S}_\ell\setminus\mathcal{N}^*} r_i u_i}{1 + (1 + g(\sum_{\ell=1}^{L} \rho_\ell Q_\ell)) \sum_{\ell\in[L]} \sum_{i\in\mathcal{S}_\ell\cap\mathcal{N}^*} u_i + \sum_{\ell\in[L]} \sum_{i\in\mathcal{S}_\ell\setminus\mathcal{N}^*} u_i} \\ \text{s.t.} \quad & |\mathcal{S}_\ell| = Q_\ell, \forall \ell \in [L]. \end{aligned} \tag{12}$$

Problem (12) can be further reformulated as a TU-constrained assortment optimization problem under the MNL model, where

$$r'_i = r_i, \quad \text{and} \quad u'_i = \begin{cases} (1 + g(\sum_{\ell=1}^{L} \rho_\ell Q_\ell)) u_i, i \in \mathcal{N}_\ell \cap \mathcal{N}^*, \\ u_i, i \in \mathcal{N}_\ell \setminus \mathcal{N}^*, \end{cases} \quad \forall \ell \in [L].$$

Therefore, for a given tuple $(Q_1, Q_2, \ldots, Q_L)$, (12) can be solved via a linear program, following the same idea in Lemma 1. Since we have in total $\binom{Q+L-1}{L-1}$ different tuples of $(Q_1, Q_2, \ldots, Q_L)$ for each $Q \in \{1, 2, \ldots, n\}$, the number of linear programs to be solved is in the order of $O(n^L)$. Hence, the optimal assortment can be obtained by solving $O(n^L)$ linear programs.

THEOREM 4. *Suppose that the focal function is of form (11) and the distortion parameters have L different values, the time complexity of finding the optimal assortment is $O(n^L \cdot \phi_{LP})$, where ϕ_{LP} is the time complexity of solving a linear program with $O(n)$ variables and $O(n)$ constraints.*

Theorem 4 shows that when L is a fixed number, the assortment optimization problem can be solved in polynomial time with respect to n. The parameter L shows how the structures of distortion parameters influence the problem's complexity. In EC.4.2.1, we show that when the distortion function takes a different form, another parameter serves as the lever that controls the problem's complexity. When the distortion function does not take any of the two structures discussed above, we evaluate the performance of the best profit-ordered assortment and provide an approximation guarantee in EC.4.2.2.

$\delta(\mathcal{S})$ \ $F(\mathcal{S})$	top-K focal model	fixed focal items
$\delta(\mathcal{S}) = \delta$	$O((n-K+1)\phi_{LP})$	ϕ_{LP}
$\delta(\mathcal{S}) = \rho_0 + \sum_{i\in F(\mathcal{S})}\rho_i$	$O(n^{K+1})$	$O((2^{\lvert\mathcal{N}^*\rvert}-1)n)$
$\delta(\mathcal{S}) = g(\lvert\mathcal{S}\rvert)$	$O(n\phi_{LP})$	$O(n\phi_{LP})$
$\delta(\mathcal{S}) = g(\sum_{\ell=1}^{L}\rho_\ell\lvert\mathcal{S}_\ell\rvert)$	N/A	$O(n^L\phi_{LP})$
$\delta(\mathcal{S}) = g(\sum_{i\in\mathcal{S}}\rho_i)$	N/A	NP-hard

Table 1 Time complexity of assortment optimization under the FLM.

In the discussions above, we consider the assortment optimization problem. In some scenarios, the seller needs to determine the assortment and price simultaneously. In that case, the decision problem becomes a joint decision over assortment and prices, and we consider such a problem in the next section.

5 Joint Assortment and Pricing Optimization

In this section, we consider the joint assortment and pricing optimization problem under the FLM, where the seller must simultaneously decide the assortment and price for each item in the assortment. To model the relation between price and attraction value, we follow the literature (Wang 2012, Miao and Chao 2021), and assume that if item $i\in\mathcal{N}$ is priced at p_i, then its attraction value is $u_i = \exp(\alpha_i - \beta_i p_i)$, where α_i is the fixed utility of item i and $\beta_i > 0$ represents the price sensitivity. For the no-purchase option, we let $\alpha_0 = \beta_0 = 0$.

Under the FLM, the attraction value of item i in the assortment $\mathcal{S}$ is given by $u_i(p_i,\mathcal{S}) = (1+\delta(\mathcal{S})\mathbf{1}\{i\in F(\mathcal{S})\})\exp(\alpha_i - \beta_i p_i)$. The feasible set of price vector is $\mathcal{P} = \{\mathbf{p} | p_i \geq c_i, \forall i\in\mathcal{S}\}$, implying that the price of each item must be higher than its cost. The joint decision problem now becomes

$$\max_{\mathcal{S}\subseteq\mathcal{N},\mathbf{p}\in\mathcal{P}} R(\mathcal{S},\mathbf{p}) := \frac{\sum_{i\in\mathcal{S}}(p_i - c_i)u_i(p_i,\mathcal{S})}{\sum_{i\in\mathcal{S}_+} u_i(p_i,\mathcal{S})}. \tag{13}$$

In the following, we solve problem (13) when the focal and distortion functions take different forms as discussed in Section 3 and Section 4. The inclusion of the new price variable brings more complexity to optimization. However, we demonstrate that these joint decision problems can still be solved efficiently, even in polynomial time.

5.1 Ranking-Based Focal Items

In this subsection, we develop an efficient algorithm for the joint assortment and pricing problem under the top-K focal model with the constant distortion function. In EC.5.2, we show that the same technique can be applied to other distortion functions discussed in Section 3, and to the arbitrarily-located focal model.

Under the top-K focal model, we assume that the customers build the focal set by including top-K items in $\mathcal{S}$ based on the *attraction value ranking*. Without loss of generality, we let $n \geq K$. Although both price and assortment are decision variables now, the next lemma indicates that the joint decision problem can be simplified to a pricing-only problem.

LEMMA 2. *Let $(\mathcal{S}^*, \mathbf{p}^*)$ denote the optimal solution of the joint assortment and pricing optimization problem under the FLM, where the focal function is given by (6) with attraction value ranking and the distortion function is a constant. Then, we have $\mathcal{S}^* = \mathcal{N}$.*

Wang (2012) shows that it is optimal to offer all items in the joint assortment and pricing optimization problem under the MNL model. Lemma 2 extends the result to the FLM and the insight is as follows. Note that adding an item to the current assortment does not change the value of the distortion function. If this new item is priced sufficiently large, it will always be excluded from the focal set but generate a high profit if some customers purchase it. Moreover, its influence on the choice probability of other items is almost negligible. Consequently, for any solution $(\mathcal{S}, \mathbf{p})$ with $\mathcal{S} \subset \mathcal{N}$, one can always include an item $i \in \mathcal{N} \backslash \mathcal{S}$ and set its price p_i sufficiently large to increase the expected profit. Therefore, providing all items is optimal.

With Lemma 2, it is left to decide the optimal price of each item in $\mathcal{N}$. To solve the pricing problem, we first fix the item in the focal set with the least attraction value and denote it by k. Then, for each fixed k, we enumerate all possible focal sets $\mathcal{F}$ of size K with $k \in \mathcal{F}$. We also impose the constraints to ensure that item k has the least attraction value in $\mathcal{F}$. Then, problem (13) becomes

$$
\begin{aligned}
\max_{k \in \mathcal{N}} \max_{\substack{\mathcal{F} \subseteq \mathcal{N}, \\ |\mathcal{F}| = K,\ k \in \mathcal{F}}} \max_{\mathbf{p}} \ & \frac{\sum_{i \in \mathcal{F}} (p_i - c_i)(1+\delta) \exp(\alpha_i - \beta_i p_i) + \sum_{j \in \mathcal{N} \setminus \mathcal{F}} (p_j - c_j) \exp(\alpha_j - \beta_j p_j)}{1 + \sum_{i \in \mathcal{F}} (1+\delta) \exp(\alpha_i - \beta_i p_i) + \sum_{j \in \mathcal{N} \setminus \mathcal{F}} \exp(\alpha_j - \beta_j p_j)} \\
\text{s.t.} \ & \alpha_i - \beta_i p_i \geq \alpha_k - \beta_k p_k, \ \forall i \in \mathcal{F} \setminus \{k\}, \\
& \alpha_j - \beta_j p_j \leq \alpha_k - \beta_k p_k, \ \forall j \in \mathcal{N} \setminus \mathcal{F}, \\
& p_\ell \geq c_\ell, \qquad \forall \ell \in \mathcal{N}.
\end{aligned}
\tag{14}
$$

Solving (14) directly is not easy due to the complexity of the objective function. However, for a given $\mathcal{F}$, we can transform the maximization problem over price vector $\mathbf{p}$ in (14) to a fixed point problem. Specifically, given $\mathcal{F}$ and price p_k of item k, we define

$$p'_i(r) := \max\{c_i, \min\{r + 1/\beta_i + c_i,\ (\alpha_i - \alpha_k + \beta_k p_k)/\beta_i\}\} \quad \forall i \in \mathcal{F} \setminus \{k\},$$
$$\tilde{p}_j(r) := \max\{r + 1/\beta_j + c_j,\ (\alpha_j - \alpha_k + \beta_k p_k)/\beta_j\} \quad \forall j \in \mathcal{N} \setminus \mathcal{F},$$

and

$$H(p_k, r) := (1+\delta)(p_k - c_k - r)\exp(\alpha_k - \beta_k p_k) + \sum_{i \in \mathcal{F}\setminus\{k\}} (1+\delta)(p'_i(r) - c_i - r)\exp(\alpha_i - \beta_i p'_i(r)) + \sum_{j \in \mathcal{N}\setminus\mathcal{F}} (\tilde{p}_j(r) - c_j - r)\exp(\alpha_j - \beta_j \tilde{p}_j(r)).$$

We also let $\phi(r) := \max_{p_k \in \mathcal{P}_k} H(p_k, r)$, where $\mathcal{P}_k := \{p_k : p_k \geq c_k\}$.

The next lemma characterizes the optimal value to the maximization problem over $\mathbf{p}$ in (14).

LEMMA 3. *With the above definitions, the following statements hold:*

(1) $\phi(r)$ *is continuous and decreasing in* r.

(2) Given $\mathcal{F}$*, the optimal value to the maximization problem over* $\mathbf{p}$ *in* (14) *is the unique solution of the following equation:*

$$r = \phi(r). \tag{15}$$

By Lemma 3, solving the maximization problem over $\mathbf{p}$ in (14) reduces to finding the unique solution of (15). Since the left-hand-side r is increasing and continuous in r and the right-hand-side $\phi(r)$ is decreasing and continuous in r, we can use the bisection algorithm to solve equation (15). To compute the value of $\phi(r)$ for a given r, we need to solve a maximization problem over p_k. In the next lemma, we show that $\phi(r)$ can be computed in polynomial time for each given r.

LEMMA 4. *For a given* r, $\phi(r)$ *can be computed in the time complexity of* $O(n^3)$.

With Lemma 4, given $\mathcal{F}$, the maximization problem over $\mathbf{p}$ in (14) can be solved in the time complexity of $\tilde{O}(n^3)$, where $\tilde{O}(\cdot)$ hides the logarithmic factor that comes from the bisection algorithm. Because we need to enumerate all possible choices of item k and the focal set $\mathcal{F}$ of size K such that $k \in \mathcal{F}$, the total time complexity to solve problem (13) is then $n \times O(n^{K-1}) \times \tilde{O}(n^3) = \tilde{O}(n^{K+3})$, as shown in Theorem 5.

THEOREM 5. *The optimal solution of problem* (13) *under the top-*K *focal model with the constant distortion function can be solved in the time complexity of* $\tilde{O}(n^{K+3})$.

In practice, customers usually focus on only a few top-ranked items in a given assortment (Feng et al. 2007, Chen et al. 2022); thus, K is usually a small number. When K is considered as a fixed number, our algorithm gives the optimal solution in polynomial time.

5.2 Fixed Focal Items

In this subsection, we consider the joint assortment and pricing optimization problem when the focal items are predetermined. In particular, the focal function $F(\mathcal{S}) = \mathcal{S}_+ \cap \mathcal{N}^*$ for all $\mathcal{S} \subseteq \mathcal{N}$, where $\mathcal{N}^*$ contains prespecified focal items. Similar to Section 4, we assume that the distortion function satisfies Assumption 1, i.e., there exists a function $g(\cdot)$ such that $\delta(\mathcal{S}) = g(|\mathcal{S}|)$ for all $\mathcal{S} \subseteq \mathcal{N}$. Then the joint decision problem (13) becomes

$$\max_{\mathcal{S}\subseteq\mathcal{N},\ \mathbf{p}\in\mathcal{P}} \frac{\sum_{\mathcal{S}\cap\mathcal{N}^*}(1+g(|\mathcal{S}|))(p_i-c_i)\exp(\alpha_i-\beta_i p_i)+\sum_{\mathcal{S}\setminus\mathcal{N}^*}(p_i-c_i)\exp(\alpha_i-\beta_i p_i)}{\sum_{\mathcal{S}_+\cap\mathcal{N}^*}(1+g(|\mathcal{S}|))\exp(\alpha_i-\beta_i p_i)+\sum_{\mathcal{S}_+\setminus\mathcal{N}^*}\exp(\alpha_i-\beta_i p_i)}. \tag{16}$$

In this case, $\mathcal{S}^* = \mathcal{N}$ does not hold, thus we need to jointly optimize $\mathcal{S}$ and $\mathbf{p}$. First, we consider the price optimization for a given assortment $\mathcal{S}$. Given $\mathcal{S} \subseteq \mathcal{N}$, we define $\mathbf{p}^{\mathcal{S}} \in \arg\max_{\mathbf{p}\in\mathcal{P}} R(\mathcal{S}, \mathbf{p})$ and let $r^{\mathcal{S}} := R(\mathcal{S}, \mathbf{p}^{\mathcal{S}})$. The following lemma characterizes the relationship between $r^{\mathcal{S}}$ and $\mathbf{p}^{\mathcal{S}}$.

LEMMA 5. *Given an assortment $\mathcal{S}$, let $\mathcal{S}_1 = \mathcal{S}_+ \cap \mathcal{N}^*$ and $\mathcal{S}_2 = \mathcal{S} \setminus \mathcal{S}_1$. We have:*

(1) The optimal solution $\mathbf{p}^{\mathcal{S}}$ satisfies $p_i^{\mathcal{S}} = r^{\mathcal{S}} + c_i + 1/\beta_i$, for all $i \in \mathcal{S}$.

(2) The optimal profit $r^{\mathcal{S}}$ satisfies

$$r^{\mathcal{S}} = J\left(r^{\mathcal{S}}, \mathcal{S}_1, \mathcal{S}_2\right), \tag{17}$$

where

$$J(r,\mathcal{S}_1,\mathcal{S}_2) := \frac{1+g(|\mathcal{S}_1|+|\mathcal{S}_2|)}{1+g(|\mathcal{S}_1|+|\mathcal{S}_2|)\mathbf{1}\{0\in\mathcal{N}^*\}}\sum_{i\in\mathcal{S}_1}\exp(\gamma_i-\beta_i r)+\frac{\sum_{i\in\mathcal{S}_2}\exp(\gamma_i-\beta_i r)}{1+g(|\mathcal{S}_1|+|\mathcal{S}_2|)\mathbf{1}\{0\in\mathcal{N}^*\}},$$

and $\gamma_i := \alpha_i - \beta_i c_i - 1 - \log\beta_i$ for all $i \in \mathcal{N}$.

Lemma 5 shows that for any given assortment $\mathcal{S}$, the optimal price for each item $i \in \mathcal{S}$ is the cost c_i plus the term $r^{\mathcal{S}} + 1/\beta_i$. Recall that β_i is the price sensitivity of item i. Thus, the more sensitive the customers are to the price change, the lower the optimal price should be. The quantity $p_i - c_i - 1/\beta_i$ is referred to as the *adjusted markup*, which is the same among items in $\mathcal{S}$ at the optimal price, i.e., when $p_i = p_i^{\mathcal{S}}$ for all $i \in \mathcal{S}$. The property of the same adjusted markup is also observed in Wang (2012).

Moreover, Lemma 5 enables us to obtain a fixed point characterization of the optimal value R^* to problem (16) and devise an efficient algorithm. The time complexity of the algorithm is given in Theorem 6. The detailed algorithm and proof are provided in EC.5.3. The algorithm is efficient since the size of $\mathcal{N}^*$ is not large in practice. When $\mathcal{N}^*$ contains only one item, such as the no-purchase option, the joint decision problem can be solved within the time complexity of $O\left(n\log^2 n\right)$.

THEOREM 6. *There exists an algorithm that returns the optimal solution of (16) in the time complexity of $O\left(n\left(\log^2 n + 2^{|\mathcal{N}^*|-1}\right)\right)$.*

In addition, we identify several structural properties when the focal effect intensifies. The details are provided in EC.6.2.

6 Numerical Experiments

This section gives several numerical studies using real-world datasets to evaluate the performance of the FLM. Specifically, we analyze Expedia's dataset and the transaction dataset from JD.com. For Expedia's dataset, we apply the top-K focal model and evaluate its prediction capability. For JD.com dataset, we use the fixed focal items model to capture the focal effect. The results from both datasets show that the FLM outperforms the MNL model in predicting customer choice behavior, and its performance is comparable to that of the general attraction model (Gallego et al. 2015). To evaluate the profitability of the optimal assortment recommended by the FLM, we conduct an experiment with a synthetic dataset in EC.9 and show that the FLM can improve the seller's profit compared to other choice models.

6.1 Expedia Dataset: Top-K Focal Model

Data description. Expedia is the world's largest online travel agency and handles millions of customer search queries daily (Kaggle 2025). When a customer submits a search request, Expedia returns a list of related hotels. This dataset provides the results of hotel search queries. Each entry in the dataset corresponds to a specific hotel shown to a certain customer in a search query. It also contains information about the displayed hotel, the search query, and the customer's ultimate booking decision. Following Aouad et al. (2023), we restrict our attention to search queries where displayed hotels are randomized.

We preprocess the dataset following Gao et al. (2021). It involves keeping search queries for single-night stays, removing incomplete features and queries, and trimming outliers. After data preprocessing, we are left with 14 different features and $30,346$ search queries. The first 2 features are the unique code for each search query and the indicator of whether the hotel is booked in the search query. The last 12 features are used in model estimation. In EC.7, we give details of our preprocessing approach and the last 12 features. We then partition the remaining Expedia search queries by site ID and focus on the 5 sites with the largest number of search queries.

Models. We compare the estimated FLM with an MNL model. For the FLM, we let the focal function be of form (6) and the distortion function be a positive constant. We assume that the focal effect comes from the price ranking. This is natural because the customers may focus on some cheapest hotels that are within their budget constraints.

Model Calibration. We randomly split each dataset into training, validation, and testing datasets, each of which, respectively, consists of 64%, 16%, and 20% of the total search queries. We use the maximum likelihood estimation (MLE) to estimate the FLM and the MNL model.

Since the total number of hotels is large, it is difficult to directly estimate the attraction value for each hotel. Thus, we follow the method in existing literature (Gao et al. 2021, Bai et al. 2023, Gao et al. 2023, Aouad et al. 2023) which uses a feature-based approach for estimation. Specifically, we assume that the attraction value of a hotel i is of the following form

$$u_i = \exp(\beta^{(0)} + \sum_{\ell=1}^{12} \beta^{(\ell)} x_i^{(\ell)}), \tag{18}$$

where $(x_i^{(1)}, x_i^{(2)}, \ldots, x_i^{(12)})$ are the values of the last 12 features of hotel i, and $(\beta^{(0)}, \beta^{(1)}, \ldots, \beta^{(12)})$ are coefficients to be estimated.

The estimation algorithm for the MNL model is from Berbeglia et al. (2022). To estimate the top-K focal model, we first determine the value of K and then construct a binary variable to represent whether each displayed hotel has the top-K cheapest prices. This attribute is then incorporated into (18), and the parameters of the top-K focal model are estimated similarly to the MNL model.

To determine the value of K, we try the value of K in the set $\{1, 3, 5, 7, 10\}$. For each fixed K, we fit a top-K focal model with the training dataset and evaluate the log-likelihood of our fitted choice model on the validation dataset. Finally, we choose the model with the highest log-likelihood on the validation dataset.

Performance. The performance of the two models is shown in Table 2. Each column in Table 2 represents a distinct site. The "Log value" in the table refers to the *normalized* log-likelihood value calculated on the training dataset. This serves as a measure for the goodness of fit on the training dataset, i.e., a model with a large log value fits the training dataset better than a model with a smaller log value. To measure the prediction accuracy on the testing dataset, we use the root mean square error (RMSE), which is widely used in the literature (e.g., Berbeglia et al. 2022, Gao et al. 2023) and is computed by the following formula:

$$\text{RMSE} = \sqrt{\frac{\sum_{t=1}^{T} \sum_{i \in \mathcal{S}_t \cup \{0\}} (\mathbf{1}(i_t = i) - \mathbb{P}(i, \mathcal{S}_t))^2}{\sum_{t=1}^{T} (|\mathcal{S}_t| + 1)}},$$

where T is the size of the testing dataset and $\mathcal{S}_t$ is the assortment presented to the t-th customer. The indicator function $\mathbf{1}(\cdot)$ indicates whether hotel i is booked by the t-th customer or not. The value $\mathbb{P}(i, \mathcal{S}_t)$ denotes the probability that a customer books hotel i when assortment $\mathcal{S}_t$ is offered and is computed by the fitted choice model. The smaller the RMSE value is, the more accurate the model predicts on the testing dataset.

For each site, we randomize data split for 20 times. Then we take the average of the normalized log-likelihood values on the training dataset and the RMSE values on the testing dataset. Table 2 shows that the FLM dominates the MNL model over all 5 datasets on both "Log value" and "RMSE" criteria. Table 2 shows that the FLM has a better fitting and prediction ability than the MNL model.

Structure of the optimal assortments. We compare the optimal assortment under the MNL model and the FLM. For each site ID, we define the full set of hotels as the displayed hotels in

Expedia Site ID	5	14	15	24	32
Max assortment cardinality	38	36	36	31	34
# search queries	19777	3468	1397	1362	853
MNL					
Log value	-1.0677	-1.0600	-1.0374	-1.0402	-0.9611
RMSE	0.1050	0.1082	0.1137	0.1120	0.0947
FLM					
Log value	**-1.0627**	**-1.0561**	**-1.0317**	**-1.0367**	**-0.9568**
RMSE	**0.1045**	**0.1078**	**0.1134**	**0.1115**	**0.0939**

Table 2 Comparison of the predictive power of the MNL model and FLM on the Expedia dataset.

the search query with the largest number of displayed hotels. Based on the estimated MNL model and FLM, we compute the optimal assortment under each fitted model, denoted by $\mathcal{S}^*_{\text{MNL}}$ and $\mathcal{S}^*_{\text{FLM}}$, respectively.

We find that $\mathcal{S}^*_{\text{FLM}} \subseteq \mathcal{S}^*_{\text{MNL}}$ on all the five sites. Moreover, we also find that $\mathcal{S}^*_{\text{FLM}}$ excludes hotels in $\mathcal{S}^*_{\text{MNL}}$ with low prices. We provide a specific example in EC.7.2. To understand this result, recall that the focal items are the K cheapest hotels in our setting. When $\mathcal{S}_{\text{FLM}} = \mathcal{S}^*_{\text{MNL}}$, according to the FLM, the choice probabilities of high-profit non-focal items will decrease because of the attraction value increase of those cheap focal items. Thus, under the FLM, the seller can remove less profitable hotels in $\mathcal{S}^*_{\text{MNL}}$ so that items with "moderately large" profits become focal. This adjustment improves the overall expected profit under the FLM.

6.2 JD.com Dataset: Choice Overload

In this subsection, we use the transaction dataset from JD.com (Shen et al. 2024). This dataset includes the transaction data associated with over 2.5 million customers and 31,868 items in one category on JD.com in March 2018.[2] The existence of choice overload in online retailing platforms has long been recognized (Nagar 2016, Long et al. 2025). Recently, Chen et al. (2022) analyze this dataset and find that over 90% of customers choose the no-purchase option. Given these clues, we conjecture that the choice overload effect exists in this dataset. Thus, we calibrate the choice overload model introduced in Example 3 and compare its performance with other choice models, including the MNL model and the GAM.

Data description. The dataset includes information about items, customers, inventory, etc. We refer readers to Shen et al. (2024) for a detailed dataset description. Since our work focuses on choice behavior, we only use the items, customers, clicks, and order information in the dataset. Each record in the dataset is of the following form: {`customer_id, item_id, attribute_1,`

[2] A category is a set of items with similar features. For example, a "shirt" category may contain thousands of shirts of different brands and prices.

`attribute_2, action_type, time_of_action`}. A customer with certain `customer_id` visits JD.com to buy an item in a certain category, say a sunscreen cream. The customer is presented with a list of sunscreen creams after entering the keywords. Each item is represented by a small picture and a description. Along with the description, key attributes of the item are also shown. For example, in the sunscreen category, the customer can see from the search page which skin type the item is designed for (`attribute_1`) and the sun protection factor (`attribute_2`). If the customer is interested in a certain item, she can click the item with certain `item_id`, and a new page containing detailed information shows up. On the item page, the customer can choose to purchase the item or leave the item page. The two actions: click and purchase, are the `action_type` values in this dataset. We note that this example is only to help understand the dataset's composition. The category information in the dataset is anonymous for confidential reasons.

Since the number of records is huge, we further split items into subcategories by their attributes: `attribute_1` takes integers 2, 3, and 4 (3 different values), and `attribute_2` takes integers 30, 40, 50, 60, 70, 80, 90, and 100 (8 different values). We obtain $3 \times 8 = 24$ subcategories and analyze each subcategory separately. Thus, each record now becomes {`customer_id, item_id, subcategory_id, action_type, time_of_action`}. Meanwhile, we randomly take seven consecutive days and analyze the customer choice behavior through the clicking data and ordering data within the seven days. Following Ruan et al. (2023), we assume that the set of items clicked by the customer is the offered assortment, and we exclude customers who purchase multiple items.

Models. We calibrate the MNL model, the FLM with the focal function $F(\mathcal{S}) = \{0\}$, and the GAM. For the FLM, we let $\delta(\mathcal{S}) = \exp(\alpha|\mathcal{S}|)$ where $\alpha > 0$ denotes the level of choice overload effect. In this case, the attraction value of the no-purchase option under a given assortment $\mathcal{S}$ is $u(0,\mathcal{S}) = 1 + \exp(\alpha|\mathcal{S}|)$, which is increasing in the cardinality of $\mathcal{S}$.

Under the GAM, in addition to the attraction value v_j for each item j, there is a shadow attraction value $w_j \in [0, v_j]$. The purchase probability of $i \in \mathcal{S}$ is given by

$$\mathbb{P}(i,\mathcal{S}) = \frac{v_i}{1 + \sum_{j\in\mathcal{N}\setminus\mathcal{S}} w_j + \sum_{j\in\mathcal{S}} v_j}.$$

The probability of the no-purchase is $\mathbb{P}(0,\mathcal{S}) = (1+\sum_{j\in\mathcal{N}\setminus\mathcal{S}} w_j)/(1+\sum_{j\in\mathcal{N}\setminus\mathcal{S}} w_j + \sum_{j\in\mathcal{S}} v_j)$, and the term $1+\sum_{j\in\mathcal{N}\setminus\mathcal{S}} w_j$ is the overall attraction value of the no-purchase option, which is a function of the items not offered in the assortment. Such a configuration is to avoid being too optimistic about recapturing the demand from unavailable items.

Model Calibration. We randomly split the dataset of each subcategory into training and testing datasets, each of which includes 70% and 30% of the observations, respectively. A validation dataset is not used because the parameters can be estimated directly and we do not need to choose from several candidates. We fit these three models by MLE. For the MNL model, we use the algorithm in Berbeglia et al. (2022). To maximize the log-likelihood function under the GAM, we use `fmincon` routine in MATLAB, following Cao et al. (2022). For the specified FLM, we present the detailed estimation algorithm in EC.8. From the high level, we maximize the log-likelihood function by alternatively optimizing over α and intrinsic utility vector $\boldsymbol{v}$. The algorithm stops when the relative increase of the log-likelihood value is less than 10^{-6}.

Subcategory	(2,30)	(2,80)	(3,30)	(3, 80)	(3,90)	(4, 50)
Number of items	69	25	14	140	80	20
Number of customers	9757	22270	1248	26021	26882	2917
MNL						
Log value	-0.2557	-0.2141	-0.1335	-0.2266	-0.2560	**-0.0485**
RMSE	0.2484	0.2182	0.1900	0.2227	0.2476	0.1292
GAM						
Log value	-0.2581	**-0.2140**	**-0.1334**	-0.2264	-0.2559	**-0.0485**
RMSE	**0.2453**	**0.2180**	0.1936	0.2228	0.2477	0.1422
FLM						
Log value	**-0.2552**	**-0.2140**	-0.1335	**-0.2263**	**-0.2558**	**-0.0485**
RMSE	0.2456	0.2181	**0.1894**	**0.2223**	**0.2470**	**0.1266**

Table 3 Comparison of the performance of the MNL model, the GAM, and the FLM on the JD.com dataset. The tuple in the "subcategory" row denotes the value of attribute 1 and attribute 2.

Performance. We still use Log value and RMSE as metrics to evaluate the fitting capability and prediction accuracy of each model. We present the results in Table 3, which shows that the GAM and FLM outperform the MNL model, and the FLM and GAM perform comparably to each other on all five subcategories. The mechanism for FLM to capture the choice overload effect is already explained. For the GAM, it gives higher weight to the no-purchase option using w_j, making it natural to capture the choice-overload effect. Therefore, GAM and FLM are both reasonable models to use in the presence of the choice overload effect.

On the other hand, the FLM offers greater flexibility than the GAM in capturing other focal effects, such as the spiked effect, by specifying different focal and distortion functions. Thus, though FLM and GAM perform comparably in this scenario, FLM serves as a more flexible modeling framework.

7 Conclusion

In this paper, we apply the FLM to various scenarios where the focal effect occurs and consider the assortment optimization problems under these scenarios. Our analysis focuses on identifying the structure of the optimal assortment and developing efficient algorithms to find it. We provide insights into how the focal effect affects the optimal assortment and how the seller should adjust it when the focal effect intensifies. On the optimization front, though the assortment optimization problem under the FLM is generally NP-hard, we show that when some practical assumptions are imposed, there exist polynomial-time algorithms to solve the problem. This polynomial-solvability can be extended to the more complex joint assortment and pricing optimization problems as well. Additionally, we conduct numerical studies on real datasets that exhibit different types of focal effects to evaluate the performance of FLM. The results show that the FLM outperforms the MNL model in prediction accuracy and performs comparably to the GAM. To evaluate the profitability of the optimal assortment recommended by the FLM, we conduct an experiment with synthetic data in EC.9. It shows that the FLM can improve the seller's profit compared to other choice models when customer choice behavior violates the regularity condition. In the following, we summarize our main managerial insights for clarity.

(1) When the focal effect arises due to item ranking, under the top-K focal model with a constant distortion function, the optimal assortment has the K-truncated-profit-ordered property. It means that both item profit and item ranking matter for the optimal assortment. Notably, this property also holds when the focal items are not the top-K items but take other positions in the ranking.

(2) When the focal items are fixed, such as the no-purchase option, we identify a floating threshold property, which states that if a non-focal item i is included in the optimal assortment S^*, then its profit r_i must exceed an item-dependent threshold. When the fixed focal item is not the no-purchase option, this property corresponds to the decoy effect and similarity effect under different assumptions on the distortion functions.

(3) We explore how the optimal assortment changes as the distortion function increases, especially for the choice overload case. We find that when the choice overload effect gets more severe, the optimal assortment size does not necessarily shrink, instead, it may expand because the seller needs to increase the purchase probability to hedge this effect. When the seller can jointly optimize the assortment and price, the seller should lower the price to mitigate the influence of choice overload. Whether to increase the size of the optimal assortment in the joint decision problem depends on the property of the distortion function.

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E-Companion for "Assortment Optimization in the Presence of Focal Effect: Operational Insights and Efficient Algorithms"

EC.1 Proofs of Technical Results in Section 3

We first provide the following lemma, making preparation for the proof of Proposition 1. We note that an identical result is also provided in Barré et al. (2024).

LEMMA EC.1. *Given $C_1 > 0$ and $C_2 > 0$, there exists a profit-ordered optimal assortment to the following optimization problem:*

$$\max_{\mathcal{S}\subseteq\mathcal{A}} \frac{\sum_{\ell\in\mathcal{S}} r_\ell u_\ell + C_1}{C_2 + \sum_{\ell\in\mathcal{S}} u_\ell}.$$

Hence, it can be solved in polynomial time.

Proof of Lemma EC.1.

Let R^* be its optimal value, and $\mathcal{S}^*$ be its optimal assortment. We have $R^* \geq \left(\sum_{\ell\in\mathcal{S}} r_\ell u_\ell + C_1\right)/(C_2 + \sum_{\ell\in\mathcal{S}} u_\ell)$ for all $\mathcal{S}\subseteq\mathcal{A}$ with equality holding at the optimal assortments. It is equivalent to $C_2 R^* - C_1 \geq \sum_{\ell\in\mathcal{S}} (r_\ell - R^*)\, u_\ell$ for all $\mathcal{S}\subseteq\mathcal{A}$ with equality holding at the optimal assortments. Thus, the optimal assortment can be recovered by solving $\max_{\mathcal{S}\subseteq\mathcal{A}} \sum_{\ell\in\mathcal{S}} (r_\ell - R^*)\, u_\ell$. It is optimal to offer each item whose profit exceeds R^*, that is, $\mathcal{S}^* = \{\ell\in\mathcal{A} : r_\ell \geq R^*\}$. In other words, there exists a $\mathcal{S}^*$ that is profit-ordered in $\mathcal{A}$. $\square$

Proof of Proposition 1.

To prove Proposition 1, we only need to prove that, when $|\mathcal{S}^*| \geq K+1$, it can be decomposed as $\{i_1, i_2, \ldots, i_K\} \cup \mathcal{S}^{RO}_{i_K}$. With this formulation, it is easy to see that set $\mathcal{S}_1 = \{i_1, i_2, \ldots, i_K\}$ is the focal set in $\mathcal{S}^*$. Given focal set $\mathcal{S}_1$, we only need to find the non-focal items in $\mathcal{S}^*$ by solving

$$\max_{\mathcal{S}_2\subseteq\mathcal{R}_{i_K}} \frac{(1+\delta)\sum_{i\in\mathcal{S}_1} r_i u_i + \sum_{i\in\mathcal{S}_2} r_i u_i}{1 + (1+\delta)\sum_{i\in\mathcal{S}_1} u_i + \sum_{i\in\mathcal{S}_2} u_i},$$

where $\mathcal{R}_{i_K} := \{j\in\mathcal{N} : j > i_K\}$. With Lemma EC.1, there exists a profit-ordered optimal assortment in $\mathcal{R}_{i_K}$ to the above problem. Hence, the optimal assortment to problem (1) under the top-K focal model is either of size no more than K or can be decomposed as

$$\mathcal{S}^* = \{i_1, i_2, \ldots, i_K\} \cup \mathcal{S}^{RO}_{i_K},$$

for some $i_1 < i_2 < \cdots < i_K$, where $\mathcal{S}^{RO}_{i_K}$ is a profit-ordered assortment in $\mathcal{R}_{i_K}$. Then the proof is complete. $\square$

Proof of Lemma 1.

The proof idea is the same as Davis et al. (2013), and we provide the proof for completeness.

By letting $u_i' = (1+\delta)u_i$ for all $i \in \mathcal{L}_k \cup \{k\}$, and $u_j' = u_j$ for all $j \in \mathcal{R}_k$, we can reformulate (8) as

$$\begin{aligned} \eta^* = \max_{\boldsymbol{x},\, \boldsymbol{y}} \quad & \frac{r_k u_k' + \sum_{i\in\mathcal{L}_k} r_i u_i' x_i + \sum_{j\in\mathcal{R}_k} r_j u_j' y_j}{1 + u_k' + \sum_{i\in\mathcal{L}_k} u_i' x_i + \sum_{j\in\mathcal{R}_k} u_j' y_j} \\ \text{s.t.} \quad & \boldsymbol{e}^T \boldsymbol{x} = K-1, \\ & x_i \in \{0,1\} \quad \forall i \in \mathcal{L}_k, \\ & y_j \in \{0,1\} \quad \forall j \in \mathcal{R}_k, \end{aligned} \tag{EC.1}$$

where $\boldsymbol{e}$ is the all-one vector. Denote the optimal solution to (EC.1) by $(\boldsymbol{x}^*, \boldsymbol{y}^*)$. Since

$$\eta^* = \frac{r_k u_k' + \sum_{i\in\mathcal{L}_k} r_i u_i' x_i^* + \sum_{j\in\mathcal{R}_k} r_j u_j' y_j^*}{1 + u_k' + \sum_{i\in\mathcal{L}_k} u_i' x_i^* + \sum_{j\in\mathcal{R}_k} u_j' y_j^*},$$

we have $\eta^* = (r_k - \eta^*)u_k' + \sum_{i\in\mathcal{L}_k}(r_i - \eta^*)u_i' x_i^* + \sum_{j\in\mathcal{R}_k}(r_j - \eta^*)u_j' y_j^*$. Now we consider the following problem,

$$\begin{aligned} \psi^* = \max_{\mathbf{s},\, \mathbf{t}} \quad & (r_k - \eta^*)u_k' + \sum_{i\in\mathcal{L}_k}(r_i - \eta^*)u_i' s_i + \sum_{j\in\mathcal{R}_k}(r_j - \eta^*)u_j' t_j \\ \text{s.t.} \quad & \boldsymbol{e}^T \mathbf{s} = K-1, \\ & 0 \le s_i \le 1 \quad \forall i \in \mathcal{L}_k, \\ & 0 \le t_j \le 1 \quad \forall j \in \mathcal{R}_k. \end{aligned} \tag{EC.2}$$

Denote the optimal solution to (EC.2) by $(\mathbf{s}^*, \mathbf{t}^*)$. Following the same idea in Davis et al. (2013), we can prove that (EC.1) is equivalent to (EC.2).

Note that the optimal solution to problem (EC.1), $(\boldsymbol{x}^*, \boldsymbol{y}^*)$, is a feasible solution to problem (EC.2). Evaluating the objective function of problem (EC.2) with $(\boldsymbol{x}^*, \boldsymbol{y}^*)$, we have $\psi^* \ge (r_k - \eta^*)u_k' + \sum_{i\in\mathcal{L}_k}(r_i - \eta^*)u_i' x_i^* + \sum_{j\in\mathcal{R}_k}(r_j - \eta^*)u_j' y_j^* = \eta^*$. On the other hand, by putting $(\mathbf{s}, \mathbf{t})$ together, the first constraint is

$$(\boldsymbol{e} \ \mathbf{0}) \begin{pmatrix} \mathbf{s} \\ \mathbf{t} \end{pmatrix} = K-1$$

Since $\boldsymbol{e}$ is TU, we obtain that $(\boldsymbol{e}, \mathbf{0})$ is also TU. Then by Theorem 4.5 in Conforti et al. (2014), we have the polyhedron $\{(\mathbf{s}, \mathbf{t}) : (\boldsymbol{e}, \mathbf{0})(\mathbf{s}, \mathbf{t})^T = K-1, \mathbf{0} \le (\mathbf{s}, \mathbf{t}) \le \mathbf{1}\}$ is integral. Since the objective function is linear in $(\mathbf{s}, \mathbf{t})$, we have that there exists an optimal solution with $(\mathbf{s}^*, \mathbf{t}^*) \in \{0,1\}^{|\mathcal{L}_k| + |\mathcal{R}_k|}$. Thus, it is a feasible solution to problem (EC.1), and we have

$$\begin{aligned} & \eta^* \ge \frac{r_k u_k' + \sum_{i\in\mathcal{L}_k} r_i u_i' s_i^* + \sum_{j\in\mathcal{R}_k} r_j u_j' t_j^*}{1 + u_k' + \sum_{i\in\mathcal{L}_k} u_i' s_i^* + \sum_{j\in\mathcal{R}_k} u_j' t_j^*} \\ \Rightarrow\ & \eta^* \ge (r_k - \eta^*)u_k' + \sum_{i\in\mathcal{L}_k}(r_i - \eta^*)u_i' s_i^* + \sum_{j\in\mathcal{R}_k}(r_j - \eta^*)u_j' t_j^* \\ \Rightarrow\ & \eta^* \ge \psi^* \end{aligned}$$

Therefore, we have the optimal solution to (EC.2) is also optimal to (EC.1).

In the following, we prove that (9) is equivalent to (EC.2), implying that (9) is equivalent to (8). Let ζ^* and $(x_k^*, \boldsymbol{x}^*, \boldsymbol{y}^*, z^*)$ be the optimal value and optimal solution to problem (9), respectively.

Given the optimal solution $(\mathbf{s}^*, \mathbf{t}^*)$ to problem (EC.2), we construct $(\hat{x}_k, \hat{\boldsymbol{x}}, \hat{\boldsymbol{y}}, \hat{z})$ by letting

$$\begin{aligned}
\hat{x}_k &= \frac{u'_k}{1 + u'_k + \sum_{i \in \mathcal{L}_k} u'_i s_i^* + \sum_{j \in \mathcal{R}_k} u'_j t_j^*}, \\
\hat{x}_i &= \frac{u'_i s_i^*}{1 + u'_k + \sum_{i \in \mathcal{L}_k} u'_i s_i^* + \sum_{j \in \mathcal{R}_k} u'_j t_j^*} \quad \forall i \in \mathcal{L}_k, \\
\hat{y}_j &= \frac{u'_j t_j^*}{1 + u'_k + \sum_{i \in \mathcal{L}_k} u'_i s_i^* + \sum_{j \in \mathcal{R}_k} u'_j t_j^*} \quad \forall j \in \mathcal{R}_k, \\
\hat{z} &= \frac{1}{1 + u'_k + \sum_{i \in \mathcal{L}_k} u'_i s_i^* + \sum_{j \in \mathcal{R}_k} u'_j t_j^*}.
\end{aligned}$$

It is easy to verify that $(\hat{x}_k, \hat{\boldsymbol{x}}, \hat{\boldsymbol{y}}, \hat{z})$ is a feasible solution to problem (9). Then, evaluating the objective function of problem (9) with $(\hat{x}_k, \hat{\boldsymbol{x}}, \hat{\boldsymbol{y}}, \hat{z})$, we have

$$\frac{r_k u'_k + \sum_{i \in \mathcal{L}_k} r_i u'_i s_i^* + \sum_{j \in \mathcal{R}_k} r_j u'_j t_j^*}{1 + u'_k + \sum_{i \in \mathcal{L}_k} u'_i s_i^* + \sum_{j \in \mathcal{R}_k} u'_j t_j^*} = \eta^* = \psi^* \le \zeta^*.$$

Given the optimal solution $(x_k^*, \boldsymbol{x}^*, \boldsymbol{y}^*, z^*)$ to problem (9), then we construct $(\hat{\mathbf{s}}, \hat{\mathbf{t}})$ by letting $\hat{s}_i = x_i^*/(u'_i z^*)$ for all $i \in \mathcal{L}_k$ and $\hat{t}_j = y_j^*/(u'_j z^*)$ for all $j \in \mathcal{R}_k$, it is easy to verify that $(\hat{\mathbf{s}}, \hat{\mathbf{t}})$ is a feasible solution to (EC.2). Then, evaluating the objective function of problem (EC.2) with $(\hat{\mathbf{s}}, \hat{\mathbf{t}})$, we have

$$\begin{aligned}
\eta^* &\ge (r_k - \eta^*) u'_k + \sum_{i \in \mathcal{L}_k} (r_i - \eta^*) u'_i \hat{s}_i + \sum_{j \in \mathcal{R}_k} (r_j - \eta^*) u'_j \hat{t}_j \\
&= \left(r_k u'_k + \sum_{i \in \mathcal{L}_k} r_i u'_i \hat{s}_i + \sum_{j \in \mathcal{R}_k} r_j u'_j \hat{t}_j \right) - \eta^* \left(u'_k + \sum_{i \in \mathcal{L}_k} u'_i \hat{s}_i + \sum_{j \in \mathcal{R}_k} u'_j \hat{t}_j \right) \\
&\ge \frac{r_k x_k^* + \sum_{i \in \mathcal{L}_k} r_i x_i^* + \sum_{j \in \mathcal{R}_k} r_j y_j^*}{z^*} - \zeta^* \left(\frac{x_k^* + \sum_{i \in \mathcal{L}_k} x_i^* + \sum_{j \in \mathcal{R}_k} y_j^*}{z^*} \right) \\
&= \frac{\zeta^*}{z^*} - \zeta^* \left(\frac{1}{z^*} - 1 \right) \\
&= \zeta^*,
\end{aligned}$$

where the second inequality holds since $\eta^* \le \zeta^*$ and the definition of $(\hat{\mathbf{s}}, \hat{\mathbf{t}})$; the second equality holds due to the first constraint in (9), implying that $(x_k^* + \sum_{i \in \mathcal{L}_k} x_i^* + \sum_{j \in \mathcal{R}_k} y_j^*)/z^* = 1/z^* - 1$.

Therefore, we have $\psi^* = \eta^*$, and the optimal solution to (8) can be recovered by letting $(\mathcal{S}_1^*, \mathcal{S}_2^*)$ with $\mathcal{S}_1^* = \{i : x_i^*/u'_i = z^*, \text{for all } i \in \mathcal{L}_k\}$ and $\mathcal{S}_2^* = \{j : y_j^*/u'_j = z^*, \text{for all } j \in \mathcal{R}_k\}$. Then the proof is complete. □

Proof of Theorem 1.

According to the above analysis, to obtain the optimal assortment, we need to solve problem (7) and problem (8) for each $k \in \mathcal{R}_{K-1} \setminus \{n\}$. Thus, the total number of linear programs to be solved is given by $1 + (n - 1 - K + 1) = n - K + 1$. □

Proof of Theorem 2.

We first prove that there exists a K-truncated-profit-ordered optimal assortment. Similar to the proof of Proposition 1, we only need to prove that, when $|\mathcal{S}^*| \geq K+1$, it can be decomposed as $\{i_1, i_2, \ldots, i_K\} \cup \mathcal{S}_{i_K}^{RO}$. It further reduces to show that, for any set $\mathcal{S}_1 = \{i_1, i_2, \ldots, i_K\}$, there exists an optimal solution of the following problem that is profit-ordered in $\mathcal{R}_{i_K}$,

$$\begin{aligned}
\max_{\mathcal{S}_2} \quad & \frac{\sum_{i\in\mathcal{S}_1}\left(1+\rho_0+\sum_{\ell\in F(\mathcal{S}_1)}\rho_\ell\right) r_i u_i + \sum_{j\in\mathcal{S}_2} r_j u_j}{1+\sum_{i\in\mathcal{S}_1}\left(1+\rho_0+\sum_{\ell\in F(\mathcal{S}_1)}\rho_\ell\right) u_i + \sum_{j\in\mathcal{S}_2} u_j} \\
\text{s.t.} \quad & \mathcal{S}_2 \subseteq \mathcal{R}_{i_K} = \{\ell : \ell > i, \forall i \in \mathcal{S}_1\}.
\end{aligned}$$

By Lemma EC.1, the above claim holds.

Now we discuss how to solve the assortment optimization problem. We construct two candidate sets $\mathcal{C}_1$ and $\mathcal{C}_2$ of the optimal assortment and show that $\mathcal{S}^*$ must be included in the union set $\mathcal{C} := \mathcal{C}_1 \cup \mathcal{C}_2$. First, we include all assortments of size no more than K into $\mathcal{C}_1$. If $|\mathcal{S}^*| \leq K$, then $\mathcal{S}^* \in \mathcal{C}_1$, and further $\mathcal{S}^* \in \mathcal{C}$. If $|\mathcal{S}^*| \geq K+1$, it is easy to see that the focal set of $\mathcal{S}^*$ is of size K. Let $\mathcal{S}_{=K}$ contain all assortments of size K. Then, for each $\mathcal{S}_1 \in \mathcal{S}_{=K}$, we assume that it is the focal set in $\mathcal{S}^*$, and aim to find the non-focal set $\mathcal{S}^* \setminus \mathcal{S}_1$. We must have $\mathcal{S}^* \setminus \mathcal{S}_1 \subseteq \mathcal{R}_{i_K}$, where i_K is the largest-labeled item in focal set $\mathcal{S}_1$. Hence, we only need to solve

$$\begin{aligned}
\max_{\mathcal{S}_2} \quad & \frac{\sum_{i\in\mathcal{S}_1}\left(1+\rho_0+\sum_{\ell\in F(\mathcal{S}_1)}\rho_\ell\right) r_i u_i + \sum_{j\in\mathcal{S}_2} r_j u_j}{1+\sum_{i\in\mathcal{S}_1}\left(1+\rho_0+\sum_{\ell\in F(\mathcal{S}_1)}\rho_\ell\right) u_i + \sum_{j\in\mathcal{S}_2} u_j} \\
\text{s.t.} \quad & \mathcal{S}_2 \subseteq \mathcal{R}_{i_K} = \{\ell : \ell > i, i \in \mathcal{S}_1\}.
\end{aligned} \tag{EC.3}$$

Denote the optimal solution of problem (EC.3) by $\mathcal{S}_2(\mathcal{S}_1)$. If the focal set of $\mathcal{S}^*$ is $\mathcal{S}_1$, then we have $\mathcal{S}^* = \mathcal{S}_1 \cup \mathcal{S}_2(\mathcal{S}_1)$. For each $\mathcal{S}_1 \in \mathcal{S}_{=K}$, we now include the assortment $\mathcal{S}_1 \cup \mathcal{S}_2(\mathcal{S}_1)$ into the candidate set $\mathcal{C}_2$. Combining the above two cases, we must obtain $\mathcal{S}^* \in \mathcal{C}$.

For a given $\mathcal{S}_1$, solving problem (EC.3) reduces to enumerating all profit-ordered assortments in $\mathcal{R}_{i_K}$, which can be done in time $O(n)$. Since we also need to enumerate assortments in $\mathcal{S}_{=K}$, the time complexity of constructing $\mathcal{C}_2$ is $O(n^{K+1})$. For each candidate optimal assortment in $\mathcal{C}_1$, its expected profit is given by

$$\frac{\sum_{i\in\mathcal{S}}\left(1+\rho_0+\sum_{j\in\mathcal{S}} c_j\right) r_i u_i}{1+\sum_{i\in\mathcal{S}}\left(1+\rho_0+\sum_{j\in\mathcal{S}} c_j\right) u_i},$$

because all items are focal in this case. Thus, the total time complexity of the above algorithm is $O(n^{K+1})$. $\square$

Proof of Proposition 3.

When $h(|\mathcal{S}|) = \tau$ for all $\mathcal{S} \subseteq \mathcal{N}$, then the focal function is $F(\mathcal{S}) = \{i : \sigma(i, \mathcal{S}) = \min\{\tau, |\mathcal{S}|\}, i \in \mathcal{S}\}$. It then reduces to prove that, if $|\mathcal{S}^*| \geq \tau + 1$ and the focal item in $\mathcal{S}^*$ is item k, then $\mathcal{S}^* =$

$\{i_1, i_2, \ldots, i_{\tau-1}, k\} \cup \mathcal{S}_k^{RO}$. For each possible focal item $k \in \mathcal{R}_\tau = \{i : i > \tau, i \in \mathcal{N}\}$, we solve the following problem:

$$\begin{aligned}
\max_{\mathcal{S}_1, \mathcal{S}_2} \quad & \frac{(1+\rho_0+c_k) r_k u_k + \sum_{i\in\mathcal{S}_1} r_i u_i + \sum_{j\in\mathcal{S}_2} r_j u_j}{1 + (1+\rho_0+c_k) u_k + \sum_{i\in\mathcal{S}_1} u_i + \sum_{j\in\mathcal{S}_2} u_j} \\
\text{s.t.} \quad & |\mathcal{S}_1| = \tau - 1, \\
& \mathcal{S}_1 \subseteq \mathcal{L}_k, \quad \mathcal{S}_2 \subseteq \mathcal{R}_k.
\end{aligned}$$

Denote by $(\mathcal{S}_1^*, \mathcal{S}_2^*)$ the optimal solution of the above problem. By Lemma EC.1, we obtain that there exists a $\mathcal{S}_2^*$ that is profit-ordered in $\mathcal{R}_k$. Then the proof is complete. $\square$

EC.2 Proofs of Technical Results in Section 4

Proof of Proposition 4.

When $\mathcal{N}^* = \{i_{\mathrm{F}}\}$ and $F(\mathcal{S}) = \mathcal{S}_+ \cap \mathcal{N}^*$ for all $\mathcal{S} \subseteq \mathcal{N}$, then the expected profit under the FLM becomes

$$f(\mathcal{S}) = \begin{cases} \frac{\sum_{i\in\mathcal{S}} r_i u_i}{\sum_{i\in\mathcal{S}_+} u_i} & \text{if } i_{\mathrm{F}} \notin \mathcal{S}, \\ \sum_{i\in\mathcal{S}} \frac{r_i u_i}{u_{i_{\mathrm{F}}}\delta(\mathcal{S}) + \sum_{i\in\mathcal{S}_+} u_i} + \frac{r_{i_{\mathrm{F}}} u_{i_{\mathrm{F}}} \delta(\mathcal{S})}{u_{i_{\mathrm{F}}}\delta(\mathcal{S}) + \sum_{i\in\mathcal{S}_+} u_i} & \text{if } i_{\mathrm{F}} \in \mathcal{S}. \end{cases} \tag{EC.4}$$

We begin with the first statement. If $i_{\mathrm{F}} \in \mathcal{S}^*$, then we have

$$\begin{aligned}
f(\mathcal{S}^*) &= \frac{\sum_{j\in\mathcal{S}^*\setminus\{i_{\mathrm{F}}\}} r_j u_j}{u_{i_{\mathrm{F}}}(1+\delta(\mathcal{S}^*)) + \sum_{j\in\mathcal{S}_+^*\setminus\{i_{\mathrm{F}}\}} u_j} + \frac{u_{i_{\mathrm{F}}}(1+\delta(\mathcal{S}^*))}{u_{i_{\mathrm{F}}}(1+\delta(\mathcal{S}^*)) + \sum_{j\in\mathcal{S}_+^*\setminus\{i_{\mathrm{F}}\}} u_j} r_{i_{\mathrm{F}}} \\
&= \frac{\sum_{j\in\mathcal{S}_+^*\setminus\{i_{\mathrm{F}}\}} u_j}{u_{i_{\mathrm{F}}}(1+\delta(\mathcal{S}^*)) + \sum_{j\in\mathcal{S}_+^*\setminus\{i_{\mathrm{F}}\}} u_j} \cdot \frac{\sum_{j\in\mathcal{S}^*\setminus\{i_{\mathrm{F}}\}} r_j u_j}{\sum_{j\in\mathcal{S}_+^*\setminus\{i_{\mathrm{F}}\}} u_j} + \frac{u_{i_{\mathrm{F}}}(1+\delta(\mathcal{S}^*))}{u_{i_{\mathrm{F}}}(1+\delta(\mathcal{S}^*)) + \sum_{j\in\mathcal{S}_+^*\setminus\{i_{\mathrm{F}}\}} u_j} r_{i_{\mathrm{F}}} \\
&= \frac{\sum_{j\in\mathcal{S}_+^*\setminus\{i_{\mathrm{F}}\}} u_j}{u_{i_{\mathrm{F}}}(1+\delta(\mathcal{S}^*)) + \sum_{j\in\mathcal{S}_+^*\setminus\{i_{\mathrm{F}}\}} u_j} \cdot f(\mathcal{S}^* \setminus \{i_{\mathrm{F}}\}) + \frac{u_{i_{\mathrm{F}}}(1+\delta(\mathcal{S}^*))}{u_{i_{\mathrm{F}}}(1+\delta(\mathcal{S}^*)) + \sum_{j\in\mathcal{S}_+^*\setminus\{i_{\mathrm{F}}\}} u_j} r_{i_{\mathrm{F}}}.
\end{aligned}$$

Note that we have $f(\mathcal{S}^*) \geq f(\mathcal{S}^* \setminus \{i_{\mathrm{F}}\})$, thus $r_{i_{\mathrm{F}}} \geq f(\mathcal{S}^*) \geq f(\mathcal{S}_{\mathrm{MNL}}^*)$. The last inequality holds due to the fact that $\mathcal{S}_{\mathrm{MNL}}^*$ is only a feasible solution. Note that

$$\mathcal{S}_{\mathrm{MNL}}^* \in \arg\max_{\mathcal{S}\subseteq\mathcal{N}\setminus\{i_{\mathrm{F}}\}} \frac{\sum_{i\in\mathcal{S}} u_i r_i}{1 + \sum_{i\in\mathcal{S}} u_i}.$$

By the definition of $f(\mathcal{S})$ in (EC.4), we have $f(\mathcal{S}_{\mathrm{MNL}}^*) = f_{\mathrm{MNL}}(\mathcal{S}_{\mathrm{MNL}}^*)$. Hence, we also have $r_{i_{\mathrm{F}}} \geq f_{\mathrm{MNL}}(\mathcal{S}_{\mathrm{MNL}}^*)$. Now we prove that, if $r_{i_{\mathrm{F}}} \geq f_{\mathrm{MNL}}(\mathcal{S}_{\mathrm{MNL}}^*)$, then we have $i_{\mathrm{F}} \in \mathcal{S}^*$. Suppose that $i_{\mathrm{F}} \notin \mathcal{S}^*$, then we have $f(\mathcal{S}^*) = f_{\mathrm{MNL}}(\mathcal{S}_{\mathrm{MNL}}^*)$. Let us construct a new assortment $\mathcal{S}' = \mathcal{S}^* \cup \{i_{\mathrm{F}}\}$. Note that

$$\begin{aligned}
f(\mathcal{S}^* \cup \{i_{\mathrm{F}}\}) &= \frac{\sum_{j\in\mathcal{S}_+^*} u_j}{u_{i_{\mathrm{F}}}\delta(\mathcal{S}^* \cup \{i_{\mathrm{F}}\}) + \sum_{j\in\mathcal{S}_+^*} u_j} \cdot \frac{\sum_{j\in\mathcal{S}^*} r_j u_j}{\sum_{j\in\mathcal{S}_+^*} u_j} + \frac{u_{i_{\mathrm{F}}}\delta(\mathcal{S}^* \cup \{i_{\mathrm{F}}\})}{u_{i_{\mathrm{F}}}\delta(\mathcal{S}^* \cup \{i_{\mathrm{F}}\}) + \sum_{j\in\mathcal{S}_+^*} u_j} \cdot r_{i_{\mathrm{F}}} \\
&= \frac{\sum_{j\in\mathcal{S}_+^*} u_j}{u_{i_{\mathrm{F}}}(\mathcal{S}^* \cup \{i_{\mathrm{F}}\}) + \sum_{j\in\mathcal{S}_+^*} u_j} \cdot f_{\mathrm{MNL}}(\mathcal{S}_{\mathrm{MNL}}^*) + \frac{u_{i_{\mathrm{F}}}(\mathcal{S}^* \cup \{i_{\mathrm{F}}\})}{u_{i_{\mathrm{F}}}(\mathcal{S}^* \cup \{i_{\mathrm{F}}\}) + \sum_{j\in\mathcal{S}_+^*} u_j} r_{i_{\mathrm{F}}} \\
&\geq f_{\mathrm{MNL}}(\mathcal{S}_{\mathrm{MNL}}^*) \\
&\geq f(\mathcal{S}^*).
\end{aligned}$$

The inequality holds due to $r_{i_\text{F}} \geq f_{\text{MNL}}(\mathcal{S}^*_{\text{MNL}}) = f(\mathcal{S}^*)$. This makes a contradiction to the optimality of $\mathcal{S}^*$. Thus, we have $i_\text{F} \in \mathcal{S}^*$.

We then prove the results for the non-focal items when $i_\text{F} \neq 0$. Suppose that $i_\text{F} \in \mathcal{S}^*$, and a non-focal item $i \in \mathcal{S}^*$, then we have $f(\mathcal{S}^* \setminus \{i\}) \leq f(\mathcal{S}^*)$, implying that

$$\begin{aligned}
&\frac{u_{i_\text{F}}\delta(\mathcal{S}^*) + \sum_{\ell \in \mathcal{S}^*_+} u_\ell}{u_{i_\text{F}}\delta(\mathcal{S}^* \setminus \{i\}) + \sum_{\ell \in \mathcal{S}^*_+} u_\ell - u_i} f(\mathcal{S}^*) + \frac{u_{i_\text{F}} r_{i_\text{F}} \left(\delta(\mathcal{S}^* \setminus \{i\}) - \delta(\mathcal{S}^*)\right) - u_i r_i}{u_{i_\text{F}}\delta(\mathcal{S}^* \setminus \{i\}) + \sum_{\ell \in \mathcal{S}^*_+} u_\ell - u_i} \leq f(\mathcal{S}^*) \\
&\Rightarrow \left(u_{i_\text{F}} \left(\delta(\mathcal{S}^*) - \delta(\mathcal{S}^* \setminus \{i\})\right) + u_i\right) f(\mathcal{S}^*) \leq u_i r_i - u_{i_\text{F}} r_{i_\text{F}} \left(\delta(\mathcal{S}^* \setminus \{i\}) - \delta(\mathcal{S}^*)\right) \\
&\Rightarrow r_i \geq f(\mathcal{S}^*) - \frac{u_{i_\text{F}} \left(\delta(\mathcal{S}^*) - \delta(\mathcal{S}^* \setminus \{i\})\right)\left(r_{i_\text{F}} - f(\mathcal{S}^*)\right)}{u_i}.
\end{aligned} \tag{EC.5}$$

Meanwhile, suppose that $i_\text{F} \in \mathcal{S}^*$, and a non-focal item $i \notin \mathcal{S}^*$, then we have $f(\mathcal{S}^* \cup \{i\}) \leq f(\mathcal{S}^*)$, implying that

$$\begin{aligned}
&u_i(f(\mathcal{S}^*) - r_i) \geq u_{i_\text{F}} \left(\delta(\mathcal{S}^* \cup \{i\}) - \delta(\mathcal{S}^*)\right)\left(r_{i_\text{F}} - f(\mathcal{S}^*)\right) \\
\Rightarrow &r_i \leq f(\mathcal{S}^*) - \frac{u_{i_\text{F}} \left(\delta(\mathcal{S}^* \cup \{i\}) - \delta(\mathcal{S}^*)\right)\left(r_{i_\text{F}} - f(\mathcal{S}^*)\right)}{u_i}.
\end{aligned} \tag{EC.6}$$

Finally, we consider the case where $i_\text{F} = 0$. If a non-focal item $i \in \mathcal{S}^*$, then we have $f(\mathcal{S}^* \setminus \{i\}) \leq f(\mathcal{S}^*)$, implying that

$$\begin{aligned}
&\frac{\sum_{i \in \mathcal{S}^* \setminus \{i\}} r_i u_i}{1 + \delta(\mathcal{S}^* \setminus \{i\}) + \sum_{i \in \mathcal{S}^* \setminus \{i\}} u_i} \leq f(\mathcal{S}^*) \\
\Rightarrow &\left(\frac{1 + \delta(\mathcal{S}^*) + \sum_{i \in \mathcal{S}^*} u_i}{1 + \delta(\mathcal{S}^* \setminus \{i\}) + \sum_{i \in \mathcal{S}^* \setminus \{i\}} u_i} - 1\right) f(\mathcal{S}^*) \leq \frac{r_i u_i}{1 + \delta(\mathcal{S}^* \setminus \{i\}) + \sum_{i \in \mathcal{S}^* \setminus \{i\}} u_i} \\
\Rightarrow &\ r_i \geq f(\mathcal{S}^*) + \frac{\delta(\mathcal{S}^*) - \delta(\mathcal{S}^* \setminus \{i\}) f(\mathcal{S}^*)}{u_i}.
\end{aligned} \tag{EC.7}$$

Recall that $r_0 = 0$. Hence, combining (EC.5) and (EC.7), we obtain that if a non-focal item $i \in \mathcal{S}^*$, then

$$r_i \geq f(\mathcal{S}^*) - \frac{u_{i_\text{F}} \left(\delta(\mathcal{S}^*) - \delta(\mathcal{S}^* \setminus \{i\})\right)\left(r_{i_\text{F}} - f(\mathcal{S}^*)\right)}{u_i},$$

Thus, the second statement is proved.

If a non-focal item $i \notin \mathcal{S}^*$, then we have $f(\mathcal{S}^* \cup \{i\}) \leq f(\mathcal{S}^*)$, implying that

$$\begin{aligned}
&\frac{\sum_{i \in \mathcal{S}^* \cup \{i\}} r_i u_i}{1 + \delta(\mathcal{S}^* \cup \{i\}) + \sum_{i \in \mathcal{S}^* \cup \{i\}} u_i} \leq f(\mathcal{S}^*) \\
\Rightarrow &\left(\frac{1 + \delta(\mathcal{S}^*) + \sum_{i \in \mathcal{S}^*} u_i}{1 + \delta(\mathcal{S}^* \cup \{i\}) + \sum_{i \in \mathcal{S}^* \cup \{i\}} u_i} - 1\right) f(\mathcal{S}^*) + \frac{u_i r_i}{1 + \delta(\mathcal{S}^* \cup \{i\}) + \sum_{i \in \mathcal{S}^* \cup \{i\}} u_i} \leq 0 \\
\Rightarrow &\ r_i \leq f(\mathcal{S}^*) + \frac{\delta(\mathcal{S}^* \cup \{i\}) - \delta(\mathcal{S}^*) f(\mathcal{S}^*)}{u_i}.
\end{aligned} \tag{EC.8}$$

Similarly, combining (EC.6) and (EC.8), we obtain that if a non-focal item $i \notin \mathcal{S}^*$, then

$$r_i \leq f(\mathcal{S}^*) - \frac{u_{i_\text{F}} \left(\delta(\mathcal{S}^* \cup \{i\}) - \delta(\mathcal{S}^*)\right)\left(r_{i_\text{F}} - f(\mathcal{S}^*)\right)}{u_i}.$$

Thus, the third statement is proved, and the proof is complete. $\square$

Proof of Theorem 3.

When $i_{\mathrm{F}} = 0$, we can choose the distortion function $\delta(\mathcal{S}) = \exp(\nu \sum_{i\in S} u_i) - 1$, where $\nu > 0$. Then, the model reduces to the generalized MNL model proposed by Goutam et al. (2019). They prove that the associated assortment optimization problem is NP-hard.

Then, we focus on the case where $i_{\mathrm{F}} \neq 0$. Without loss of generality, we let $u_0 = 1$. We first restate problem (1) as follows.

Decision Problem of (1): Given a threshold K, does there exist a subset $\mathcal{S} \subseteq \mathcal{N}$ that generates an objective value of K or more in problem (1)?

Note that if one can answer the above decision problem in polynomial time for any given $K \in \mathbb{R}_+$, then the optimal value to problem (1) can be found in polynomial time as well. In what follows, we map any instance of the *partition problem* to an instance of the decision problem of (1). First, we introduce the partition problem.

Partition Problem (Hartmanis 1982): Given a sequence of non-negative integers $\{c_1, c_2, \ldots, c_{n-1}\}$ with $\sum_{i=1}^{n-1} c_i = 2U$, does there exist a subset $\mathcal{S}$ such that $\sum_{i\in\mathcal{S}} c_i = U$? It is well-known that the partition problem is NP-complete.

Given a partition problem instance defined by a sequence $\{c_1, c_2, \ldots, c_{n-1}\}$, we construct a decision problem as follows. Let

$$r_i = 1\ ,\ u_i = c_i, \quad \forall i = 1, 2, \ldots, n-1.$$

and

$$r_n = 1\ ,\ u_n = U.$$

We let the focal function be $F(\mathcal{S}) = \mathcal{S}_+ \cap \{n\}$ for all $\mathcal{S} \subseteq \mathcal{N}$, that is, item n is the focal item. We also let the distortion function be

$$\delta(\mathcal{S}) = \begin{cases} 0, & \text{if } n \notin \mathcal{S}, \\ 0, & \text{if } n \in \mathcal{S} \text{ and } \sum_{i\in\mathcal{S}\setminus\{n\}} u_i < U, \\ \frac{-\left(\sum_{i\in\mathcal{S}\setminus\{n\}} u_i\right)^2/2 + (U-1)\sum_{i\in\mathcal{S}\setminus\{n\}} u_i + 2U}{u_n \cdot \mathbf{1}\{n\in\mathcal{S}\}}, & \text{otherwise}\ , \end{cases}$$

and set the target objective value to be $K = (U^2 + 6U)/(U^2 + 6U + 2)$.

In the following, we turn back to problem (1), and analyze when it will obtain the optimal value. We first claim that item n must be in the optimal assortment. To see this, we apply Proposition 4-(1) and arrive at the following property:

$$n \in \mathcal{S}^* \Leftrightarrow r_n \geq \max_{\mathcal{S}\subseteq\mathcal{N}\setminus\{n\}} \frac{\sum_{i\in\mathcal{S}} c_i}{1 + \sum_{i\in\mathcal{S}} c_i}.$$

Note that $\mathcal{N}\setminus\{n\}$ itself is the optimal solution of the above problem, and we have

$$r_n = 1 > \frac{\sum_{i\in\mathcal{N}\setminus\{n\}} c_i}{1+\sum_{i\in\mathcal{N}\setminus\{n\}} c_i}.$$

Thus, item n must be in the optimal assortment. Furthermore, problem (1) reduces to determine the subset $\mathcal{S}\subseteq\mathcal{N}\setminus\{n\}$ such that the profit of $\mathcal{S}\cup\{n\}$ is maximized, i.e.,

$$\max_{\mathcal{S}\subseteq\mathcal{N}\setminus\{n\}} g(\mathcal{S}) := \frac{\sum_{i\in\mathcal{S}} u_i + (1+\delta(\mathcal{S}\cup\{n\}))u_n}{1+\sum_{i\in\mathcal{S}} u_i + (1+\delta(\mathcal{S}\cup\{n\}))u_n}.$$

It is easy to see that we only need to maximize $\sum_{i\in\mathcal{S}} u_i + \delta(\mathcal{S}\cup\{n\})u_n$. By the definition of $\delta(\cdot)$, we have

$$\sum_{i\in\mathcal{S}} u_i + \delta(\mathcal{S}\cup\{n\})u_n = \begin{cases} \sum_{i\in\mathcal{S}} u_i & \text{if } \sum_{i\in\mathcal{S}} u_i < U \\ -\left(\sum_{i\in\mathcal{S}} u_i\right)^2/2 + U\cdot\sum_{i\in\mathcal{S}} u_i + 2U & \text{if } \sum_{i\in\mathcal{S}} u_i \geq U \end{cases}.$$

Now we discuss function $v(x) = x + (-x^2/2 + (U-1)x + 2U)\cdot \mathbf{1}\{x\geq U\}$. It is clear that $v(x)$ is increasing in $[0,U)$, and decreasing in $[U,2U]$. The maximum is obtained at $v=U$, and the maximal value is $U^2/2+2U$, and this corresponds to the target value $K=(U^2+6U)/(U^2+6U+2)$. This implies that, only when $\sum_{i\in\mathcal{S}} u_i = U$, i.e., $\sum_{i\in\mathcal{S}} c_i = U$, the function $g(\mathcal{S})$ has the value of $K=(U^2+6U)/(U^2+6U+2)$.

In other words, there exists an assortment $\mathcal{S}\subseteq\mathcal{N}$ whose expected profit is at least K if and only if there exists an assortment $\mathcal{S}'\subseteq\mathcal{N}\setminus\{n\}$ that satisfies $\sum_{i\in\mathcal{S}'} c_i = U$. Thus, the decision problem with target value K reduces to the partition problem, which is NP-complete. This further implies the assortment optimization problem is NP-hard. The proof is then complete. □

EC.3 Proofs of Technical Results in Section 5

Proof of Lemma 2.

We prove by discussing the cardinality of $\mathcal{S}^*$. First, we show that when $|\mathcal{S}^*|\geq K$, it must be the case that $\mathcal{S}^*=\mathcal{N}$. Suppose that $|\mathcal{S}^*|<n$. Let $(\mathcal{S}^*,\mathbf{p}^*)$ be the optimal solution of the joint optimization problem. We then construct a new assortment $\mathcal{S}'=\mathcal{S}^*\cup\{j\}$, where item $j\notin\mathcal{S}^*$, and a new price vector $\mathbf{p}'=[\mathbf{p}^*;p_j]$. Note that we consider the attraction value ranking and let p_j be sufficiently large such that $\exp(\alpha_\ell-\beta_\ell p_\ell)\geq\exp(\alpha_j-\beta_j p_j)$, $\forall \ell\in\mathcal{S}^*$. Hence, the sets containing top-K focal items in $\mathcal{S}'$ and $\mathcal{S}^*$ are the same, and we denote them by $\mathcal{F}$. This implies that the expected profit corresponding to $(\mathcal{S}',\mathbf{p}')$ can be represented by

$$\begin{aligned} &R(\mathcal{S}',\mathbf{p}') \\ =&\frac{\sum_{\ell\in\mathcal{S}^*\setminus\mathcal{F}}(p^*_\ell-c_\ell)\exp(\alpha_\ell-\beta_\ell p^*_\ell)+(1+\delta)\sum_{i\in\mathcal{F}}(p^*_i-c_i)\exp(\alpha_i-\beta_i p^*_i)+(p_j-c_j)\exp(\alpha_j-\beta_j p_j)}{1+\sum_{\ell\in\mathcal{S}^*\setminus\mathcal{F}}\exp(\alpha_\ell-\beta_\ell p^*_\ell)+(1+\delta)\sum_{i\in\mathcal{F}}\exp(\alpha_i-\beta_i p^*_i)+\exp(\alpha_j-\beta_j p_j)} \\ =&\frac{\sum_{\ell\in\mathcal{S}^*}(p^*_\ell-c_\ell)\exp(\alpha_\ell-\beta_\ell p^*_\ell)+\delta\sum_{i\in\mathcal{F}}(p^*_i-c_i)\exp(\alpha_i-\beta_i p^*_i)+(p_j-c_j)\exp(\alpha_j-\beta_j p_j)}{1+\sum_{\ell\in\mathcal{S}^*}\exp(\alpha_\ell-\beta_\ell p^*_\ell)+\delta\sum_{i\in\mathcal{F}}\exp(\alpha_i-\beta_i p^*_i)+\exp(\alpha_j-\beta_j p_j)}. \end{aligned}$$

Furthermore, we have

$$R(\mathcal{S}',\mathbf{p}') = \frac{1+\sum_{\ell\in\mathcal{S}^*}\exp(\alpha_\ell-\beta_\ell p_\ell^*)+\delta\sum_{i\in\mathcal{F}}\exp(\alpha_i-\beta_i p_i^*)}{1+\sum_{\ell\in\mathcal{S}^*}\exp(\alpha_\ell-\beta_\ell p_\ell^*)+\delta\sum_{i\in\mathcal{F}}\exp(\alpha_i-\beta_i p_i^*)+\exp(\alpha_j-\beta_j p_j)}\cdot R(\mathcal{S}^*,\mathbf{p}^*)$$
$$+\frac{\exp(\alpha_j-\beta_j p_j)}{1+\sum_{\ell\in\mathcal{S}^*}\exp(\alpha_\ell-\beta_\ell p_\ell^*)+\delta\sum_{i\in\mathcal{F}}\exp(\alpha_i-\beta_i p^*)+\exp(\alpha_j-\beta_j p_j)}\cdot(p_j-c_j).$$

Hence, as long as $p_j-c_j>R(\mathcal{S}^*,\mathbf{p}^*)$, we have $R(\mathcal{S}',\mathbf{p}')>R(\mathcal{S}^*,\mathbf{p}^*)$, which contradicts to the optimality of $(\mathcal{S}^*,\mathbf{p}^*)$. Therefore, we obtain $\mathcal{S}^*=\mathcal{N}$.

When $|\mathcal{S}^*|<K$, we can prove that adding a sufficiently high-priced item will increase the expected profit by a similar argument, implying a contradiction. Thus, the proof is complete. □

Proof of Lemma 3.

Given $\mathcal{F}$, we first rewrite the maximization problem over $\mathbf{p}$ in (14). Let

$$r=\frac{\sum_{i\in\mathcal{F}}(1+\delta)(p_i-c_i)\exp(\alpha_i-\beta_i p_i)+\sum_{j\in\mathcal{N}\setminus\mathcal{F}}(p_j-c_j)\exp(\alpha_j-\beta_j p_j)}{1+\sum_{i\in\mathcal{F}}(1+\delta)\exp(\alpha_i-\beta_i p_i)+\sum_{j\in\mathcal{N}\setminus\mathcal{F}}\exp(\alpha_j-\beta_j p_j)},$$

then the maximization problem over $\mathbf{p}$ in (14) becomes

$$\begin{aligned}\max_{\mathbf{p}}\quad & \left\{r: r=\sum_{i\in\mathcal{F}}(1+\delta)(p_i-c_i-r)\exp(\alpha_i-\beta_i p_i)+\sum_{j\in\mathcal{N}\setminus\mathcal{F}}(p_j-c_j-r)\exp(\alpha_j-\beta_j p_j)\right\}\\ \text{s.t.}\quad & \alpha_i-\beta_i p_i\ge\alpha_k-\beta_k p_k\quad \forall i\in\mathcal{F}\setminus\{k\},\\ & \alpha_j-\beta_j p_j\le\alpha_k-\beta_k p_k\quad \forall j\in\mathcal{N}\setminus\mathcal{F},\\ & p_\ell\ge c_\ell\ ,\ \forall\ell\in\mathcal{N}.\end{aligned}$$

Note that r is continuous and increasing in r, and $(p_\ell-c_\ell-r)\exp(\alpha_\ell-\beta_\ell p_\ell)$, $\forall\ell\in\mathcal{N}$, is continuous and decreasing in r. Thus, the above problem can be further simplified into the following one-dimensional equation of r,

$$\begin{aligned}r=\quad & \max_{\mathbf{p}}\left\{\sum_{i\in\mathcal{F}}(1+\delta)(p_i-c_i-r)\exp(\alpha_i-\beta_i p_i)+\sum_{j\in\mathcal{N}\setminus\mathcal{F}}(p_j-c_j-r)\exp(\alpha_j-\beta_j p_j)\right\}\\ & \text{s.t. } \alpha_i-\beta_i p_i\ge\alpha_k-\beta_k p_k\quad \forall i\in\mathcal{F}\setminus\{k\},\\ & \alpha_j-\beta_j p_j\le\alpha_k-\beta_k p_k\quad \forall j\in\mathcal{N}\setminus\mathcal{F},\\ & p_\ell\ge c_\ell\ ,\ \forall\ell\in\mathcal{N}.\end{aligned}\tag{EC.9}$$

It is easy to show that $(p_\ell-c_\ell-r)\exp(\alpha_\ell-\beta_\ell p_\ell)$ is unimodal in p_ℓ, $\forall\ell\in\mathcal{N}\setminus\{k\}$, and the maximum is obtained at $p_\ell^*=r+c_\ell+1/\beta_\ell$. Given p_k, to maximize $(p_i-c_i-r)\exp(\alpha_i-\beta_i p_i)$ with the constraint $\alpha_i-\beta_i p_i\ge\alpha_k-\beta_k p_k$ and $p_i\ge c_i$, we have its maximizer $p_i'(r)=\max\{c_i,\min\{r+1/\beta_i+c_i,(\alpha_i-\alpha_k+\beta_k p_k)/\beta_i\}\}$ for $i\in\mathcal{F}\setminus\{k\}$. Similarly, denoted by $\tilde{p}_j$ the maximizer of $(p_j-c_j-r)\exp(\alpha_j-\beta_j p_j)$ with the constraint $\alpha_j-\beta_j p_j\le\alpha_k-\beta_k p_k$ and $p_j\ge c_j$, we obtain $\tilde{p}_j(r)=\max\{r+1/\beta_j+c_j,(\alpha_j-$

$\alpha_k + \beta_k p_k)/\beta_j\}$ for $j \in \mathcal{N} \setminus \mathcal{F}$. Thus, we have that the optimal profit, denoted by r^*, is the solution of the equation with respect to r,

$$r = \max_{p_k \in \mathcal{P}_k} H(p_k, r), \tag{EC.10}$$

where

$$\begin{aligned} H(p_k, r) &= (1+\delta)(p_k - c_k - r)\exp(\alpha_k - \beta_k p_k) \\ &\quad + \sum_{i \in \mathcal{F}\setminus\{k\}} (1+\delta)(p_i'(r) - c_i - r)\exp(\alpha_i - \beta_i p_i'(r)) + \sum_{j \in \mathcal{N}\setminus\mathcal{F}} (\tilde{p}_j(r) - c_j - r)\exp(\alpha_j - \beta_j \tilde{p}_j(r)), \end{aligned}$$

and $\mathcal{P}_k = \{p_k : p_k \geq c_k\}$.

Next, we show that function $\phi(r) := \max_{p_k \in \mathcal{P}_k} H(p_k, r)$ is continuous and decreasing in r. We first show that $H(p_k, r)$ is decreasing and continuous in r for a given p_k. Note that given p_k, $H(p_k, r)$ is the objective value of the optimization problem in the RHS of (EC.9) only over $\mathbf{p}_{-k}$, where $\mathbf{p}_{-k}$ represents the price vector of the item set $\mathcal{N}$ without p_k. Suppose that $r_2 \geq r_1$, $\mathbf{p}^*_{-k,1}$ and $\mathbf{p}^*_{-k,2}$ is the optimal solution of the optimization problem in (EC.9) only over $\mathbf{p}_{-k}$ with respect to r_1 and r_2, respectively. Then we have

$$H(p_k, r_2) = H(p_k, r_2; \mathbf{p}^*_{-k,2}) \leq H(p_k, r_1; \mathbf{p}^*_{-k,2}) \leq H(p_k, r_1; \mathbf{p}^*_{-k,1}) = H(p_k, r_1).$$

Note that here we explicitly show $H(p_k, r)$'s dependence on $\mathbf{p}^*_{-k}$. The first inequality holds due to the objective function is decreasing in r, and the second inequality follows from the definition of $\mathbf{p}^*_{-k,1}$. Therefore, $H(p_k, r)$ is decreasing in r when p_k is given. With a similar argument, we can easily show that $\phi(r)$ is also decreasing in r. Meanwhile, since $p_i'(r)$ and $\tilde{p}_j(r)$ is continuous in r, we obtain that $H(p_k, r)$ is continuous in r, which further implies the continuity of $\phi(r)$ in r.

Finally, since the LHS r in (EC.10) is increasing and continuous in r, and the RHS in (EC.10) is decreasing and continuous in r, there is a unique solution of the equation (EC.10), which is exactly the optimal value to the maximization problem over $\mathbf{p}$ in (14) given $\mathcal{F}$. This completes the proof. □

Proof of Lemma 4.

For a given r, we let $p_k^* \in \arg\max_{p_k \in \mathcal{P}_k} H(p_k, r)$ and construct the set of linear functions in p_k,

$$\left\{ \frac{\beta_k}{\beta_\ell} p_k + \frac{\alpha_\ell - \alpha_k - 1}{\beta_\ell} - c_\ell - r \right\}_{\ell \in \mathcal{N}\setminus\{k\}},$$

and denote their intersection points by $\{\hat{p}_{k,m}\}_{m \in [M]}$, where $M = O(n^2)$ is the total number of intersection points. Without loss of generality, we assume that $\hat{p}_{k,0} = c_k \leq \hat{p}_{k,1} < \hat{p}_{k,2} < \cdots < \hat{p}_{k,M} < +\infty = \hat{p}_{k,M+1}$. The set of points $\{\hat{p}_{k,m}\}_{m \in [M]}$ divides the feasible region of p_k into several non-overlapping intervals, i.e., $\{p_k : p_k \geq c_k\} = [\hat{p}_{k,0}, \hat{p}_{k,1}) \cup [\hat{p}_{k,1}, \hat{p}_{k,2}) \cup \ldots \cup [\hat{p}_{k,M}, \hat{p}_{k,M+1})$. For each

$[\hat{p}_{k,m}, \hat{p}_{k,m+1})$, we identify the index set $I_{k,m}$, which contains all $\ell \in \mathcal{N} \setminus \{k\}$ satisfying $\beta_k p_k/\beta_\ell + (\alpha_\ell - \alpha_k - 1)/\beta_\ell - c_\ell - r \geq 0$. This further implies that for all $\ell \in I_{k,m}$, $(\alpha_\ell - \alpha_k + \beta_k p_k)/\beta_\ell \geq r + 1/\beta_\ell + c_\ell$, which helps identify the values of maximizers $p_i'(r)$ and $\tilde{p}_j(r)$. Moreover, we denote by $I_{k,m}^c$ the set $(\mathcal{N} \setminus \{k\}) \setminus I_{k,m}$. Then for each $m \in [M]$, we can maximize $H(p_k, r)$ over $p_k \in [\hat{p}_{k,m}, \hat{p}_{k,m+1})$, that is,

$$
\begin{aligned}
\max_{p_k \in [\hat{p}_{k,m}, \hat{p}_{k,m+1})} \; & H_m(p_k, r) \\
:= & \sum_{j \in (\mathcal{N} \setminus \mathcal{F}) \cap I_{k,m}^c} (1+\delta)(p_k - c_k - r) \exp(\alpha_k - \beta_k p_k) \\
& + \sum_{i \in (\mathcal{F} \setminus \{k\}) \cap I_{k,m}} (1+\delta) \frac{\exp(\alpha_i - \beta_i r - \beta_i c_i - 1)}{\beta_i} \\
& + \sum_{i \in (\mathcal{F} \setminus \{k\}) \cap I_{k,m}^c} (1+\delta) \left(\max\left\{ \frac{\alpha_i - \alpha_k + \beta_k p_k}{\beta_i}, c_i \right\} - c_i - r \right) \exp\left(\min\left\{\alpha_k - \beta_k p_k, \alpha_i - \beta_i c_i\right\}\right) \\
& + \sum_{j \in (\mathcal{N} \setminus \mathcal{F}) \cap I_{k,m}} \left(\frac{\alpha_j - \alpha_k + \beta_k p_k}{\beta_j} - c_j - r \right) \exp(\alpha_k - \beta_k p_k) \\
& + \sum_{j \in (\mathcal{N} \setminus \mathcal{F}) \cap I_{k,m}^c} \frac{\exp(\alpha_j - \beta_j r - \beta_j c_j - 1)}{\beta_j}.
\end{aligned}
\tag{EC.11}
$$

Since $H_m(p_k, r)$ is a finite sum of terms of the form $\mathscr{L}(p_k)\exp(\alpha_k - \beta_k p_k)$, where $\mathscr{L}(\cdot)$ is a linear function, it can be shown that $H_m(p_k, r)$ is an unimodal function in p_k. Hence, the above one-dimensional optimization problem can be solved efficiently by some classic optimization algorithms. Let $p_{k,m}^*$ be its optimal solution. Finally, we choose p_k^* from $\left\{p_{k,m}^*\right\}_{m \in [M]}$ such that $H(p_k^*, r)$ has the largest value. In other words, we have

$$
p_k^* = \underset{p_k \in \left\{p_{k,m}^*\right\}_{m \in [M]}}{\arg\max} \; H(p_k, r). \tag{EC.12}
$$

We now give the time complexity of the above procedure. Problem (EC.11) can be solved in the time complexity of $O(n)$ by computing the stationary point of $H_m(p_k, r)$ and comparing it with $\hat{p}_{k,m}$ and $\hat{p}_{k,m+1}$. Since we also need to compute M intersection points and construct the index set $I_{k,m}$, the total time complexity is $O(n^3)$. $\square$

Proof of Lemma 5.

Since

$$
r^{\mathcal{S}} = \max_{\mathbf{p} \in \mathcal{P}} R(\mathcal{S}, \mathbf{p}), \tag{EC.13}
$$

we have

$$
r^{\mathcal{S}} = R(\mathcal{S}, \mathbf{p}^{\mathcal{S}}) \geq R(\mathcal{S}, \mathbf{p}), \; \forall \mathbf{p} \in \mathcal{P}. \tag{EC.14}
$$

By the definition of $R(\mathcal{S},\mathbf{p})$ in (16), we obtain

$$\sum_{i\in\mathcal{S}\cap\mathcal{N}^*}\left(p_i-c_i-r^{\mathcal{S}}\right)\exp\left(\alpha_i-\beta_i p_i\right)+\sum_{\ell\in\mathcal{S}\setminus\mathcal{N}^*}\left(1+g(|\mathcal{S}|)\right)\left(p_\ell-c_\ell-r^{\mathcal{S}}\right)\exp\left(\alpha_\ell-\beta_\ell p_\ell\right)$$
$$-r^{\mathcal{S}}\left(1+g(|\mathcal{S}|)\mathbf{1}\{0\in\mathcal{N}^*\}\right)\le 0\,,\,\forall\mathbf{p}\in\mathcal{P},$$

where the equality holds when $\mathbf{p}=\mathbf{p}^*$ according to (EC.14). In other words, let $G(\mathbf{p})$ denote the LHS of the above inequality, then

$$G(\mathbf{p})\le 0\,,\,\forall\mathbf{p}\in\mathcal{P}\text{ and }G(\mathbf{p}^*)=0,$$

thus the maximum of $G(\mathbf{p})$ is achieved when $\mathbf{p}=\mathbf{p}^*$.

We now consider the following constrained optimization problem,

$$\begin{aligned}\max_{\mathbf{p}}\quad & G(\mathbf{p})\\ \text{s.t.}\quad & p_i\ge c_i\,,\,\forall i\in\mathcal{S}.\end{aligned}$$

By the Karush–Kuhn–Tucker (KKT) conditions, we have

$$\begin{cases}\nabla\left(G(\mathbf{p})+\sum_{i\in\mathcal{S}}\lambda_i(p_i-c_i)\right)=0,\\ \lambda_i(p_i-c_i)=0,\forall i\in\mathcal{S},\\ \lambda_i\ge 0,p_i\ge c_i,\forall i\in\mathcal{S},\end{cases}\tag{EC.15}$$

where λ_i is the Lagrange multiplier corresponding to the constraint $p_i\ge c_i$. For all $i\in\mathcal{S}\setminus\mathcal{N}^*$, from the first inequality, we have

$$\beta_i\exp\left(\alpha_i-\beta_i p_i\right)\left(r^{\mathcal{S}}+c_i+\frac{1}{\beta_i}-p_i\right)+\lambda_i^*=0$$
$$\Rightarrow\quad\lambda_i^*=-\beta_i\exp\left(\alpha_i-\beta_i p_i\right)\left(r^{\mathcal{S}}+c_i+\frac{1}{\beta_i}-p_i^*\right).$$

Since $\lambda_i^*\ge 0$ for all $i\in\mathcal{S}$, we obtain $p_i^*\ge r^{\mathcal{S}}+1/\beta_i+c_i$. Then by the second equality in (EC.15), we must have $\lambda_i^*=0$. Hence,

$$p_i^*=r^{\mathcal{S}}+c_i+\frac{1}{\beta_i},\forall i\in\mathcal{S}\setminus\mathcal{N}^*.$$

Similarly, we have $p_i^*=r^{\mathcal{S}}+c_i+1/\beta_i$ for all $i\in\mathcal{S}\cap\mathcal{N}^*$. Furthermore, we have

$$\begin{aligned}r^{\mathcal{S}}&=R(\mathcal{S},\mathbf{p}^{\mathcal{S}})\\ &=\frac{\sum_{i\in\mathcal{S}\cap\mathcal{N}^*}\left(1+g(|\mathcal{S}|)\right)(p_i^{\mathcal{S}}-c_i)\exp\left(\alpha_i-\beta_i p_i^{\mathcal{S}}\right)+\sum_{\ell\in\mathcal{S}\setminus\mathcal{N}^*}(p_\ell^{\mathcal{S}}-c_\ell)\exp\left(\alpha_\ell-\beta_\ell p_\ell\right)}{\sum_{i\in\mathcal{S}_+\cap\mathcal{N}^*}\left(1+g(|\mathcal{S}|\right)\exp\left(\alpha_i-\beta_i p_i^{\mathcal{S}}\right)+\sum_{\ell\in\mathcal{S}_+\setminus\mathcal{N}^*}\exp\left(\alpha_\ell-\beta_\ell p_\ell^{\mathcal{S}}\right)}\\ \Rightarrow\quad r^{\mathcal{S}}&=\frac{\sum_{i\in\mathcal{S}\cap\mathcal{N}^*}(1+g(|\mathcal{S}|))\exp\left(\gamma_i-\beta_i r^{\mathcal{S}}\right)+\sum_{\ell\in\mathcal{S}\setminus\mathcal{N}^*}\exp\left(\gamma_\ell-\beta_\ell r^{\mathcal{S}}\right)}{1+g(|\mathcal{S}|)\mathbf{1}\{0\in\mathcal{N}^*\}}.\end{aligned}$$

where $\gamma_i:=\alpha_i-\beta_i c_i-1-\log\beta_i$ for all $i\in\mathcal{N}$.

Now we decompose the assortment $\mathcal{S}$ into the subset $\mathcal{S}_1 \subseteq \mathcal{N}^* \setminus \{0\}$ and the subset $\mathcal{S}_2 \subseteq \mathcal{N} \setminus \mathcal{N}^*$. We define

$$J(r, \mathcal{S}_1, \mathcal{S}_2) := \frac{\sum_{i\in\mathcal{S}_1} (1 + g(|\mathcal{S}_1| + |\mathcal{S}_2|) \exp(\gamma_i - \beta_i r)) + \sum_{\ell\in\mathcal{S}_2} \exp(\gamma_\ell - \beta_\ell r)}{1 + g(|\mathcal{S}_1| + |\mathcal{S}_2|)\mathbf{1}\{0 \in \mathcal{N}^*\}},$$

where $\mathcal{S}_1 \subseteq \mathcal{N}^* \setminus \{0\}$, and $\mathcal{S}_2 \subseteq \mathcal{N} \setminus \mathcal{N}^*$. Then $J(r, \mathcal{S}_1, \mathcal{S}_2)$ becomes

$$\frac{1 + g(|\mathcal{S}_1| + |\mathcal{S}_2|)}{1 + g(|\mathcal{S}_1| + |\mathcal{S}_2|)\mathbf{1}\{0 \in \mathcal{N}^*\}} \sum_{i\in\mathcal{S}_1} \exp(\gamma_i - \beta_i r) + \frac{\sum_{i\in\mathcal{S}_2} \exp(\gamma_i - \beta_i r)}{1 + g(|\mathcal{S}_1| + |\mathcal{S}_2|)\mathbf{1}\{0 \in \mathcal{N}^*\}}.$$

Moreover, we obtain

$$r^{\mathcal{S}} = J(r^{\mathcal{S}}, \mathcal{S}_1, \mathcal{S}_2).$$

Then the proof of Lemma 5 is complete. □

EC.4 Algorithms and More Analysis for Assortment Optimization

EC.4.1 Multiple Arbitrarily-Located Focal Items Model

We generalize the results in Section 3.2 to the multiple focal items case. In particular, we assume that the customers form a focal set based on K predetermined ranks $(\tau_1, \tau_2, \ldots, \tau_K)$, satisfying $1 \le \tau_1 < \tau_2 < \cdots < \tau_K \le n$ and $\tau_k \in \mathbb{Z}_{++}$ for all $k \in [K]$. We also let the distortion function be a positive constant δ. For simplicity, we assume $K = 2$ and define the focal set as,

$$F(\mathcal{S}) = \{(i_1, i_2, \ldots, i_K) : \sigma(i_k, \mathcal{S}) = \min\{|\mathcal{S}|, \tau_k\} \text{ for all } k \in [K]\}.$$

It covers the top-K focal model as a special case. We also remark that it is straightforward to extend the following results to the case where τ_k is a function of $|\mathcal{S}|$ and $K \ge 3$.

To solve the associated assortment optimization problem, we construct the candidate set $\mathcal{C}_1$, $\mathcal{C}_2$ and $\mathcal{C}_3$, and show that $\mathcal{S}^*$ must belong to the union set $\mathcal{C}_1 \cup \mathcal{C}_2 \cup \mathcal{C}_3$. Suppose that $|\mathcal{S}^*| \le \tau_1$, then the focal item is the largest-labeled item in $\mathcal{S}^*$. To cover this case, we enumerate all possible choices of the focal item in $\mathcal{S}^*$, denoted by k. Then, we solve

$$\begin{aligned} \max_{\mathcal{S}\subseteq\mathcal{L}_k} \quad & \frac{(1+\delta) r_k u_k + \sum_{i\in\mathcal{S}} r_i u_i}{1 + (1+\delta) u_k + \sum_{i\in\mathcal{S}} u_i} \\ \text{s.t.} \quad & |\mathcal{S}| \le \tau_1 - 1, \end{aligned}$$

and include the corresponding optimal solution into set $\mathcal{C}_1$ for each $k \in \mathcal{N}$. Given k, this problem can be solved by a linear program since it is a cardinality-constrained assortment optimization problem under the MNL model. If $\tau_1 < |\mathcal{S}^*| \le \tau_2$, then the second focal item i_2 is the largest-labeled item in $\mathcal{S}^*$. To cover this case, we enumerate all possible choices of the focal set $\{i_1, i_2\}$, and solve

$$\begin{aligned} \max_{\mathcal{S}_1\subseteq\mathcal{L}_{i_1}, \mathcal{S}_2\subseteq\mathcal{R}_{i_1}\cap\mathcal{L}_{i_2}} \quad & \frac{(1+\delta)(r_{i_1} u_{i_1} + r_{i_2} u_{i_2}) + \sum_{i\in\mathcal{S}_1} r_i u_i + \sum_{i\in\mathcal{S}_2} r_i u_i}{1 + (1+\delta)(u_{i_1} + u_{i_2}) + \sum_{i\in\mathcal{S}_1} u_i + \sum_{i\in\mathcal{S}_2} u_i} \\ \text{s.t.} \quad & |\mathcal{S}_1| = \tau_1 - 1, \\ & |\mathcal{S}_2| \le \tau_2 - \tau_1 - 1. \end{aligned} \tag{EC.16}$$

For $\mathcal{S}_1$, we restrict it to be the subset of size $\tau_1 - 1$ belonging to $\mathcal{L}_{i_1}$ to ensure that $\sigma(i,\mathcal{S}^*) < \sigma(i_1,\mathcal{S}^*)$ for all $i \in \mathcal{S}_1$ and $\sigma(i_1,\mathcal{S}^*) = \tau_1$. Note that we require that $i_1 \in \{i : \sigma(i,\mathcal{N}) \geq \tau_1\}$, and $i_2 \in \{i : \sigma(i,\mathcal{N}) > \sigma(i_1,\mathcal{N})\}$. Similar to Lemma 1, problem (EC.16) can be solved via a linear program with $O(n)$ variables and constraints. Denote by $(\mathcal{S}_1^*(i_1,i_2), \mathcal{S}_2^*(i_1,i_2))$ the optimal solution of the above problem, then we include set $\mathcal{S}_1^*(i_1,i_2) \cup \mathcal{S}_2^*(i_1,i_2) \cup \{i_1,i_2\}$ into $\mathcal{C}_2$ for each focal set $\{i_1,i_2\}$. Furthermore, if $|\mathcal{S}^*| > \tau_2$, we decompose the decision variable $\mathcal{S}$ into three non-overlapping subsets $\mathcal{S}_1$, $\mathcal{S}_2$ and $\mathcal{S}_3$, and solve the following problem via a linear program for each focal set $\{i_1,i_2\}$,

$$\begin{aligned}
\max_{\mathcal{S}_1 \subseteq \mathcal{L}_{i_1}, \mathcal{S}_2 \subseteq \mathcal{R}_{i_1} \cap \mathcal{L}_{i_2}, \mathcal{S}_3 \subseteq \mathcal{R}_{i_2}} \quad & \frac{(1+\delta)\,(r_{i_1}u_{i_1} + r_{i_2}u_{i_2}) + \sum_{i\in\mathcal{S}_1} r_i u_i + \sum_{i\in\mathcal{S}_2} r_i u_i + \sum_{i\in\mathcal{S}_3} r_i u_i}{1 + (1+\delta)\,(u_{i_1} + u_{i_2}) + \sum_{i\in\mathcal{S}_1} u_i + \sum_{i\in\mathcal{S}_2} u_i + \sum_{i\in\mathcal{S}_3} u_i} \\
\text{s.t.} \quad & |\mathcal{S}_1| = \tau_1 - 1, \\
& |\mathcal{S}_2| = \tau_2 - \tau_1 - 1.
\end{aligned} \tag{EC.17}$$

Denote by $(\mathcal{S}_1^*(i_1,i_2), \mathcal{S}_2^*(i_1,i_2), \mathcal{S}_3^*(i_1,i_2))$ the optimal solution of the above problem, then we include set $\mathcal{S}_1^*(i_1,i_2) \cup \mathcal{S}_2^*(i_1,i_2) \cup \mathcal{S}_3^*(i_1,i_2) \cup \{i_1,i_2\}$ into $\mathcal{C}_3$ for each focal set $\{i_1,i_2\}$. Then, it must hold that $\mathcal{S}^* \in \mathcal{C}_1 \cup \mathcal{C}_2 \cup \mathcal{C}_3$, and we need to solve at most $O(n^2)$ linear programs with $O(n)$ variables and constraints.

EC.4.2 Fixed Focal Items Model

EC.4.2.1 Another Tractable Form of Distortion Function

In this subsection, we provide another direction to study the transition between the tractability and NP-hardness of assortment optimization under the FLM. Specifically, we sequentially modify the distortion function in the following way:

$$\delta(S) = g\left(\max_{\ell\in S} \rho_\ell\right) \to \delta(S) = g\left(\sum_{\ell=1}^{\min\{L,|S|\}} \rho_{[\ell]}\right) \to \delta(S) = g\left(\sum_{\ell\in S} \rho_\ell\right),$$

where ρ_ℓ is the distortion parameter of item ℓ, $\rho_{[\ell]}$ is the ℓ-th largest distortion parameter in assortment $\mathcal{S}$, and L is a predetermined positive integer. We still consider the focal function $F(\mathcal{S}) = \mathcal{S}_+ \cap \mathcal{N}^*$ for all $\mathcal{S} \subseteq \mathcal{N}$. Without loss of generality, we assume that $\{\rho_\ell\}_{\ell\in\mathcal{N}}$ are different.

We first consider the assortment optimization problem under $\delta(S) = g(\max_{\ell\in S} \rho_\ell)$. To solve this problem, we can enumerate all possible choices of $\max_{\ell\in S} \rho_\ell$, in a total number of n, and solve the corresponding subproblem. For the ℓ-th subproblem, we assume that item ℓ has the largest distortion parameter. For any $\ell \in \mathcal{N}$, we define the set $\mathcal{N}_\ell := \{k \in \mathcal{N} : \rho_k < \rho_\ell\}$ with a slight abuse of notation. Then we need to solve

$$\max_{\mathcal{S}\subseteq\mathcal{N}_\ell} \frac{(1+g(\rho_\ell)\mathbf{1}\{\ell\in\mathcal{N}^*\})r_\ell u_\ell + (1+g(\rho_\ell))\sum_{i\in S\cap\mathcal{N}^*} r_i u_i + \sum_{j\in S\setminus\mathcal{N}^*} r_j u_j}{1 + g(\rho_\ell)\mathbf{1}\{0\in\mathcal{N}^*\} + (1+g(\rho_\ell)\mathbf{1}\{\ell\in\mathcal{N}^*\})u_\ell + (1+g(\rho_\ell))\sum_{i\in S\cap\mathcal{N}^*} u_i + \sum_{j\in S\setminus\mathcal{N}^*} u_j}$$

In this case, the problem can be mapped to an MNL model with modified attraction values, i.e.,

$$\begin{cases} u'_k = (1+g(\rho_\ell)\mathbf{1}\{\ell \in \mathcal{N}^*\})u_\ell, \\ u'_0 = 1+g(\rho_\ell)\mathbf{1}\{0 \in \mathcal{N}^*\}, \\ u'_i = (1+g(\rho_\ell))u_i \ \forall i \in \mathcal{N}_\ell \cap \mathcal{N}^*, \\ u'_j = u_j \ \forall j \in \mathcal{N}_\ell \setminus \mathcal{N}^*. \end{cases}$$

Then according to Lemma EC.1, the above subproblem can be solved in the time complexity of $O(n)$, implying the total time complexity is $O(n^2)$.

Next, we consider the assortment optimization problem under $\delta(S) = g\left(\sum_{\ell=1}^{\min\{L,|S|\}} \rho_{[\ell]}\right)$. We discuss whether $|S^\star| \le L$. In the first case, we consider all the assortments that have cardinality no more than L. For each such assortment S, the corresponding expected profit is given by

$$\frac{\left(1+g\left(\sum_{\ell\in S}\rho_\ell\right)\right)\sum_{i\in S\cap\mathcal{N}^*} r_i u_i + \sum_{j\in S\setminus\mathcal{N}^*} r_j u_j}{1+g\left(\sum_{\ell\in S}\rho_\ell\right)\mathbf{1}\{0\in\mathcal{N}^*\} + \left(1+g\left(\sum_{\ell\in S}\rho_\ell\right)\right)\sum_{i\in S\cap\mathcal{N}^*} u_i + \sum_{j\in S\setminus\mathcal{N}^*} u_j}.$$

The corresponding time complexity is of the order $O(n^L)$. In the second case, we consider all the assortments that have cardinality larger than L. Now we fix a set $\{i_1, i_2, \ldots, i_L\}$ such that $\rho_{i_1} > \rho_{i_2} > \cdots > \rho_{i_L}$, and assume that they are the items with top-L distortion parameters in S. To maximize the profit under this case, similarly, we let $\mathcal{N}_{i_L} := \{\ell \in \mathcal{N} : \rho_\ell < \rho_{i_L}\}$ and solve

$$\max_{S\subseteq\mathcal{N}_{i_L}} \frac{\left(1+g\left(\sum_{\ell=1}^{L}\rho_{i_\ell}\right)\right)\sum_{i\in S\cap\mathcal{N}^*} r_i u_i + \sum_{j\in S\setminus\mathcal{N}^*} r_j u_j}{1+g\left(\sum_{\ell=1}^{L}\rho_{i_\ell}\right)\mathbf{1}\{0\in\mathcal{N}^*\} + \left(1+g\left(\sum_{\ell=1}^{L}\rho_{i_\ell}\right)\right)\sum_{i\in S\cap\mathcal{N}^*} u_i + \sum_{j\in S\setminus\mathcal{N}^*} u_j}$$

which again can be solved in a time complexity of $O(n)$. Then the time complexity under the second case is $O(n^{L+1})$, implying that the total time complexity is $O(n^{L+1})$, which is fixed-parameter tractable.

Finally, if $\delta(S) = g\left(\sum_{i\in S}\rho_i\right)$, we have proven that the assortment optimization is NP-hard in Theorem 3. Therefore, the parameter L shows how the number of items that the distortion function depends on influences the problem's complexity.

EC.4.2.2 Performance of the Optimal Profit-ordered Assortment

PROPOSITION EC.1. *When $F(\mathcal{S}) = \mathcal{S}_+ \cap \{i_F\}$ for all $\mathcal{S} \subseteq \mathcal{N}$, the following statements hold:*

(1) If $i_F = 0$, then the optimal profit-ordered assortment achieves at least

$$\frac{1+\delta(\mathcal{S}^*)+\sum_{j\in\mathcal{S}_\ell} u_j}{1+\delta(\mathcal{S}_\ell)+\sum_{j\in\mathcal{S}_\ell} u_j}$$

of the optimal profit, where $\mathcal{S}_\ell = \{1, 2, \ldots, \ell\}$ and ℓ is the largest index of items in $\mathcal{S}^*$.

(2) If $i_F \neq 0$, then the optimal profit-ordered assortment achieves at least

$$\frac{\sum_{j\in\mathcal{S}_{i_F}\setminus\{i_F\}} u_j + (1+\delta(\mathcal{S}_{i_F}))u_{i_F}}{1+\sum_{j\in\mathcal{S}_{i_F}\setminus\{i_F\}} u_j + (1+\delta(\mathcal{S}_{i_F}))u_{i_F}}$$

of the optimal profit, where $\mathcal{S}_{i_F} = \{1, 2, \ldots, i_F\}$.

Proof of Proposition EC.1.

We begin with the first statement. Suppose that $\mathcal{S}^*$ is not a profit-ordered assortment. We label the items such that $r_1 \geq r_2 \geq \cdots \geq r_n$ and denote by ℓ the largest label among all items in $\mathcal{S}^*$. Thus, item ℓ has the smallest profit among $\mathcal{S}^*$. We first show that $r_\ell \geq f(\mathcal{S}^*)$. To see this, we compute

$$\begin{aligned}
f(\mathcal{S}^*) &= \frac{\sum_{j\in\mathcal{S}^*} r_j u_j}{1+\delta(\mathcal{S}^*)+\sum_{j\in\mathcal{S}^*} u_j} \\
&= \frac{1+\delta(\mathcal{S}^*\setminus\{\ell\})+\sum_{j\in\mathcal{S}^*\setminus\{\ell\}} u_j}{1+\delta(\mathcal{S}^*)+\sum_{j\in\mathcal{S}^*} u_j}\cdot\frac{\sum_{j\in\mathcal{S}^*\setminus\{\ell\}} r_j u_j}{1+\delta(\mathcal{S}^*\setminus\{\ell\})+\sum_{j\in\mathcal{S}^*\setminus\{\ell\}} u_j} \\
&\quad + r_\ell\cdot\frac{u_\ell}{1+\delta(\mathcal{S}^*)+\sum_{j\in\mathcal{S}^*} u_j} \\
&= \frac{1+\delta(\mathcal{S}^*\setminus\{\ell\})+\sum_{j\in\mathcal{S}^*\setminus\{\ell\}} u_j}{1+\delta(\mathcal{S}^*)+\sum_{j\in\mathcal{S}^*} u_j} f(\mathcal{S}^*\setminus\{\ell\}) + \frac{u_\ell}{1+\delta(\mathcal{S}^*)+\sum_{j\in\mathcal{S}^*} u_j} r_\ell.
\end{aligned}$$

Note that $f(\mathcal{S}^*\setminus\{\ell\}) \leq f(\mathcal{S}^*)$, and

$$\frac{1+\delta(\mathcal{S}^*\setminus\{\ell\})+\sum_{j\in\mathcal{S}^*\setminus\{\ell\}} u_j + u_\ell}{1+\delta(\mathcal{S}^*)+\sum_{j\in\mathcal{S}^*} u_j} \leq 1,$$

due to the fact that $\delta(\mathcal{S})$ is an increasing function over $\mathcal{S}$ defined in Section 4.2. Thus, we must have $r_\ell \geq f(\mathcal{S}^*)$. Note that $\mathcal{S}^* \subseteq \mathcal{S}_\ell = \{1,2,\ldots,\ell\}$, and this further implies that

$$\begin{aligned}
f(\mathcal{S}_\ell) &= \frac{\sum_{j\in\mathcal{S}_\ell} r_j u_j}{1+\delta(\mathcal{S}_\ell)+\sum_{j\in\mathcal{S}_\ell} u_j} \\
&\geq f(\mathcal{S}^*)\cdot\frac{1+\delta(\mathcal{S}^*)+\sum_{j\in\mathcal{S}^*} u_j}{1+\delta(\mathcal{S}_\ell)+\sum_{j\in\mathcal{S}_\ell} u_j} + r_\ell\cdot\frac{\sum_{j\in\mathcal{S}_\ell\setminus\mathcal{S}^*} u_j}{1+\delta(\mathcal{S}_\ell)+\sum_{j\in\mathcal{S}_\ell} u_j} \\
&\geq f(\mathcal{S}^*)\cdot\frac{1+\delta(\mathcal{S}^*)+\sum_{j\in\mathcal{S}_\ell} u_j}{1+\delta(\mathcal{S}_\ell)+\sum_{j\in\mathcal{S}_\ell} u_j},
\end{aligned}$$

where the first inequality holds since $r_j \geq r_\ell$ for all $j \in \mathcal{S}_\ell \setminus \mathcal{S}^*$.

Now we prove the second statement. Recall that we only focus on the case that $i_{\mathrm{F}} \in \mathcal{S}^*$. Hence, we have $r_{i_{\mathrm{F}}} \geq f(\mathcal{S}^*)$. Then, we consider the profit-ordered assortment $\mathcal{S}_{i_{\mathrm{F}}} = \{1,2,\ldots,i_{\mathrm{F}}\}$, and obtain

$$\begin{aligned}
f(\mathcal{S}_{i_{\mathrm{F}}}) = \frac{\sum_{j\in\mathcal{S}_{i_{\mathrm{F}}}\setminus\{i_{\mathrm{F}}\}} r_j u_j + (1+\delta(\mathcal{S}_{i_{\mathrm{F}}}))u_{i_{\mathrm{F}}} r_{i_{\mathrm{F}}}}{1+\sum_{j\in\mathcal{S}_{i_{\mathrm{F}}}\setminus\{i_{\mathrm{F}}\}} u_j + (1+\delta(\mathcal{S}_{i_{\mathrm{F}}}))u_{i_{\mathrm{F}}}} &\geq \frac{\sum_{j\in\mathcal{S}_{i_{\mathrm{F}}}\setminus\{i_{\mathrm{F}}\}} u_j + (1+\delta(\mathcal{S}_{i_{\mathrm{F}}}))u_{i_{\mathrm{F}}}}{1+\sum_{j\in\mathcal{S}_{i_{\mathrm{F}}}\setminus\{i_{\mathrm{F}}\}} u_j + (1+\delta(\mathcal{S}_{i_{\mathrm{F}}}))u_{i_{\mathrm{F}}}} r_{i_{\mathrm{F}}} \\
&\geq \frac{\sum_{j\in\mathcal{S}_{i_{\mathrm{F}}}\setminus\{i_{\mathrm{F}}\}} u_j + (1+\delta(\mathcal{S}_{i_{\mathrm{F}}}))u_{i_{\mathrm{F}}}}{1+\sum_{j\in\mathcal{S}_{i_{\mathrm{F}}}\setminus\{i_{\mathrm{F}}\}} u_j + (1+\delta(\mathcal{S}_{i_{\mathrm{F}}}))u_{i_{\mathrm{F}}}} f(\mathcal{S}^*).
\end{aligned}$$

The first inequality holds by $r_1 \geq r_2 \geq \cdots \geq r_{i_{\mathrm{F}}}$. Then the proof is complete. □

Proposition EC.1 shows that the optimal profit-ordered assortment achieves an instance-dependent approximation ratio of the optimal expected profit. It also provides a generalized result of Berbeglia and Joret (2020) when the regularity condition does not hold. Proposition EC.1-(1) gives the approximation ratio under the choice overload model. When the distortion function satisfies $\delta(\mathcal{S}_\ell) \le 2\delta(\mathcal{S}^*)$, the approximation ratio must be more than $1/2$. Claim (2) provides the approximation ratio when $i_{\mathrm{F}} \neq 0$. As long as the attraction value of any item in $\mathcal{S}_{i_{\mathrm{F}}} \setminus \{i_{\mathrm{F}}\}$ or the amplified attraction value of i_{F} is larger than the one of the no-purchase option, the approximation ratio of $\mathcal{S}_{i_{\mathrm{F}}}$ is always greater than $1/2$.

EC.5 Algorithms and Analysis for Joint Assortment and Pricing Optimization

In this section, we discuss the algorithms for solving the joint assortment and pricing optimization problem under the FLM in detail.

EC.5.1 Top-K Focal Model

We start from the algorithm under the top-K focal model. Since we already discuss the sketch of the algorithm, we now only provide the details in Algorithm 2. The upper bound of optimal value Q in Algorithm 2 can be set as $\max_{i\in\mathcal{N}}\max_{p_i\ge c_i}(1+\delta)(p_i-c_i)\exp(\alpha_i-\beta_i p_i)$.

EC.5.2 Arbitrarily-Located Focal Model

Now we consider the joint assortment and optimization problem under the arbitrarily-located focal model. To demonstrate, we only discuss the case that $h(|\mathcal{S}|)=\tau$ for all $\mathcal{S}\subseteq\mathcal{N}$. The corresponding algorithm can be easily extended to the general formulation of $h(|\mathcal{S}|)$.

We still adopt the attraction value ranking, that is, the ranking σ based on $\{\exp(\alpha_i-\beta_i p_i)\}_{i=1}^n$. Hence, the focal item of a given assortment $\mathcal{S}$ is,

$$F(\mathcal{S})=\{i:\sigma(i,\mathcal{S})=\min\{|\mathcal{S}|,\tau\}, i\in\mathcal{S}\}.$$

In the following lemma, we characterize the property of the optimal assortment $\mathcal{S}^*$.

LEMMA EC.2. *For the joint assortment and pricing optimization under the arbitrarily-located focal model, the optimal assortment $\mathcal{S}^*$ satisfies either $\mathcal{S}^*=\mathcal{N}$ or $|\mathcal{S}^*|<\tau$.*

The proof is similar to the one of Lemma 2, and we omit it here. With Lemma EC.2, we are able to solve the joint optimization problem by considering two different cases based on $\mathcal{S}^*$. The first case is $|\mathcal{S}^*|<\tau$, and the other case is $\mathcal{S}^*=\mathcal{N}$. However, the above two cases can be unified into a pricing-only problem with different whole item sets. The first case corresponds to the pricing problem with the whole item set $\mathcal{N}_\tau\subset\mathcal{N}$ for each $\mathcal{N}_\tau$ satisfying that $|\mathcal{N}_\tau|\le\tau-1$, and the other case is related to the pricing problem with the whole item set $\mathcal{N}$. Therefore, the techniques used in Section 5 can be extended here. We omit the detailed algorithm since it is almost the same as Algorithm 2.

Algorithm 2: Algorithm for Joint Assortment and Pricing Optimization under the Top-K Focal Model with the Constant Distortion Function

Input: $\{\alpha_i\}_{i=1}^n$, $\{\beta_i\}_{i=1}^n$, $\{c_i\}_{i=1}^n$, the focal function $F(\mathcal{S})$ specified in (6), the constant distortion function $\delta(\mathcal{S})=\delta$, the tolerance level $\epsilon>0$, and the upper bound of the optimal value Q

Initialize: $\mathcal{SOL}\leftarrow\emptyset$, $OPT\leftarrow 0$

for $k=1,2,\ldots,n$ **do**
  $\underline{r}\leftarrow L$, $\overline{r}\leftarrow Q$
  for $\mathcal{S}_1\in\{\mathcal{S}:|\mathcal{S}|=K-1,k\notin\mathcal{S}\}$ **do**
    $\mathcal{F}\leftarrow\mathcal{S}_1\cup\{k\}$
    while *true* **do**
      $L\leftarrow 0$
      $r=(L+Q)/2$
      Find the intersection points $\{\hat{p}_{k,m}\}_{m\in[M]}$ of $\left\{\frac{\beta_k}{\beta_\ell}p_k+\frac{\alpha_\ell-\alpha_k-1}{\beta_\ell}-c_\ell-r\right\}_{\ell\in\mathcal{N}\setminus\{k\}}$
      Solve (EC.11) for each interval $[\hat{p}_{k,m},\hat{p}_{k,m+1})$
      Find p_k^* by (EC.12), and obtain $\mathbf{p}_{-k}^*$ and $H(p_k^*,r)$
      if $|H(p_k^*,r)-r|\le\epsilon$ **then**
        break
      else if $H(p_k^*,r)>r$ **then**
        $L\leftarrow r$
      else
        $Q\leftarrow r$
    if $R(\mathcal{N},[\mathbf{p}_{-k}^*;p_k^*])\ge OPT$ **then**
      $OPT\leftarrow R(\mathcal{N},[\mathbf{p}_{-k}^*;p_k^*])$
      $\mathcal{SOL}\leftarrow\left(\mathcal{N},[\mathbf{p}_{-k}^*;p_k]\right)$

Output: The optimal solution $(\mathcal{S}^*,\mathbf{p}^*)\in\arg\max_{\mathcal{S}\in\mathcal{SOL}}R(\mathcal{S},\mathbf{p})$ with optimal value $R(\mathcal{S}^*,\mathbf{p}^*)$.

EC.5.3 Fixed Focal Items Model

In this subsection, we give the algorithm for joint assortment and pricing optimization problem under the fixed focal items model. Recall that Lemma 5 shows that for any given assortment $\mathcal{S}$, the optimal price for each item $i\in\mathcal{S}$ is the summation of the cost c_i and the term $r^{\mathcal{S}}+1/\beta_i$. With this result, we can obtain the following one-dimensional fixed point characterization of the optimal value R^* to problem (16).

LEMMA EC.3. *Let R^* denote the optimal value to (16) when the focal function and the distortion function are specified as above. It satisfies the following equation,*

$$R^*=H\left(R^*\right), \tag{EC.18}$$

where

$$H(r) = \max_{\mathcal{S}_1 \subseteq \mathcal{N}^* \backslash \{0\}, \mathcal{S}_2 \subseteq \mathcal{N} \backslash \mathcal{N}^*} J(r, \mathcal{S}_1, \mathcal{S}_2). \tag{EC.19}$$

Moreover, $H(r)$ is continuous and decreasing in r.

Proof. With (17) in Lemma 5, problem (16) is equivalent to the problem of finding the assortments $\mathcal{S}_1$ and $\mathcal{S}_2$ such that the associated solution of (17) is maximized, i.e.,

$$\begin{aligned} \max_{r, \mathcal{S}_1 \subseteq \mathcal{N}^* \backslash \{0\}, \mathcal{S}_2 \subseteq \mathcal{N} \backslash \mathcal{N}^*} \quad & r \\ \text{s.t.} \quad & r = J\left(r, \mathcal{S}_1, \mathcal{S}_2\right). \end{aligned} \tag{EC.20}$$

Clearly, R^* is also the optimal value to the above problem. We let $(R^*, \mathcal{S}_1^*, \mathcal{S}_2^*)$ be its optimal solution. Then we have

$$R^* = J(R^*, \mathcal{S}_1^*, \mathcal{S}_2^*) = \max_{\mathcal{S}_1 \subseteq \mathcal{N}^* \backslash \{0\}, \mathcal{S}_2 \subseteq \mathcal{N} \backslash \mathcal{N}^*} J(R^*, \mathcal{S}_1, \mathcal{S}_2) = H(R^*). \tag{EC.21}$$

The first equality comes from the constraint in (EC.20). To see why the second equality holds, suppose to the contrary that there exists an assortment pair $(\mathcal{S}_1', \mathcal{S}_2') \neq (\mathcal{S}_1^*, \mathcal{S}_2^*)$ such that $J(R^*, \mathcal{S}_1', \mathcal{S}_2') > J(R^*, \mathcal{S}_1^*, \mathcal{S}_2^*) = R^*$. Let r' be the associated fixed point for $(\mathcal{S}_1', \mathcal{S}_2')$ satisfying $J(r', \mathcal{S}_1', \mathcal{S}_2') = r'$. One can check that the function $J(r, \mathcal{S}_1', \mathcal{S}_2') - r$ is strictly decreasing in r. Together with the fact that $J(R^*, \mathcal{S}_1', \mathcal{S}_2') - r' > 0$ and $J(r', \mathcal{S}_1', \mathcal{S}_2') - r' = 0$, we must have $R^* < r'$, which violates that R^* is the optimal value to (EC.20). Hence, we have $J(R^*, \mathcal{S}_1^*, \mathcal{S}_2^*) = \max_{\mathcal{S}_1 \subseteq \mathcal{N}^* \backslash \{0\}, \mathcal{S}_2 \subseteq \mathcal{N} \backslash \mathcal{N}^*} J(R^*, \mathcal{S}_1, \mathcal{S}_2)$.

In the following, we show that function $H(r)$ in (EC.19) is decreasing in r. Suppose that $r_2 \geq r_1 \geq 0$. There exist assortment pairs $\left(\mathcal{S}_{1,1}^*, \mathcal{S}_{1,2}^*\right)$ and $\left(\mathcal{S}_{2,1}^*, \mathcal{S}_{2,2}^*\right)$ such that

$$\begin{aligned} \left(\mathcal{S}_{1,1}^*, \mathcal{S}_{1,2}^*\right) &\in \arg \max_{\mathcal{S}_1 \subseteq \mathcal{N}^* \backslash \{0\}, \mathcal{S}_2 \subseteq \mathcal{N} \backslash \mathcal{N}^*} J(r_1, \mathcal{S}_1, \mathcal{S}_2), \\ \text{and } \left(\mathcal{S}_{2,1}^*, \mathcal{S}_{2,2}^*\right) &\in \arg \max_{\mathcal{S}_1 \subseteq \mathcal{N}^* \backslash \{0\}, \mathcal{S}_2 \subseteq \mathcal{N} \backslash \mathcal{N}^*} J(r_2, \mathcal{S}_1, \mathcal{S}_2). \end{aligned}$$

Then, we have

$$H(r_2) = J(r_2, \mathcal{S}_{2,1}^*, \mathcal{S}_{2,2}^*) \leq J(r_1, \mathcal{S}_{2,1}^*, \mathcal{S}_{2,2}^*) \leq J(r_1, \mathcal{S}_{1,1}^*, \mathcal{S}_{1,2}^*) = H(r_1).$$

The first inequality holds because $J(r, \mathcal{S}_1, \mathcal{S}_2)$ is decreasing in r and the second inequality holds by the definition of $H(r)$. Therefore, $H(r)$ is decreasing in r. Note that $H(r) = \max_{\mathcal{S}_1 \subseteq \mathcal{N}^* \backslash \{0\}, \mathcal{S}_2 \subseteq \mathcal{N} \backslash \mathcal{N}^*} J(r, \mathcal{S}_1, \mathcal{S}_2)$, hence it is continuous in r. Hence, we conclude the proof of Lemma EC.3. $\square$

With Lemma EC.3, we can design an algorithm based on equation (EC.18). We begin with the optimization problem that defines $H(r)$. The problem may be complicated since there are two variables $\mathcal{S}_1$ and $\mathcal{S}_2$. However, we show in the next two lemmas that for a given $r > 0$, the value of $H(r)$ can be computed efficiently. We denote the optimal solution of the problem in (EC.19) by $(\mathcal{S}_1(r), \mathcal{S}_2(r))$.

LEMMA EC.4. *For a given $r>0$, we label the items in the set $\mathcal{N}\setminus\mathcal{N}^*$ such that $\gamma_1-\beta_1 r\geq\gamma_2-\beta_2 r\geq\cdots\geq\gamma_{n'}-\beta_{n'}r$, where $n'=|\mathcal{N}\setminus\mathcal{N}^*|$. Given $\mathcal{S}_1$, the optimal $\mathcal{S}_2(r)$ to the problem in (EC.19) is $\{1,2,\ldots,\ell\}$ for some $\ell\in\{1,2,\ldots,n'\}$. Conversely, given an ordering of $\{\gamma_i-\beta_i r\}_{i\in\mathcal{N}\setminus\mathcal{N}^*}$, there exist at most $n'\cdot\left(2^{|\mathcal{N}^*|-1}\right)$ possible triples $(\mathcal{S}_1(r),\mathcal{S}_2(r),r)$ such that $(\mathcal{S}_1(r),\mathcal{S}_2(r))=\arg\max_{\mathcal{S}_1\subseteq\mathcal{N}^*\setminus\{0\},\mathcal{S}_2\subseteq\mathcal{N}\setminus\mathcal{N}^*}J(r,S_1,S_2)$, and r induces the given ordering.*

Proof. We prove the first claim by contradiction. Suppose that $\mathcal{S}_2(r)\neq\{1,2,\ldots,\ell\}$, then there must exist item $j\notin\mathcal{S}_2(r)$ and item $i\in\mathcal{S}_2(r)$ such that $\gamma_i-\beta_i r\leq\gamma_j-\beta_j r$. By swapping item i and item j, we obtain a new assortment $\mathcal{S}_2'(r)=\mathcal{S}_2(r)\cup\{j\}\setminus\{i\}$. If $\gamma_i-\beta_i r<\gamma_j-\beta_j r$, we then have

$$J(r,\mathcal{S}_1,\mathcal{S}_2'(r))>J(r,\mathcal{S}_1,\mathcal{S}_2(r)),$$

which contradicts the optimality of $\mathcal{S}_2(r)$. If $\gamma_i-\beta_i r=\gamma_j-\beta_j r$, then $J(r,\mathcal{S}_1,\mathcal{S}_2'(r))=J(r,\mathcal{S}_1,\mathcal{S}_2(r))$, implying that $\mathcal{S}_2'(r)$ is also optimal. Hence, one can swap item i and item j to obtain the new assortment of form $\{1,2,\ldots,\ell\}$ for some $\ell\in[n']$. Moreover, if there is more than one item pair like (i,j), then one can swap such item pairs finite times to obtain a new assortment of form $\{1,2,\ldots,\ell\}$ whose expected profit is no less than the previous one. Hence, the optimal assortment $\mathcal{S}_2'=\{1,2,\ldots,\ell\}$ for some $\ell\in[n']$.

Now we prove the second claim. Given an ordering of $\{\gamma_i-\beta_i r\}_{i\in\mathcal{N}\setminus\mathcal{N}^*}$, then for any r such that the ordering holds, to solve $\max_{\mathcal{S}_1\subseteq\mathcal{N}^*\setminus\{0\},\mathcal{S}_2\subseteq\mathcal{N}\setminus\mathcal{N}^*}J(r,\mathcal{S}_1,\mathcal{S}_2)$, there are only n' possible candidates of $\mathcal{S}_2(r)$, and $2^{|\mathcal{N}^*|-1}$ possible candidates of $\mathcal{S}_1(r)$. Then the number of assortment pair $(\mathcal{S}_1,\mathcal{S}_2)$ is $n'\cdot(2^{|\mathcal{N}^*|-1})$, and the proof is complete. □

The lemma indicates that the optimal $\mathcal{S}_2(r)$ to the problem in (EC.19) has an ordered-structure in terms of the value $\{\gamma_i-\beta_i r\}_{i\in\mathcal{N}\setminus\mathcal{N}^*}$. This significantly simplifies the problem, and we only need to enumerate all possible pairs of $(\mathcal{S}_1(r),\mathcal{S}_2(r))$, in a total of $n'\cdot\left(2^{|\mathcal{N}^*|-1}\right)$, to compute $H(r)$ for a given r. The details are presented in Algorithm 3, whose polynomial time complexity is given in Lemma EC.5. Furthermore, once the ordering of $\{\gamma_i-\beta_i r\}_{i\in\mathcal{N}\setminus\mathcal{N}^*}$ with respect to R^* is known, the above lemma shows that there are $n'\cdot\left(2^{|\mathcal{N}^*|-1}\right)$ possible choices of $(\mathcal{S}_1(r),\mathcal{S}_2(r))$ to the problem $\max_{\mathcal{S}_1\subseteq\mathcal{N}^*\setminus\{0\},\mathcal{S}_2\subseteq\mathcal{N}\setminus\mathcal{N}^*}\ J(r,\mathcal{S}_1,\mathcal{S}_2)$ for all r including R^* that satisfies the given ordering. In other words, we obtain all the possible choices of the optimal assortment to (16). Then, by Lemma 5, we are able to compute the optimal profit R^*.

LEMMA EC.5. *Under the fixed focal items model, for given $r>0$, Algorithm 3 returns the value of $H(r)$. The time complexity of Algorithm 3 is $O\left(n\left(\log n+\cdot 2^{|\mathcal{N}^*|-1}\right)\right)$.*

Proof. With Lemma EC.4, given $\mathcal{S}_1$ and r, one can maximize over $\mathcal{S}_2$ by enumerate all its n' possible candidates. To maximize over $\mathcal{S}_1$, one can enumerate all its $2^{|\mathcal{N}^*|-1}$ possible choices of $\mathcal{S}_1$. Then the assortment pair $(\mathcal{S}_1,\mathcal{S}_2)$ that brings the largest value of $J(r,\mathcal{S}_1,\mathcal{S}_2)$ is the optimal solution of the

Algorithm 3: Algorithm for Computing $H(r)$ for a Given $r>0$

Input: $r>0$, $\{\gamma_i\}_{i=1}^n$, $\{\beta_i\}_{i=1}^n$ and $\mathcal{N}^*$

$H(r)\leftarrow 0$

Label the items such that $\gamma_1-\beta_1 r\geq\gamma_2-\beta_2 r\geq\cdots\geq\gamma_{n'}-\beta_{n'}r$

Construct the power set of $\mathcal{N}^*$

for $\mathcal{S}_1\in 2^{\mathcal{N}^*}\setminus\emptyset$ **do**
  for $\ell=1,2,\ldots,n'$ **do**
    $\mathcal{S}_2\leftarrow\{1,2,\ldots,\ell\}$
    Compute $J(r,\mathcal{S}_1,\mathcal{S}_2)$
    if $J(r,\mathcal{S}_1,\mathcal{S}_2)\geq H(r)$ **then**
      $H(r)\leftarrow J(r,\mathcal{S}_1,\mathcal{S}_2)$

Output: $H(r)$

problem $\max_{\mathcal{S}_1\subseteq\mathcal{N}^*\setminus\{0\},\mathcal{S}_2\subseteq\mathcal{N}\setminus\mathcal{N}^*} J(r,\mathcal{S}_1,\mathcal{S}_2)$, and its optimal value is $H(r)$. Hence, Algorithm 3 can compute $H(r)$.

The time complexity consists of the ordering complexity $O(n\log n)$ and the complexity brought by the optimization over $\mathcal{S}_1$ and $\mathcal{S}_2$, i.e., enumeration of $(\mathcal{S}_1,\mathcal{S}_2)$. This complexity is $O(n'\cdot 2^{|\mathcal{N}^*|-1})$. Hence, the total time complexity is within $O\left(n\left(\log n+2^{|\mathcal{N}^*|-1}\right)\right)$. □

With the above preparations, we now discuss how to find R^* that satisfies the fixed point equation (EC.18). This can be completed by searching R^* in the region $\{r:r>0\}$. To this end, we divide the region into multiple intervals, i.e., $\{r:r>0\}=(\hat r_0,\hat r_1]\cup(\hat r_1,\hat r_2]\cup\ldots\cup(\hat r_K,\hat r_{K+1})$, where $\hat r_0=0$, $\hat r_{K+1}=\infty$, $\{\hat r_k\}_{k=1}^K$ are the intersection points among functions $\{\gamma_i-\beta_i r\}_{i\in\mathcal{N}\setminus\mathcal{N}^*}$. The merit of such partition is that the ordering of $\{\gamma_i-\beta_i r\}_{i\in\mathcal{N}\setminus\mathcal{N}^*}$ is unchanged within each interval, and the enumeration of all possible assortment in each interval is at most $n'\cdot\left(2^{|\mathcal{N}^*|-1}\right)$ due to Lemma EC.4. Moreover, the monotone decreasing property of $H(r)$ implies that

$$r>H(r),\forall r>R^* \quad\text{and}\quad r<H(r),\forall r<R^*. \tag{EC.22}$$

It suffices to check the endpoint $\hat r_k$ of each interval. If we find the index k^* such that $H(\hat r_{k^*})>\hat r_{k^*}$ and $H(\hat r_{k^*+1})\leq\hat r_{k^*+1}$, then it must be the case that $R^*\in(\hat r_{k^*},\hat r_{k^*+1}]$ by the property (EC.22). This can be done by a bisection method due to the monotone property of $H(r)$. We present the detailed search procedure in Algorithm 4, whose polynomial time complexity is given in Theorem EC.1.

THEOREM EC.1. *Algorithm 4 returns the optimal solution of (16) in time* $O\left(n\left(\log^2 n+2^{|\mathcal{N}^*|-1}\right)\right)$.

Proof of Theorem EC.1.

For a given r, by Lemma EC.5, we can compute $H(r)$ in $O\left(n\left(\log n+2^{|\mathcal{N}^*|-1}\right)\right)$. Since we run the bisection method to locate the index k^*, the complexity to locate k^* is $O\left(\log K\cdot n\left(\log n+2^{|\mathcal{N}^*|-1}\right)\right)$.

Algorithm 4: Algorithm for Joint Assortment and Pricing Optimization Problem Under Fixed Focal Items

Input: $\{\alpha_i\}_{i=1}^n$, $\{\beta_i\}_{i=1}^n$, $\{c_i\}_{i=1}^n$, focal function $F(\mathcal{S})$ specified in (11) and distortion function $\delta(\mathcal{S})$ satisfying Assumption 1.

Initialize: $\gamma_i \leftarrow \alpha_i - \beta_i c_i - 1 - \log(\beta_i)$ for all $i \in \mathcal{N}$

Find the intersection points $\{\hat{r}_k\}_{k=1}^K$ of $\{\gamma_i - \beta_i r\}_{i\in\mathcal{N}\setminus\mathcal{N}^*}$

$\hat{r}_0 \leftarrow 0$, $\hat{r}_{K+1} \leftarrow \infty$

$\underline{k} \leftarrow 0$, $\overline{k} \rightarrow K+1$

while *True* **do**

$k = \lfloor(\underline{k}+\overline{k})/2\rfloor$ (Bisection)

if $H(\hat{r}_k) \le \hat{r}_k$ **then**

$\overline{k} \leftarrow k$

else if $H(\hat{r}_{k+1}) > \hat{r}_{k+1}$ **then**

$\underline{k} \leftarrow k+1$

else

break

Find the ordering $\sigma^k = (\sigma_1^k, \sigma_2^k, \dots, \sigma_{n'}^k)$ of $\{\gamma_i - \beta_i r\}_{i\in\mathcal{N}\setminus\mathcal{N}^*}$ in the interval $(\hat{r}_k, \hat{r}_{k+1}]$

Construct the set $\mathcal{Q} = \left\{\mathcal{Q}_\ell : \mathcal{Q}_\ell = \{\sigma_1^k, \sigma_2^k, \dots, \sigma_\ell^k\}, \ell = 1, 2, \dots, n'\right\}$

Let

$$R^* = \max\{r : r \in (\hat{r}_k, \hat{r}_{k+1}], r = J(r, \mathcal{S}_1, \mathcal{S}_2), \mathcal{S}_1 \subseteq \mathcal{N}^* \setminus \{0\}, \mathcal{S}_2 \subseteq \mathcal{Q}\}$$

and

$$(\mathcal{S}_1^*, \mathcal{S}_2^*) = \arg\max\{r : r \in (\hat{r}_k, \hat{r}_{k+1}], r = J(r, \mathcal{S}_1, \mathcal{S}_2), \mathcal{S}_1 \subseteq \mathcal{N}^* \setminus \{0\}, \mathcal{S}_2 \subseteq \mathcal{Q}\}$$

$\mathcal{S}^* \leftarrow \mathcal{S}_1^* \cup \mathcal{S}_2^*$

$p_i^* \leftarrow R^* + c_i + \frac{1}{\beta_i}, \forall i \in \mathcal{S}^*$

Output: The optimal solution $(\mathcal{S}^*, \boldsymbol{p}^*)$ with optimal value R^* to problem (16).

After find the ordering of $\{\gamma_i - \beta_i r\}_{i\in\mathcal{N}\setminus\mathcal{N}^*}$ under R^*, by Lemma EC.4, we need to solve $n' \cdot \left(2^{|\mathcal{N}^*|} - 1\right)$ equations to find R^*. Hence, the total time complexity is within

$$\begin{aligned} &O\left(\log K \cdot n\left(\log n + 2^{|\mathcal{N}^*|-1}\right)\right) + O\left(n' \cdot \left(2^{|\mathcal{N}^*|} - 1\right)\right) \\ =&O\left(n\left(\log^2 n + 2^{|\mathcal{N}^*|-1}\right)\right). \end{aligned}$$

The equality holds due to $K = O(n^2)$. □

EC.6 Several Structural Properties under Choice Overload

EC.6.1 Assortment Optimization

In this subsection, we present the following proposition that generalizes Example 5 and shows that the enlarged optimal assortment property holds for a class of distortion functions.

PROPOSITION EC.2. *Suppose that the focal function $F(\mathcal{S}) = \{0\}, \forall \mathcal{S} \subseteq \mathcal{N}$. Let $\mathcal{S}_1^*$, $\mathcal{S}_2^*$ be the optimal assortment under distortion functions $\delta_1(\mathcal{S})$ and $\delta_2(\mathcal{S})$, respectively, and $f(\mathcal{S}_1^*;\delta_1(\mathcal{S})), f(\mathcal{S}_2^*;\delta_2(\mathcal{S}))$ be the corresponding expected profit. If $\delta_2(\mathcal{S}) \geq \delta_1(\mathcal{S})$ for all $\mathcal{S} \subseteq \mathcal{N}$, then $f(\mathcal{S}_1^*;\delta_1(\mathcal{S})) \geq f(\mathcal{S}_2^*;\delta_2(\mathcal{S}))$.*

Furthermore, if $\delta_1(\mathcal{S})$ and $\delta_2(\mathcal{S})$ satisfy

(1) $\delta_1(\mathcal{S}) = g_1(|\mathcal{S}|)$, $\delta_2(\mathcal{S}) = g_2(|\mathcal{S}|)$, and $g_2(|S|) - g_1(|S|) \leq g_2(|S|-1) - g_1(|S|-1)$ for all $\mathcal{S} \subseteq \mathcal{N}$;

(2) $g_1(|\mathcal{S}|)$ and $g_2(|\mathcal{S}|)$ are increasing functions and satisfy the increasing marginal value property, i.e., $g_i(|\mathcal{S}|+1) - g_i(|\mathcal{S}|) \geq g_i(|\mathcal{S}|) - g_i(|\mathcal{S}|-1)$ for all $\mathcal{S} \subseteq \mathcal{N}$, where $i \in \{1,2\}$.

Then we have $|\mathcal{S}_1^*| \leq |\mathcal{S}_2^*|$.

Proof of Proposition EC.2.

When $F(\mathcal{S}) = \{0\}$ for all $\mathcal{S} \subseteq \mathcal{N}$, we have $f(\mathcal{S};\delta_1(\mathcal{S})) \geq f(\mathcal{S};\delta_2(\mathcal{S}))$ since $\delta_2(\mathcal{S}) \geq \delta_1(\mathcal{S}), \forall \mathcal{S} \subseteq \mathcal{N}$. Thus, we must have $f(\mathcal{S}_1^*;\delta_1(\mathcal{S})) \geq f(\mathcal{S}_2^*;\delta_2(\mathcal{S}))$.

We now prove $|\mathcal{S}_1^*| \leq |\mathcal{S}_2^*|$. Suppose that $|\mathcal{S}_1^*| > |\mathcal{S}_2^*|$. Then there exists item $i \in \mathcal{S}_1^*$ but $i \notin \mathcal{S}_2^*$. By Proposition 4, we have

$$u_i\left(r_i - f(\mathcal{S}_1^*;\delta_1(\mathcal{S}))\right) \geq \left(g_1(|\mathcal{S}_1^*|) - g_1(|\mathcal{S}_1^*|-1)\right) f(\mathcal{S}_1^*;\delta_1(\mathcal{S})), \tag{EC.23}$$

and

$$u_i\left(r_i - f(\mathcal{S}_2^*;\delta_2(\mathcal{S}))\right) \leq \left(g_2(|\mathcal{S}_2^*|+1) - g_2(|\mathcal{S}_2^*|)\right) f(\mathcal{S}_2^*;\delta_2(\mathcal{S})). \tag{EC.24}$$

Note that

$$g_1(|\mathcal{S}_1^*|) - g_1(|\mathcal{S}_1^*|-1) \geq g_2(|\mathcal{S}_1^*|) - g_2(|\mathcal{S}_1^*|-1) \geq g_2(|\mathcal{S}_2^*|+1) - g_2(|\mathcal{S}_2^*|).$$

Then combining (EC.23) and $f(\mathcal{S}_1^*;\delta_1(\mathcal{S})) \geq f(\mathcal{S}_2^*;\delta_2(\mathcal{S}))$, we have

$$\begin{aligned} u_i\left(r_i - f(\mathcal{S}_2^*;\delta_2(\mathcal{S}))\right) \geq u_i\left(r_i - f(\mathcal{S}_1^*;\delta_1(\mathcal{S}))\right) &\geq \left(g_1(|\mathcal{S}_1^*|) - g_1(|\mathcal{S}_1^*|-1)\right) f(\mathcal{S}_1^*;\delta_1(\mathcal{S})) \\ &\geq \left(g_2(|\mathcal{S}_2^*|+1) - g_2(|\mathcal{S}_2^*|)\right) f(\mathcal{S}_2^*;\delta_2(\mathcal{S})). \end{aligned}$$

This contradicts (EC.24). Thus, $|\mathcal{S}_1^*| \leq |\mathcal{S}_2^*|$. $\square$

The above proposition shows that when the distortion function exhibits the increasing marginal value property, an increase from $\delta_1(\mathcal{S})$ to $\delta_2(\mathcal{S})$, where the marginal increase is decreasing, leads to an expanded optimal assortment. In this case, the rise in the attraction value of the no-purchase option brought by including more items will diminish. This negative effect will be further offset by the enhanced total purchase probability. Hence, expanding the assortment can still be advantageous for maximizing the total expected profit.

EC.6.2 Joint Assortment and Pricing Optimization

In the following, we examine the impact of the increased focal effect on the optimal value and the optimal solution of the joint decision problem (16).

PROPOSITION EC.3. *Suppose that the distortion functions $\delta_1(\mathcal{S}) = g_1(|\mathcal{S}|)$ and $\delta_2(\mathcal{S}) = g_2(|\mathcal{S}|)$ for some functions $g_1(\cdot)$ and $g_2(\cdot)$. Moreover, $g_2(|\mathcal{S}|) \geq g_1(|\mathcal{S}|)$ for all $\mathcal{S} \subseteq \mathcal{N}$, and both $g_1(\cdot)$ and $g_2(\cdot)$ are increasing functions. Let R_1^* and R_2^* be the optimal value to problem (16) with respect to $g_1(\cdot)$ and $g_2(\cdot)$, and $(\mathcal{S}_1^*, \mathbf{p}_1^*)$ and $(\mathcal{S}_2^*, \mathbf{p}_2^*)$ be the corresponding optimal solutions.*

(1) Suppose that $0 \notin \mathcal{N}^*$. Then $R_2^* \geq R_1^*$, and $\mathcal{S}_1^* = \mathcal{S}_2^* = \mathcal{N}$. Moreover, $p_{2i}^* \geq p_{1i}^*$ for $i \in \mathcal{N}$.

(2) Suppose that $\mathcal{N}^* = \{0\}$. Then $R_2^* \leq R_1^*$. If item $i \in \mathcal{S}_1^* \cap \mathcal{S}_2^*$, then $p_{2i}^* \leq p_{1i}^*$.

(3) Suppose that $\mathcal{N}^* = \{0\}$ and $\beta_1 = \beta_2 = \cdots = \beta_n$. When $(1 + g_2(|\mathcal{S}|)) \,/\, (1 + g_1(|\mathcal{S}|))$ is decreasing in $|\mathcal{S}|$, then $|\mathcal{S}_1^*| \leq |\mathcal{S}_2^*|$. Otherwise, $|\mathcal{S}_2^*| \leq |\mathcal{S}_1^*|$.

Proof of Proposition EC.3.

We begin with the first statement. As we will prove in Lemma EC.3 later in EC.5.3, the optimal expected profit R^* satisfies the following equation

$$r = H(r; g(\cdot)),$$

where $H(r; g(\cdot)) = \max_{\mathcal{S}_1 \subseteq \mathcal{N}^* \setminus \{0\}, \mathcal{S}_2 \subseteq \mathcal{N} \setminus \mathcal{N}^*} \; J(r, \mathcal{S}_1, \mathcal{S}_2; g(\cdot))$. Moreover,

$$R^* = J(R^*, \mathcal{S}_1^*, \mathcal{S}_2^*) = \max_{\mathcal{S}_1 \subseteq \mathcal{N}^* \setminus \{0\}, \mathcal{S}_2 \subseteq \mathcal{N} \setminus \mathcal{N}^*} J(R^*, \mathcal{S}_1, \mathcal{S}_2) = H(R^*). \tag{EC.25}$$

Here, we use the notations $H(r; g(\cdot))$ and $J(r, \mathcal{S}_1, \mathcal{S}_2; g(\cdot))$ to show their dependence on function $g(\cdot)$ and to make the proof clear.

When $0 \notin \mathcal{N}^*$, we have

$$H(r; g(\cdot)) = \max_{\mathcal{S}_1 \subseteq \mathcal{N}^* \setminus \{0\}, \mathcal{S}_2 \subseteq \mathcal{N} \setminus \mathcal{N}^*} (1 + g(|\mathcal{S}_1| + |\mathcal{S}_2|)) \sum_{i \in \mathcal{S}_1} \exp(\gamma_i - \beta_i r) + \sum_{i \in \mathcal{S}_2} \exp(\gamma_i - \beta_i r).$$

Denote by $(\mathcal{S}_1^*(r), \mathcal{S}_2^*(r))$ the optimal solution of the above problem. Since $g(\cdot)$ is an increasing function, we have $\mathcal{S}_1^*(r) = \mathcal{N}^* \setminus \{0\}$ and $\mathcal{S}_2^*(r) = \mathcal{N} \setminus \mathcal{N}^*$. Hence, under each r, we need to offer all items. This also holds under the optimal value R^*. Thus, it is optimal to offer all items in $\mathcal{N}$. Therefore, we obtain that $\mathcal{S}_1^* = \mathcal{S}_2^* = \mathcal{N}$. Meanwhile, we have

$$R_k^* = H(R_k^*; g_k(\cdot)), k \in \{1, 2\}.$$

Suppose that $R_1^* > R_2^*$, then

$$R_2^* = H(R_2^*; g_2(\cdot)) > H(R_1^*; g_2(\cdot)) \geq H(R_1^*; g_1(\cdot)) = R_1^*,$$

where the first inequality holds due to that $H(r; g_2(\cdot))$ is decreasing in r, and the second inequality follows from the fact that $g_2(|\mathcal{S}|) \geq g_1(|\mathcal{S}|)$ for all $\mathcal{S} \subseteq \mathcal{N}$. This makes a contradiction. Thus, we have $R_2^* \geq R_1^*$. Furthermore, by Lemma 5, we obtain

$$\begin{aligned} p_{1i}^* &= r_{\mathcal{S}_1^*} + c_i + 1/\beta_i, \\ p_{2i}^* &= r_{\mathcal{S}_2^*} + c_i + 1/\beta_i, \end{aligned} \tag{EC.26}$$

and

$$\begin{aligned} r_{\mathcal{S}_1^*} &= \max_{\mathbf{p} \in \mathcal{P}} R(\mathcal{S}_1^*, \mathbf{p}; g_1(\cdot)) = R(\mathcal{S}_1^*, \mathbf{p}_1^*; g_1(\cdot)) = R_1^*, \\ r_{\mathcal{S}_2^*} &= \max_{\mathbf{p} \in \mathcal{P}} R(\mathcal{S}_2^*, \mathbf{p}; g_2(\cdot)) = R(\mathcal{S}_2^*, \mathbf{p}_2^*; g_2(\cdot)) = R_2^*. \end{aligned}$$

Here we also use the notation $R(\mathcal{S}, \mathbf{p}; g(\cdot))$ to show the dependence of $R(\mathcal{S}, \mathbf{p})$ on function $g(\cdot)$. Together with (EC.26) and $R_2^* \geq R_1^*$, we have $p_{2i}^* \geq p_{1i}^*$.

Now we come to the case that $\mathcal{N}^* = \{0\}$. The joint decision problem (16) becomes

$$\max_{\mathcal{S} \subseteq \mathcal{N}, \mathbf{p} \in \mathcal{P}} R(\mathcal{S}, \mathbf{p}; g(\cdot)) := \frac{\sum_{i \in \mathcal{S}} (p_i - c_i) \exp(\alpha_i - \beta_i p_i)}{(1 + g(|\mathcal{S}|)) + \sum_{i \in \mathcal{S}} \exp(\alpha_i - \beta_i p_i)}. \tag{EC.27}$$

Meanwhile, $H(r; g(\cdot)) = \max_{\mathcal{S} \subseteq \mathcal{N}} J(r, S; g(\cdot))$ with

$$J(r, \mathcal{S}; g(\cdot)) = \frac{\sum_{i \in \mathcal{S}} \exp(\gamma_i - \beta_i r)}{1 + g(|\mathcal{S}|)}.$$

Let $\mathcal{S}^*(r; g(\cdot)) = \arg\max_{\mathcal{S} \subseteq \mathcal{N}} J(r, \mathcal{S}; g(\cdot))$. For given $g_1(\cdot)$ and $g_2(\cdot)$, we have

$$\begin{aligned} (\mathcal{S}_1^*, \mathbf{p}_1^*) &\in \arg\max_{\mathcal{S} \subseteq \mathcal{N}, \mathbf{p} \in \mathcal{P}} R(\mathcal{S}, \mathbf{p}; g_1(\cdot)), \\ \text{and } (\mathcal{S}_2^*, \mathbf{p}_2^*) &\in \arg\max_{\mathcal{S} \subseteq \mathcal{N}, \mathbf{p} \in \mathcal{P}} R(\mathcal{S}, \mathbf{p}; g_2(\cdot)). \end{aligned}$$

Then we have

$$R_2^* = R(\mathcal{S}_2^*, \mathbf{p}_2^*; g_2(\cdot)) \leq R(\mathcal{S}_2^*, \mathbf{p}_2^*; g_1(\cdot)) \leq R(\mathcal{S}_1^*, \mathbf{p}_1^*; g_1(\cdot)) = R_1^*.$$

The first inequality holds since $g_2(|\mathcal{S}|) \geq g_1(|\mathcal{S}|)$ for all $\mathcal{S} \subseteq \mathcal{N}$ and $R(\mathcal{S}, \mathbf{p}; g(\cdot))$ is decreasing in $g(\cdot)$ for fixed $\mathcal{S}$ and $\mathbf{p}$ by its definition in (EC.27). The second inequality follows from the definition of $(\mathcal{S}_1^*, \mathbf{p}_1^*)$. Hence, we obtain $R_2^* \leq R_1^*$. Furthermore, if item $i \in \mathcal{S}_1^* \cap \mathcal{S}_2^*$, by the similar argument to prove the first statement, we have $p_{2i}^* \leq p_{1i}^*$.

Now we prove the last claim. Given $r > 0$ and two assortments $\mathcal{S}_1$, $\mathcal{S}_2$, we define the following function,

$$\begin{aligned} \phi(r, \mathcal{S}_1, \mathcal{S}_2; g(\cdot)) &:= \frac{J(r, \mathcal{S}_1; g(\cdot))}{J(r, \mathcal{S}_2; g(\cdot))} \\ &= \frac{\sum_{i \in \mathcal{S}_1} \exp(\gamma_i - \beta_i r)}{\sum_{j \in \mathcal{S}_2} \exp(\gamma_j - \beta_j r)} \cdot \frac{1 + g(|\mathcal{S}_2|)}{1 + g(|\mathcal{S}_1|)} \\ &= \frac{\sum_{i \in \mathcal{S}_1} \exp(\gamma_i)}{\sum_{j \in \mathcal{S}_2} \exp(\gamma_j)} \cdot \frac{1 + g(|\mathcal{S}_2|)}{1 + g(|\mathcal{S}_1|)}. \end{aligned}$$

The last equality holds due to the assumption that $\{\beta_i\}_{i=1}^n$ are homogeneous among items. Although the function ϕ is independent of r, we keep r in the parentheses for a clear proof. Denote by R_1^* and R_2^* the optimal value to problem (16) corresponding to $g_1(\cdot)$ and $g_2(\cdot)$, respectively. By (EC.25), we have $\mathcal{S}_k^* \in \arg\max_{\mathcal{S}\subseteq\mathcal{N}} J(R_k^*, \mathcal{S}; g_k(\cdot)), \forall k=1,2$. Thus,

$$\phi(R_1^*, \mathcal{S}, \mathcal{S}_1^*; g_1(\cdot)) \le 1 \quad \text{and} \quad \phi(R_2^*, \mathcal{S}_2^*, \mathcal{S}; g_2(\cdot)) \ge 1 \quad, \forall \mathcal{S} \subseteq \mathcal{N}. \tag{EC.28}$$

When $(1+g_2(|\mathcal{S}|))\,/\,(1+g_1(|\mathcal{S}|))$ is decreasing in $|\mathcal{S}|$, we have

$$\frac{(1+g_2(|\mathcal{S}_2|))\,/\,(1+g_1(|\mathcal{S}_2|))}{(1+g_2(|\mathcal{S}_1|))\,/\,(1+g_2(|\mathcal{S}_1|))} \le 1. \tag{EC.29}$$

for all $\mathcal{S}_1$, $\mathcal{S}_2 \subseteq \mathcal{N}$ satisfy $|\mathcal{S}_1| \le |\mathcal{S}_2|$. Suppose that $|\mathcal{S}_2^*| < |\mathcal{S}_1^*|$ in this case, then we have

$$\begin{aligned}
1 \le \phi(R_2^*, \mathcal{S}_2^*, \mathcal{S}_1^*; g_2(\cdot)) &= \frac{\sum_{j\in\mathcal{S}_2^*} \exp(\gamma_j)}{\sum_{i\in\mathcal{S}_1^*} \exp(\gamma_i)} \cdot \frac{1+g_2(|\mathcal{S}_1^*|)}{1+g_2(|\mathcal{S}_2^*|)} \\
&= \frac{\sum_{j\in\mathcal{S}_2^*} \exp(\gamma_j)}{\sum_{i\in\mathcal{S}_1^*} \exp(\gamma_i)} \cdot \frac{1+g_1(|\mathcal{S}_1^*|)}{1+g_1(|\mathcal{S}_2^*|)} \cdot \frac{(1+g_1(|\mathcal{S}_2^*|))\,(1+g_2(|\mathcal{S}_1^*|))}{(1+g_1(|\mathcal{S}_1^*|))\,(1+g_2(|\mathcal{S}_2^*|))} \\
&= \frac{\sum_{j\in\mathcal{S}_2^*} \exp(\gamma_j)}{\sum_{i\in\mathcal{S}_1^*} \exp(\gamma_i)} \cdot \frac{1+g_1(|\mathcal{S}_1^*|)}{1+g_1(|\mathcal{S}_2^*|)} \cdot \frac{(1+g_2(|\mathcal{S}_1^*|))/(1+g_1(|\mathcal{S}_1^*|))}{(1+g_2(|\mathcal{S}_2^*|))/(1+g_1(|\mathcal{S}_2^*|))} \\
&< \phi(R_2^*, \mathcal{S}_2^*, \mathcal{S}_1^*; g_1(\cdot)) \\
&= \phi(R_1^*, \mathcal{S}_2^*, \mathcal{S}_1^*; g_1(\cdot)).
\end{aligned}$$

The second inequality holds by (EC.29). This makes a contradiction to (EC.28), implying that $|\mathcal{S}_1^*| \le |\mathcal{S}_2^*|$.

With a similar argument, we can prove the second statement. When $(1+g_2(|\mathcal{S}|))\,/\,(1+g_1(|\mathcal{S}|))$ is increasing in $|\mathcal{S}|$, we have

$$\frac{(1+g_2(|\mathcal{S}_1|))\,/\,(1+g_1(|\mathcal{S}_1|))}{(1+g_2(|\mathcal{S}_2|))\,/\,(1+g_2(|\mathcal{S}_2|))} \le 1, \tag{EC.30}$$

for all $\mathcal{S}_1$, $\mathcal{S}_2 \subseteq \mathcal{N}$ satisfy $|\mathcal{S}_1| \le |\mathcal{S}_2|$. Suppose that $|\mathcal{S}_1^*| < |\mathcal{S}_2^*|$, then we have

$$\begin{aligned}
1 \le \phi(R_2^*, \mathcal{S}_2^*, \mathcal{S}_1^*; g_2(\cdot)) &= \frac{\sum_{j\in\mathcal{S}_2^*} \exp(\gamma_j)}{\sum_{i\in\mathcal{S}_1^*} \exp(\gamma_i)} \cdot \frac{1+g_1(|\mathcal{S}_1^*|)}{1+g_1(|\mathcal{S}_2^*|)} \cdot \frac{(1+g_2(|\mathcal{S}_1^*|))/(1+g_1(|\mathcal{S}_1^*|))}{(1+g_2(|\mathcal{S}_2^*|))/(1+g_1(|\mathcal{S}_2^*|))} \\
&< \phi(R_2^*, \mathcal{S}_2^*, \mathcal{S}_1^*; g_1(\cdot)) \\
&= \phi(R_1^*, \mathcal{S}_2^*, \mathcal{S}_1^*; g_1(\cdot)).
\end{aligned}$$

This also makes a contradiction to (EC.28), and thus $|\mathcal{S}_2^*| \le |\mathcal{S}_1^*|$. □

Claim (1) shows that when the focal set excludes the no-purchase option, offering all items becomes the optimal strategy since including more items reduces the market share of the no-purchase option and increases the demand for the focal items. This finding aligns with the results for the MNL model and the top-K focal model. Moreover, if the focal effect increases, the seller should raise the price of each item. The rationale behind is that increasing the price of non-focal items will decrease their demand, thus redirecting more consumer attention towards the focal items. By simultaneously increasing the prices of these focal items, the seller can leverage the heightened demand and ultimately achieve higher expected profit.

However, it may not be optimal to provide all items when $\mathcal{N}^* = \{0\}$ due to choice overload. Claim (2) indicates that the optimal expected profit decreases as choice overload becomes more severe. Furthermore, if an item i is in both $\mathcal{S}_1^*$ and $\mathcal{S}_2^*$, then the optimal price for item i in $\mathcal{S}_2^*$ should be lower than that in $\mathcal{S}_1^*$. This suggests that the seller can use price as a way to compensate for the item's attraction value loss caused by choice overload, as indicated in Malone and Lusk (2019).

Claim (3) further shows that when all items share the same price sensitivity and the distortion function $g_2(\cdot)$ increases more slowly than $g_1(\cdot)$ under the choice overload case, the optimal assortment will expand. Otherwise, the optimal assortment will shrink. Hence, in the joint decision problem, the optimal assortment size is influenced not only by the magnitude of the distortion function but also by its rate of increase.

EC.7 Details and Additional Numerical Experiments of Expedia's Dataset

EC.7.1 Data Preprocessing

In this subsection, we provide the details about preprocessing the Expedia dataset. We only concentrate on search queries for a single-night stay and exclude the other ones. Then, we focus on the search queries whose displayed hotels are determined by Expedia's ranking algorithm. Additionally, we eliminate features with entries missing for more than 25% of the rows. With the remaining features, we remove search queries with missing entries in any of these features. Subsequently, we exclude search queries containing entries outside the 0.5-th to 99.5-th percentile of entries in the corresponding column. After dataset preprocessing, we obtain 14 different features and $689,883$ rows representing $30,346$ search queries. The first two features are the unique code for each search query and the indicator of whether the hotel is booked in the search query.

The remaining 12 features provide information on the hotel's star rating, average review score, an indicator of whether the hotel is part of a chain, two location desirability scores, average hotel price over the last trading period, displayed price, promotion indicator, the number of days until the day of stay, the number of adults and children in the search query, and an indicator for whether the stay is over the weekend.

EC.7.2 An Example for Price Ranking

In this subsection, we provide an example in Expedia's dataset to show that $\mathcal{S}^*_{\text{FLM}} \subseteq \mathcal{S}^*_{\text{MNL}}$ as mentioned in Section 6.1. Let us take site 32 as an example. There are 34 hotels in total, and we assume that the price of each hotel is the profit. When estimating the top-K focal model, we try the value of K in the set $\{1,3,5,7,10\}$, and finally let $K=3$ based on validation results. Then, we obtain $\mathcal{S}^*_{\text{MNL}} = \{1,2,\dots,34\}$ while $\mathcal{S}^*_{\text{FLM}} = \{4,5,\dots,34\}$, where the hotels are indexed such that $r_1 < r_2 < \cdots < r_{34}$. The detailed profits are provided in Table EC.1. Notably, $\mathcal{S}^*_{\text{FLM}}$ includes hotel $1,2,3$ in $\mathcal{S}^*_{\text{MNL}}$ since hotel 1, 2 and 3 have profits 69.370, 71.220 and 71.220, respectively, which are much less than the profit 85.09 of hotel 4.

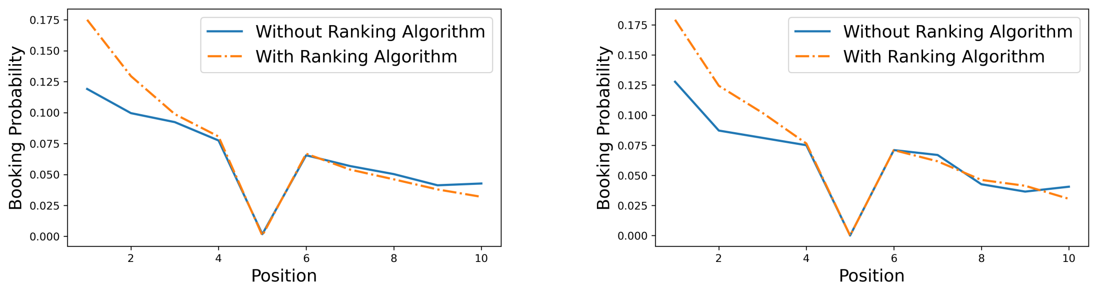
hotel	profit	hotel	profit	hotel	profit	hotel	profit
1	69.37	11	119.31	21	149.83	31	179.43
2	71.22	12	119.31	22	149.83	32	200.70
3	71.22	13	119.31	23	154.46	33	249.72
4	85.09	14	120.24	24	155.38	34	259.90
5	97.11	15	122.09	25	164.63		
6	109.14	16	127.64	26	164.63		
7	110.99	17	129.49	27	164.63		
8	115.61	18	139.66	28	170.18		
9	118.39	19	139.66	29	173.88		
10	119.31	20	140.58	30	179.43		

Table EC.1 Hotels' profit.

EC.7.3 Position Ranking

In this subsection, we consider position ranking under the top-K focal model, that is, the focal items of a given assortment are the items with the top-K positions. This phenomenon has been widely observed in real life and is called the *position bias*. For example, Feng et al. (2007) show that customers' attention to an item decreases exponentially with its distance to the top, and the items near the top of the list are more likely to be focal.

This model slightly generalizes our top-K focal model since each customer may face a different ranking. In our analysis, we restrict our attention to search queries whose displayed hotels are determined by Expedia's algorithm because the position bias is more significant, which we will demonstrate later. We utilize the same preprocessing approach introduced in Section 6.1 and obtain $1,015,068$ entries with 15 different features. The one more feature included represents the displayed positions of hotels in each search query. We focus on the 5 most popular sites based on the number of search queries and use the same estimation method as Section 6.1 to estimate the MNL model and FLM.



(a) Site 5 (b) Site 14

Figure EC.1 The empirical booking probability of hotels in the top 10 positions of 2 representative sites with and without Expedia's ranking algorithm. Position 5 displays the opaque hotel (Ursu 2018), and almost no customer books or even clicks this hotel.

Site id	Algorithm	1	2	3	4	5	6	7	8	9	10
5	with ranking algorithm	4609	3410	2602	2125	18	1759	1422	1214	999	842
	without ranking algorithm	329	275	255	214	5	181	157	139	114	118
14	with ranking algorithm	811	563	461	346	0	321	279	209	187	138
	without ranking algorithm	63	43	40	37	0	35	33	21	18	20

Table EC.2 The number of bookings occurring in the top-10 positions of 2 representative sites. The last ten columns provide the number of bookings on position i, $i = 1, 2, \ldots, 10$.

Position Bias Verification. We verify the existence of the position bias in the Expedia dataset. In Figure EC.1, we plot the empirical booking probability of hotels occupying the top 10 positions in 2 sites. The orange dash-dot line illustrates the booking probability when Expedia's ranking algorithm is employed, while the blue solid line depicts the results with random sorting. We provide the number of bookings occurring in each top 10 position in Table EC.2. It shows that customers are more likely to book hotels at the top positions. Additionally, the position bias is more significant when the display ranking is determined by Expedia's ranking algorithm due to the engagement of the platform.

To further test the significance of the influence of display positions of hotels on customers' choices, we construct a new binary variable and perform a significance test on this new variable. For demonstration, we set $K = 10$, and introduce a binary variable `is_top_10` to represent whether the hotel is displayed in top-10 position in the search query. For all five sites, we conduct a likelihood ratio test by estimating two MNL models—one with `is_top_10` and one without it. Additionally, we also compute McFadden's adjusted R^2 before and after adding the binary variable `is_top_10`. McFadden's adjusted R^2 is the change in terms of log-likelihood from the intercept-only model to the current model. The larger this value is, the better the model fits the data.

We report the test results in Table EC.3. The first column represents different sites, and the second column gives the p-values of the null hypothesis that there does not exist position bias. The third column provides McFadden's adjusted R^2 of the estimated MNL models. The numbers outside

Site	p-value	McFadden's adjusted R^2
1	10^{-4}	0.091 (0.032)
2	10^{-4}	0.093 (0.036)
3	10^{-4}	0.087 (0.019)
4	10^{-4}	0.072 (0.012)
5	10^{-4}	0.079 (0.018)

Table EC.3 The position bias verification results of the 5 most popular sites of the Expedia dataset.

Expedia Site	5	14	15	24	32
Max assortment cardinality	38	35	36	31	35
# search queries	27365	4681	1735	1495	822
MNL					
Log value	-1.8916	-1.8490	-1.9127	-1.9640	-1.9251
RMSE	0.1895	0.1930	0.1897	0.1938	**0.1867**
FLM					
Log value	**-1.8113**	**-1.7825**	**-1.8085**	**-1.8570**	**-1.8300**
RMSE	**0.1889**	**0.1923**	**0.1891**	**0.1934**	0.1869

Table EC.4 Comparison of the predictive power of the MNL model and FLM on the Expedia dataset.

the brackets correspond to the MNL model with the variable `is_top_10`, while the numbers in the brackets are related to the MNL model without the variable. The results show that the p-values are all below the significance level of 0.05 for all 5 sites and McFadden's adjusted R^2 with the MNL model incorporating `is_top_10` is larger. Therefore, we should reject the null hypothesis. It suggests that the variable `is_top_10` significantly affects customers' choices, thereby establishing the presence of position bias.

When calibrating these two models, we conduct the experiments 20 times and take the average of the normalized log-likelihood values on the training dataset and the RMSE values on the testing dataset for each site. Table EC.4 shows that the FLM outperforms the MNL model over almost all 5 datasets on both "Log value" and "RMSE" criteria. Thus, FLM has a better fitting and prediction ability than the MNL model.

In addition to the comparison of log-likelihood and RMSE, we also compute the optimal assortment and position of each item under the FLM with position ranking. This problem can be transformed into a linear program and hence can be solved efficiently. We also remark that the joint assortment and position optimization problem has multiple solutions, as it only identifies the items placed in the first K positions, i.e., the focal items. Exchanging the positions between the focal items or the positions between the non-focal items does not affect the expected profit. We find that under the FLM, the optimal position ranking will place hotels with both large profit and large attraction value in the first K position.

We still take site 32 as an example, which includes a total of 34 hotels. We try the value of K in the set $\{1, 3, 5, 7, 10\}$, and finally let $K = 3$ based on validation results. The detailed profits and the estimated attraction values under the FLM are provided in Table EC.5. We find that hotel 31, hotel 23 and hotel 8 will be placed in the top-3 positions and hence are focal items. To understand this result, we find that hotel 31 and hotel 23 have relatively large profit, while hotel 8 has the largest attraction value. Additionally, these three hotels achieve the highest product of price and attraction value. Therefore, it can be seen that the FLM with position ranking will place items of either large attraction value or large profit to the top positions.

hotel	$(r_i, u_i, \texttt{FLM})$	hotel	$(r_i, u_i, \texttt{FLM})$
1	(69.37, 0.0063)	18	(139.66, 0.0065)
2	(71.22, 0.0046)	19	(139.66, 0.0107)
3	(71.22, 0.0057)	20	(140.58, 0.0056)
4	(85.09, 0.0084)	21	(149.83, 0.0066)
5	(97.11, 0.0100)	22	(149.83, 0.0073)
6	(109.14, 0.0077)	**23**	**(154.46, 0.0123)**
7	(110.99, 0.0100)	24	(155.38, 0.0069)
8	**(115.61, 0.0166)**	25	(164.63, 0.0079)
9	(118.39, 0.0113)	26	(164.63, 0.0092)
10	(119.31, 0.0064)	27	(164.63, 0.0088)
11	(119.31, 0.0070)	28	(170.18, 0.0079)
12	(119.31, 0.0050)	29	(173.88, 0.0031)
13	(119.31, 0.0062)	30	(179.43, 0.0074)
14	(120.24, 0.0049)	**31**	**(179.43, 0.0091)**
15	(122.09, 0.0055)	32	(200.70, 0.0050)
16	(127.64, 0.0064)	33	(249.72, 0.0064)
17	(129.49, 0.0011)	34	(259.90, 0.0040)

Table EC.5 Hotels' profits and estimated attraction values under the FLM with position ranking. $(r_i, u_i, \texttt{FLM})$ stands for the profit of hotel i and the estimated attraction value under the FLM with the position ranking.

EC.8 Parameter Estimation of the Choice Overload Case

In this section, we give the parameter estimation method for the choice overload model used in Section 6.2. Under this model, the purchase probability of item $i \in \mathcal{S}$ is given by

$$\frac{\exp(v_i)}{\exp(v_0)\left(1+\exp(\alpha|\mathcal{S}|)\right)+\sum_{i\in\mathcal{S}}\exp(v_i)},$$

and the purchase probability of the no-purchase option is

$$\frac{\exp(v_0)\left(1+\exp(\alpha|\mathcal{S}|)\right)}{\exp(v_0)\left(1+\exp(\alpha|\mathcal{S}|)\right)+\sum_{i\in\mathcal{S}}\left(\exp(v_i)\right)},$$

where v_i is the utility of the item i, $i \in \mathcal{N}_+$. Without loss of generality, we let $v_0 = 0$.

Suppose we have access to the dataset of customers' purchase history $\mathcal{D} = \{(\mathcal{S}_t, i_t) : t = 1, 2, \ldots, T\}$, where T is the number of the customers, $\mathcal{S}_t$ is the assortment offered to customer t, and i_t is the item purchased by customer t. We use the maximum likelihood estimation to fit our model and maximize the logarithm of the corresponding likelihood function. Assuming that the purchase decisions of different customers are independent of each other, the log-likelihood function of the dataset $\mathcal{D}=\{(\mathcal{S}_t,i_t):t=1,2,\dots,T\}$ is given by

$$L(\boldsymbol{v},\alpha)=\sum_{t\in T_0}\log\left(1+\exp\left(\alpha|\mathcal{S}_t|\right)\right)+\sum_{t\in T_1}v_{i_t}-\sum_{t=1}^{T}\log\left(\exp\left(\alpha|\mathcal{S}_t|\right)+\sum_{i\in\mathcal{S}_t\cup\{0\}}\exp\left(v_i\right)\right),$$

where T_0 represents the subset of index set $[T]$ where the customers choose the no-purchase option, and $T_1=T\setminus T_0$ is the index set that the customers purchase other items. Then we need to solve the following optimization problem:

$$\max_{\boldsymbol{v},\alpha>0} L(\boldsymbol{v},\alpha).$$

Since $L(\boldsymbol{v},\alpha)$ is not jointly concave in $(\boldsymbol{v},\alpha)$, we use the alternating optimization algorithm to solve the problem above. The idea is inspired by Goutam et al. (2019). When we fix α, the maximization problem over $\boldsymbol{v}$ becomes

$$\max_{\boldsymbol{v}} \quad \sum_{t\in T_1}v_{i_t}-\sum_{t=1}^{T}\log\left(M_t+\sum_{i\in\mathcal{S}_t\cup\{0\}}\exp\left(v_i\right)\right), \tag{EC.31}$$

where $M_t:=\exp\left(\alpha|\mathcal{S}_t|\right)$ for all $t\in[T]$. The objective function is a sum of the linear function of $\boldsymbol{v}$ and the negative log-sum-exp function, and thus concave. The problem (EC.31) is a convex program and can be solved efficiently by some commercial solvers such as Mosek. On the other hand, when we fix $\boldsymbol{v}$, the maximization problem over α becomes

$$\max_{\alpha>0} \quad \sum_{t\in T_0}\log\left(1+\exp\left(\alpha|\mathcal{S}_t|\right)\right)-\sum_{t=1}^{T}\log\left(\exp\left(\alpha|\mathcal{S}_t|\right)+N_t\right), \tag{EC.32}$$

where $N_t:=\sum_{i\in\mathcal{S}_t}\exp\left(v_i\right)$ for all $t\in[T]$. This is a one-dimensional possibly non-convex optimization problem, and we use the MATLAB built-in function `fmincon`, which is an optimization routine that only needs to evaluate the value of the objective function, to solve the above problem. Therefore, one can apply an iterative algorithm to keep maximizing the log-likelihood function over $\boldsymbol{v}$ and α alternatively until some certain convergence criterion is satisfied. To start the algorithm, a reasonable initial point of the utility vector $\boldsymbol{v}$ can be set as the maximum likelihood estimator obtained by fitting the MNL model. The detail of the algorithm is presented in Algorithm 5.

EC.9 Profit Performance

In this section, we use synthetic datasets to compare the profit performance of the MNL model, the GAM and the FLM. Instead of restricting our numerical study to the case where the focal effect exists, we examine the performance of these models in a broader scenario where the regularity condition is violated in customer choice behavior.

Algorithm 5: The Alternating Algorithm for Parameter Estimation

Input: The purchase history dataset $\mathcal{D} = \{(\mathcal{S}_t, i_t) : t = 1, 2, \ldots, T\}$

Use the maximum likelihood estimation to fit the MNL model to the given dataset $\mathcal{D}$, and let $\boldsymbol{v}^{\text{MNL}} \in \mathbb{R}^n$ be the estimated utility vector of items

$\boldsymbol{v}^{(0)} \leftarrow \boldsymbol{v}^{\text{MNL}}$

$\alpha^{(0)} \leftarrow 0$

for $k = 1, 2, \ldots$ **do**

Solve problem (EC.32) with $\boldsymbol{v} = \boldsymbol{v}^{(k-1)}$ and let $\alpha^{(k)}$ be its optimal solution

Solve problem (EC.31) with $\alpha = \alpha^{(k)}$ and let $\boldsymbol{v}^{(k)}$ be its optimal solution

if *some convergence criterion is satisfied* **then**

break

return *The estimated parameters* $\boldsymbol{v}^{(k)}$ *and* $\alpha^{(k)}$

Experiment Setup. We choose the generalized stochastic preference (GSP) model (Berbeglia and Venkataraman 2018) as the ground-truth model. This model incorporates non-rationality into the stochastic preference model and can capture regularity violations.

The GSP model assumes that there are multiple customer types, and each customer type is described by a pair $\omega = (\boldsymbol{\ell}, k)$, where $\boldsymbol{\ell}$ is an ordering of n items, and the choice index k is an integer between 1 and n. We use Ω to denote the customer type set. Given an assortment $\mathcal{S}$, the customers of type $\omega = (\boldsymbol{\ell}, k)$ first construct a subsequence $s(\boldsymbol{\ell}, \mathcal{S})$ of $\boldsymbol{\ell}$ by removing all items that are not in $\mathcal{S}$. Then, they choose the item at the k-th position in the subsequence $s(\boldsymbol{\ell}, \mathcal{S})$ if it exists. Otherwise, the customer will select the last item in $s(\boldsymbol{\ell}, \mathcal{S})$. We then define the binary choice function $C_\omega(j, \mathcal{S})$ as follows, which states whether customer of type ω chooses items $j \in \mathcal{S}$,

$$C_\omega(j, \mathcal{S}) := \mathbf{1}\{\text{pos}(j, \mathcal{S}; \boldsymbol{\ell}) = \min(k, |\mathcal{S}|)\} = \begin{cases} 1 & \text{if } j \text{ is at position } \min\{k, |\mathcal{S}|\} \text{ in } s(\boldsymbol{\ell}, \mathcal{S}) \\ 0 & \text{otherwise} \end{cases},$$

where $\text{pos}(j, \mathcal{S}; \boldsymbol{\ell})$ is the position of item $j \in \mathcal{S}$ in the subsequence $s(\boldsymbol{\ell}, \mathcal{S})$. Hence, the choice probability of item $j \in \mathcal{S}$ is given by $\mathbb{P}(i, \mathcal{S}) = \sum_{\omega \in \Omega} C_\omega(j, \mathcal{S}) \cdot \lambda(\omega)$, where $\lambda(\cdot)$ is the probability distribution over the set of customer types Ω. Let $[n] := \{1, \ldots, n\}$

In the numerical study, we let $|\Omega| = m$. For each customer type w, the choice index $k \in [2]$ is generated with the same probability 0.5. We restrict $k \in [2]$ because Berbeglia and Venkataraman (2018) show that the GSP model with small choice indices is rich enough to capture irrational choice behavior in practice. To generate the ranking of each customer type, we first assume that items have an inherent ordering such that each item j has a higher quality and a higher price than item $j-1$ for all $j \in \{2, 3, \ldots, n\}$. Then, we follow Cao et al. (2022) and assume that the ranking $\boldsymbol{\ell}$ of each customer type w has the form $i_w \succ i_w + 1 \succ \cdots \succ j_w - 1 \succ j_w \succ 0$. This indicates that each

$\mathcal{S}$	$\hat{P}(1,\mathcal{S})$	$\hat{P}(2,\mathcal{S})$	$\hat{P}(3,\mathcal{S})$	$\hat{P}(4,\mathcal{S})$	$\hat{P}(5,\mathcal{S})$	$\hat{P}(6,\mathcal{S})$	$\hat{P}(7,\mathcal{S})$	$\hat{P}(8,\mathcal{S})$	$\hat{P}(9,\mathcal{S})$	$\hat{P}(10,\mathcal{S})$
[1]	0.1383	0	0	0	0	0	0	0	0	0
[2]	0.0670	0.2049	0	0	0	0	0	0	0	0
[3]	0.0664	0.1360	0.1965	0	0	0	0	0	0	0
[4]	0.0664	**0.1362**	0.1355	0.2033	0	0	0	0	0	0
[5]	**0.0682**	**0.1367**	0.1342	0.0672	0.1946	0	0	0	0	0
[6]	0.0680	0.1349	0.1338	0.0634	**0.1988**	0.2010	0	0	0	0
[7]	0.0680	0.1327	0.1324	**0.0640**	**0.1997**	0.0633	0.2019	0	0	0
[8]	0.0650	0.1327	0.1271	**0.0704**	0.1983	**0.0651**	**0.2090**	0.0674	0	0
[9]	**0.0688**	0.1312	**0.1364**	0.0623	**0.2040**	0.0646	0.2034	0.0665	0.0628	0
[10]	0.0643	**0.1333**	0.1316	**0.0689**	0.1990	**0.0693**	0.2006	**0.0678**	0	0.0652

Table EC.6 Choice probabilities for one realization of the ground-truth choice model. The choice probability of each item in $[n]$ that is larger than the one in $[n-1]$ is bolded for each $n=2,3,\dots,10$.

type of customer has a maximum affordable price and a minimum required quality. Therefore, the customer of type w does not consider items with prices higher than item j_w and does not consider items with quality lower than item i_w. For each w, i_w is generated according to the discrete uniform distribution on $[n-1]$, and then j_w is generated according to the discrete uniform distribution on $\{i_w+1,\dots,n\}$.

Then, we generate the i.i.d. choice dataset according to the GSP model. For each $t\in[T]$, the type of customer t is generated according to the discrete uniform distribution on $[m]$, where T denotes the number of customers in the dataset. Then we randomly generate assortment $\mathcal{S}_t$. Specifically, we generate a positive integer κ as the cardinality of $\mathcal{S}_t$ according to the discrete uniform distribution on $[n]$, then we uniformly pick an assortment of size κ as $\mathcal{S}_t$. Given $\mathcal{S}_t$, the customer t will follow the GSP model and purchase the k-th cheapest item in the subsequence $s(\boldsymbol{\ell},\mathcal{S}_t)$, denoted by j_t. Thus, the choice dataset consists of the pairs $\{(\mathcal{S}_t,j_t): t=1,2,\dots,T\}$. To facilitate the estimation of the FLM, we also generate the validation dataset of length $T/4$ with the same procedure.

Table EC.6 shows the empirical choice probabilities of items under a realization of the ground-truth choice model with $n=10$ items for different assortments $\mathcal{S}$ and different items j. The first column represents different assortments. The remaining columns correspond to the choice probability, and $\hat{P}(j,\mathcal{S})$ is the empirical choice probability of item j when assortment $\mathcal{S}$ is offered. Since the assortment is expanding as the row number increases, if the regularity condition holds, the choice probability of each item should decrease as the row number increases. However, the bold numbers indicate a choice probability increase. Thus, the regularity condition is violated.

Model. We consider the arbitrarily-located focal model, whose focal function is of form (10), and the MNL model. We let $h(|\mathcal{S}|)=\tau$ in (10), i.e., the focal item occupies the τ-th position in $\mathcal{S}$ with respect to some ranking. We also let the distortion function be a positive constant.

Model Calibration. We estimate the two models by MLE. For the MNL model, we directly estimate the attraction value of each item and use the algorithm in Berbeglia et al. (2022).

When estimating the FLM, we let $\tau \in \{1,2\}$. We use the price ranking from the cheapest to the most expensive $1 \succ 2 \succ \cdots \succ n \succ 0$, meaning that the customers prefer the cheaper item. For each combination of $\tau \in \{1,2\}$ and ranking σ, we use MLE to fit the FLM to the training dataset and evaluate the log-likelihood of the fitted choice model on the validation dataset. Finally, we choose the pair (τ,σ) that yields the largest log-likelihood on the validation dataset.

We consider 5 different training set sizes, and let $T \in \{1000, 2000, 3000, 4000, 5000\}$. For each T, we fix $n = 10$ and $m = 15$, and generate 10 independent ground-truth choice models and the corresponding training datasets. We also generate another batch of $T/2$ customers as the testing dataset for each T and each ground-truth choice model. Then, we use the training datasets to compare the two choice models in terms of the normalized log-likelihood values and calculate the RMSE values of the two choice models on the testing datasets.

Prediction Performance. The result is presented in Table EC.7. For each T, we average the normalized log-likelihood values and the RMSE values over 10 independent tests. It is evident that the FLM outperforms the MNL model in terms of the log value and RMSE across all training dataset sizes. While the GAM performs better in terms of "Log value", its RMSE is consistently higher than that of the FLM. This suggests that the GAM may overfit the data, resulting in weaker generalization ability compared to the FLM.

Training dataset size	1000	2000	3000	4000	5000
MNL					
Log value	-1.5945	-1.5975	-1.5988	-1.6018	-1.6018
RMSE	0.3323	0.3319	0.3315	0.3314	0.3311
GAM					
Log value	**-1.5698**	**-1.5741**	**-1.5761**	**-1.5797**	**-1.5799**
RMSE	0.3313	0.3308	0.3303	0.3301	0.3298
FLM					
Log value	-1.5804	-1.5840	-1.5858	-1.5879	-1.5878
RMSE	**0.3309**	**0.3304**	**0.3299**	**0.3297**	**0.3294**

Table EC.7 Comparison of the predictive power of the MNL model, the GAM and the FLM on the synthetic dataset $(n = 10, m = 15)$.

Assortment profit Performance. We compare the MNL model, the GAM and the FLM in terms of the profit performance of their chosen assortments. For each training set size, we generate 10 profit samples under each ground-truth choice model generated before, and thus obtain 100 profit samples. For each profit sample k, we generate n i.i.d. profits according to the uniform distribution over $(0,100)$, and then sort them to ensure that $r_1^{(k)} < r_2^{(k)} < \cdots < r_n^{(k)}$. After fitting these three models with the generated choice dataset, we compute the optimal assortments under each model and let $R_z^{(k)}$ be the optimal profit under these three models, where $z \in \{\text{MNL}, \text{GAM}, \text{FLM}\}$. To compute the optimal assortment under our arbitrarily-located focal model, we employ the algorithm proposed by Section 3. To compute the optimal assortment under the GAM, we use the algorithm proposed in Gallego et al. (2015). As Berbeglia and Venkataraman (2018) do not provide any algorithm for solving the assortment optimization problem under the GSP model, we use the brute-force search to find the optimal assortment and denote its optimal profit by $R_{\text{GSP}}^{(k)}$. We compute the ratios $R_z^{(k)}/R_{\text{GSP}}^{(k)}$, and average these ratios over the profit samples $k = 1, 2, \ldots, 100$.

The result is summarized in Table EC.8. The first row represents different training dataset sizes. The other rows provide the profit ratios of the MNL model, the GAM and the FLM, respectively. It demonstrates that the FLM has consistently superior profit performance compared to the MNL model. As the training dataset size increases, the performance of FLM becomes better and outperforms the GAM.

Training dataset size	1000	2000	3000	4000	5000
MNL	80.53%	81.54%	82.67%	87.80%	87.91%
GAM	**81.49%**	**82.58%**	83.05%	88.15%	88.17%
FLM	81.00%	82.23%	**83.10%**	**88.22%**	**88.30%**

Table EC.8 Comparison of the profit performance of the MNL model, the GAM and the FLM on the synthetic dataset $(n = 10, m = 15)$.